Zeros of polyanalytic polynomials

A consolidated research note,
literature review, and proof audit

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Abstract. We study distinct zeros of $f(z, \bar{z})$ with prescribed analytic and conjugate degrees. We give the bound $\min\{m^2 + n^2, (4m + 1)n - 5m\}$ and determine its optimal asymptotic coefficient. Sharp finite-degree constructions attain $m^2 + n^2$ for $m \geq 2$, $m < n \leq 2m + 2$, and for $n = 2m + 3$ when $3 \mid m$. Further bounds account for the Newton support and for the gap in total degree. For harmonic polynomials we give degree-uniform bounds of order mn, an exact count for a monomial-head family, and a Chebyshev construction. We also determine extremal counts in real-valued and cyclic subfamilies, disprove a published sparsity conjecture, and describe exact rational certificates. The note distinguishes complete proofs, computational inputs, classical results, and the finite-degree questions that remain unresolved.

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1 The problem, the literature, and the scope of the conclusions

Allowing a polynomial to depend on both z and $\bar{z}$ changes its zero set fundamentally. It may have no zeros, finitely many isolated zeros, or an entire real algebraic arc of zeros. Even when the zero set is finite, an analytic degree is not by itself the total degree, and different coefficient spaces have different extremal counts. The purpose of this note is to put the successive results of the research discussion into one proof structure, with those distinctions made explicit.

1.1 Notation and the original question

Throughout, the independent analytic variable is written first:

$$f \in \mathbb{C}[z,w], \qquad F_f(z) = f(z,\bar{z}), \qquad f^*(z,w) = \overline{f(\bar{w},\bar{z})}, \qquad N(f) = \#F_f^{-1}(0).$$

The ordered bidegree is $(\deg_w f, \deg_z f) = (m,n)$: the conjugate degree is listed first, although z is written first in $f(z,w)$. For a multivariate polynomial, an unqualified deg denotes total degree. The notation $N(F)$ is used interchangeably with $N(f)$ when $F = F_f$. Every zero count in this note counts distinct points, unless a local algebraic multiplicity is explicitly mentioned. We write $f^{\#}(w) = \overline{f(\bar{w})}$ only for a univariate polynomial.

The original rectangular problem is the following.

Problem 1.1. *Fix $m \geq 1$. For polynomials of bidegree (m,n), $n > m$, with $N(f) < \infty$, does there exist a bound $N(f) \leq C_1(m)n + C_2(m)$ independent of the coefficients and of n?*

As of the source check for this note, Eremenko's maintained problem list marks the non-holomorphic-equation estimate as solved using AI by reduction to the Thom genus inequality [Ere26b]. That status notice is not a replacement for the proof below, and does not certify the stronger finite-degree assertions in this research record.

This is the polynomial conjecture on page 1 of Eremenko's note dated September 7, 2026 [Ere26, p. 1], with the finiteness assumption stated explicitly. It is not the elliptic-system question on the following pages of that note. Two corrections are necessary when quoting the source literally. First, without finiteness, $z^{n-1}(zw-1)$ is a counterexample: its real-slice zero set contains the unit circle. Second, the rough n^2 estimate printed there is not valid for a general rectangle of bidegrees. For $0 < a < 1$,

$$f(z,w) = (z^2 - a^2)w - z$$

has bidegree $(1,2)$ and exactly the five zeros

$$0, \quad \pm\sqrt{1+a^2}, \quad \pm i\sqrt{1-a^2}.$$

For a nonzero zero, division by z gives $|z|^2 - 1 = a^2\bar{z}/z$; its real and imaginary parts give precisely the listed possibilities. The correct rectangular intersection bound is m^2+n^2. The repeated symbol C_1 in the constant term of Eremenko's display is immaterial to the qualitative growth question; we use the two-constant notation of Problem 1.1.

To avoid transferring a sharp result to a larger class without proof, we use the following separate quantities:

Notation	Coefficient class over which the maximum is taken
$M_{m,n}$	Exact bidegree (m,n) and finite zero set. No bound on total degree other than $m+n$.
$E_{m,n}$	$cz^n + \sum_{i+j\le n-1,\ j\le m} c_{ij} z^i \bar{z}^j$, where $c \neq 0$. Its highest homogeneous part is purely analytic.
$M^{\mathrm{harm}}_{m,n}$	$p(z) + \overline{q(z)}$, where $\deg p = n > \deg q = m$.
$S_{m,n}$	The restricted harmonic family $\varepsilon z^n + q(z) + \overline{q(z)}$, where $\varepsilon \neq 0$ and $\deg q = m$.
$A_{m,n}$	$\varepsilon z^n + R(x,y)$, where $R \in \mathbb{R}[x,y]$ has total degree $m < n$.
$K_{m,n,r}$	The support-restricted space in Section 6, with $1 \le m < r \le n$.
s_n	The least number of nonzero monomials among total-degree-n polynomials with n^2 distinct zeros.

The notation $\mathcal{E}_{m,n}$ denotes the coefficient space defining $E_{m,n}$. Its conjugate degree is at most m; all exact constructions below can be arranged to have degree exactly m. The total-degree class $p(z) + Q(z,\bar{z})$, $\deg p = n$, $\deg_{\mathrm{tot}} Q \le m$, is different from both the rectangular class and $\mathcal{E}_{m,n}$.

1.2 Established results and exact source locations

The harmonic problem was raised by Sheil-Small in 1992; the contemporary discussion is in Khavinson–Świątek [KS03, pp. 409–410]. Wilmshurst proved the quadratic upper bound and constructed extremizers when the smaller degree is one less than the larger [Wil98, pp. 2077–2080]. Bshouty, Hengartner, and Suez gave an independent extremal construction [BHS95, pp. 207–218]. Peretz and Schmid independently treated the quadratic zero bound by algebraic methods [PS97, pp. 203–208]; the original page 204 is also explicitly cited in the proof discussion of [SZ24, p. 6]. A direct copy of the Peretz–Schmid article was not available in this audit; this attribution and its locator are reported through those authors' published references, not represented as a page-by-page inspection.

For conjugate degree one, Khavinson–Świątek proved $3n-2$ as an upper bound [KS03, Theorem 1, p. 410], and Geyer proved sharpness [Gey08, Corollary 2, p. 549]. For a rational analytic term of degree n, Khavinson–Neumann established $5n-5$ [KN06, Theorem 1, author version, p. 1]; Rhie's construction gives sharp examples [Rhi03, pp. 3–4]. Singular zeros and degree-dependent refinements were subsequently treated by Luce–Sète–Liesen [LSL15, Theorem 1.1, author version, p. 2] and Liesen–Zur [LZ18, Theorem 1.1 and Theorem 3.5, author version, pp. 2, 7]. Lee–Makarov's prescribed-pole construction [LM14, Theorem C, p. 3] supplies the sharp degree-dependent endpoint recorded in Section 6.6. Those endpoint results are not new here.

Wilmshurst's proposed refinement $3n-2+m(m-1)$ is false in general. Lee, Lerario, and Lundberg constructed counterexamples and explicitly proposed fixed-m linear growth [LLL15, author version, pp. 1–4]. Their stronger conjecture bounds the number of orientation-reversing zeros by $m(n-1)$ [LLL15, author version, p. 3]. Neither that conjecture nor the exact unrestricted harmonic maximum is proved below. Khavinson, Lee, and Saez give further constructions and warn, in effect, against assigning at most one zero to each orientation-reversing component [KLS18, Theorems 2–3, p. 2]. Lee and Saez analyze the regime $m = n-2$ in detail [LS17, unnumbered Theorem, author version, p. 1]; their count is $n^2 - 2n + 2 + 4k_{\max}(n)$ with the trigonometric integer $k_{\max}$ defined there, not just n^2-2n+2. Their theorem explicitly counts multiplicities, a convention that

should not silently be replaced by distinct zeros.

Certified numerical constructions were developed by Hauenstein, Lerario, Lundberg, and Mehta [HLLM15, Section 1.3, author version, p. 3]. Lundberg's probabilistic construction gives examples with at least $\lceil n\sqrt{m} \rceil$ zeros [Lun23, Theorem 1.1, author version, p. 2]. The concrete deterministic question relevant here is in his Section 4.2, page 9 of arXiv:2201.00788v2: can a strategic choice of the coefficients in the monomial-head family produce order mn instead? The exact restricted-family result in Section 8 answers that question, while Section 9 changes the analytic head and improves the constructed linear coefficient.

Average-case asymptotics should be kept separate from worst-case valence. Lerario–Lundberg establish $n^{3/2}$ growth for the truncated Gaussian model with $m = \alpha n$, $0 < \alpha < 1$, and asymptotic n for fixed m [LL16, Theorem 1, author version, p. 3]. For independent identically distributed Kac coefficients, Lu–Wang prove bounds of order n when $m = \alpha n$ [LW25, Theorem 2.1 and inequalities (2.1)–(2.4), author version, p. 4]. The latter article appeared in 2025, although its accepted author manuscript was posted to arXiv in September 2026. Neither ensemble estimate is a coefficient-uniform upper bound on the maximal number of zeros.

For general polyanalytic polynomials, Khavinson, Lundberg, and Perry formulate the bidegree problem and give the coprime intersection estimate $m^2 + n^2$ [KLP24, Problem 1.9 and Proposition 1.10, author version, pp. 3–4]. In the distinct logharmonic subclass with one linear factor, their $3n - 1$ estimate is attained by Lazebnik–Lundberg [LL26, Theorem 1.3, author version, p. 2]. Sète–Zur give a systematic account of finiteness, total-degree bounds, possible zero counts, and constructions [SZ24, Theorems 2.7, 2.12, 2.16 and 4.1, pp. 5–8, 12–14]. Their Remark 4.4 on page 16 conjectures that at least n nonzero coefficients are needed for n^2 zeros; Section 11 gives a three-term quartic counterexample and proves that degree four is the first failure. Their Section 5 concerns a total-degree-restricted mixed class, not the full rectangle.

The algebraic trace method in Section 7 builds on Furchì's Hermitian trace form [Fur25, Theorem 3.13, Corollary 3.14, and Lemma 4.3, author version, pp. 9–10, 16–17]. We prove the additional trace identity and isotropic-space estimate needed here. The surface arguments use the Thom genus theorem of Kronheimer–Mrowka [KM94, Theorem 1, author version, p. 1]; the version allowing only negative ordinary double points is stated explicitly by Orevkov [Ore24, Theorem 3.1, author version, p. 8]. Finally, separated-coefficient arguments have antecedents in the real-rootedness and tropical Descartes theory of Forsgård–Novikov–Shapiro [FNS17, Proposition 1 and Section 2, author proof, pp. 2, 4]. That proposition attributes the sign-independent real-rootedness criterion to Passare–Rojas–Shapiro [PRS11, Definition 1 and Theorem 1, author version, p. 2]. We give the elementary sign argument rather than presuppose independence of the restrictions to different lines.

All references marked "author version" use the specific version in the bibliography; other page references use the published pagination. Publication metadata and mathematical locators were checked separately. Standard textbook citations identify the chapters or propositions actually used, but this audit does not claim visual inspection of every page of every cited book. The literature review concerns the zero-counting questions in this research record; it is not a proof of priority against every unpublished manuscript.

1.3 The principal conclusions

The first conclusion settles Problem 1.1 and identifies its optimal leading constant.

Theorem 1.2. *For $n > m \geq 1$ and every polynomial of bidegree (m, n) with finite zero set,*

$$N(f) \leq \min\{m^2 + n^2, (4m+1)n - 5m\}. \tag{1.1}$$

For every fixed $m \geq 1$, $\lim_{n\to\infty} M_{m,n}/n = 4m + 1$.

The finite-degree equality range is substantially larger than the first constructions in the discussion. The proof uses one exact, finite rational certificate for a degree-eight starting polynomial; its subsequent iterations are analytic arguments.

Theorem 1.3. *For $m \geq 2$ and $m < n \leq 2m + 2$,*

$$E_{m,n} = n^2, \qquad M_{m,n} = m^2 + n^2. \tag{1.2}$$

The same equalities hold when $n = 2m + 3$ and $3 \mid m$. Each maximum is attained over $\mathbb{Q}(i)$ with exact conjugate degree m and all zeros nondegenerate.

For harmonic polynomials, the strongest general estimates established in the record are most compactly stated together. The endpoints $m = 1$ and $m = n - 1$ are classical, as explained above.

Theorem 1.4. *For $1 \leq m < n$, put $g = \gcd(m, n)$. Then*

$$(m+2)n - m - g \leq M^{\mathrm{harm}}_{m,n} \leq \min\left\{n^2 - 2\left\lfloor \frac{(n-m)^2}{4} \right\rfloor, (3m+2)n - m^2 - 4m - g\right\}. \tag{1.3}$$

Moreover, $M^{\mathrm{harm}}_{1,n} = 3n - 2$ and $M^{\mathrm{harm}}_{n-1,n} = n^2$.

In particular $mn \leq M^{\mathrm{harm}}_{m,n} \leq 4mn$ uniformly in both degrees. For fixed $m \geq 2$, the lower coefficient in the constructions is $m + 2$, whereas the upper coefficient is $3m + 2 < 4m + 1$. These are not matching constants. The exact formulas for $S_{m,n}$, $A_{m,n}$, and the cyclic family are separate theorems, not formulas for the unrestricted harmonic maximum.

The proof organization follows dependencies rather than the order of the chat. Sections 2–5 establish intersection bounds, regularization, the surface argument, and the matching asymptotic construction. Sections 6–7 refine the upper bounds. Sections 8–11 treat harmonic, real-valued, and sparse constructions. Sections 13–17 prove the exact finite-degree ranges. The final sections collect exact certificates, an audit of superseded assertions, and the questions not settled by these arguments.

> **Scope of "sharp."** An exact maximum for a specified subspace, an optimal asymptotic coefficient, and a full formula for every bidegree are different claims. This note proves the first two in the places explicitly stated. It does not prove $M_{m,n} = \min\{m^2+n^2, (4m+1)n-5m\}$ for every pair, and it does not determine $M_{2,7}$.

2 Intersection bounds and nondegenerate zeros

The elementary intersection statements in this section will also certify completeness of constructions. They are established before regularization so that later extremal examples do not depend on the surface argument.

At a real-slice point, the real Jacobian is

$$J_f = |f_z|^2 - |f_w|^2 = \det \begin{pmatrix} f_z & f_w \\ \overline{f_w} & \overline{f_z} \end{pmatrix}. \tag{2.1}$$

A zero is *nondegenerate* if this number is nonzero. We denote the numbers of positive and negative nondegenerate zeros by $N_+(f)$ and $N_-(f)$, and put $N_{\mathrm{nd}} = N_+ + N_-$. Rank zero of the real differential is equivalent to $f_z = f_w = 0$.

The following total-degree statement is particularly useful. It also follows from [SZ24, Corollary 2.14, p. 7] together with local intersection multiplicity.

Lemma 2.1. *If $f = z^N + q(z, w)$ with $\deg_{\mathrm{tot}} q < N$, then (f, f^*) is a zero-dimensional complete intersection of length N^2. In particular, $N(f) \le N^2$. If N^2 distinct real-slice zeros occur, every one is nondegenerate.*

Proof. In a graded monomial order, the leading monomials are z^N and w^N. They are relatively prime, so the two polynomials form a Gröbner basis. The residue classes of $z^i w^j$, $0 \le i, j < N$, form a basis of the quotient, proving the length assertion and finiteness. Equivalently, the projective curves have no common point at infinity. If N^2 distinct real points occur, each local intersection has length one; the two gradients are independent. Its determinant is (2.1), which proves the last assertion. □

For the rectangle we remove the common-factor hypothesis explicitly.

Proposition 2.2. *If f has bidegree (m, n) and finite real-slice zero set, then $N(f) \le m^2 + n^2$.*

Proof. Let h be an irreducible factor, of bidegree (a, b). If h^* is not proportional to h, their curves in $\mathbb{CP}^1 \times \mathbb{CP}^1$ are coprime and have intersection number $a^2 + b^2$, as in [KLP24, Proposition 1.10, author version, p. 4]. Every real-slice zero is an intersection.

If h^* is proportional to h, then $a = b = d \ge 1$. Multiplication by a nonzero constant makes $h(z, \bar{z})$ real-valued. At an isolated zero its real gradient vanishes, so $h = h_z = h_w = 0$. In characteristic zero, irreducibility and the strict drop in w-degree imply $\gcd(h, h_w) = 1$. Their bidegrees are (d, d) and at most $(d - 1, d)$, and their intersection number is at most $2d^2 - d \le 2d^2$.

Write $f = c \prod h_\nu^{e_\nu}$ with distinct irreducible factors of bidegrees (a_ν, b_ν). Each factor has a finite real-slice zero set. Thus

$$N(f) \le \sum_\nu (a_\nu^2 + b_\nu^2) \le \Big(\sum_\nu a_\nu\Big)^2 + \Big(\sum_\nu b_\nu\Big)^2 \le m^2 + n^2.$$

Univariate factors satisfy the same bound directly, including degree one. This also covers repeated factors. □

A related estimate requires no global finiteness assumption and is useful when a certificate has already supplied many simple zeros.

Proposition 2.3. *Let $f \in \mathbb{C}[z, w]$ have bidegree (m, n), where $n \ge m \ge 1$, and set $D = \gcd(f, f^*)$. There is an integer $0 \le \delta \le m$ such that $\deg_z D = \deg_w D = \delta$, and*

$$N_{\mathrm{nd}}(f) \le (n - \delta)^2 + (m - \delta)^2. \tag{2.2}$$

Proof. Unique factorization implies D^* is a constant multiple of D, proving equality of its two degrees. Write $f = Du$, $f^* = Dv$, with u, v coprime. At $D = 0$ their differentials are multiples of dD, so (2.1) vanishes. Nondegenerate real-slice zeros therefore lie outside $D = 0$ and are intersections of $u = 0$ and $v = 0$. Their bidegrees are $(m - \delta, n - \delta)$ and $(n - \delta, m - \delta)$; the intersection product gives (2.2). □

In particular, more than $(n-1)^2 + (m-1)^2$ nondegenerate zeros force f, f^* to be coprime. Their entire complex intersection is then finite. This is stronger than checking only that a list of real zeros is long.

Saturation also excludes factorization of f itself.

Corollary 2.4. *If $n > m \geq 1$ and $N(f) = m^2 + n^2 < \infty$, then f is irreducible over $\mathbb{C}$.*

Proof. For $f = uv$ with bidegrees (a, b) and $(m - a, n - b)$, Proposition 2.2 gives a strict deficit $2\{a(m-a) + b(n-b)\}$ unless the factor degrees are $(m, 0)$ and $(0, n)$, in either order. In that exceptional case the actual univariate counts give $N(f) \leq m + n < m^2 + n^2$. Thus no nontrivial factorization is possible. □

For comparison, a polynomial of arbitrary total degree D with finite zero set has $N(f) \leq D^2$ [SZ24, Theorem 2.12, p. 6]. This estimate uses D, not its analytic degree. The examples in Section 1 show why the distinction cannot be omitted.

3 Removing degenerate zeros

For a polynomial $Q \in \mathbb{C}[z, w]$, write Q_w and Q_z for its complex partial derivatives. The real differential of $F_Q(z) = Q(z, \bar{z})$ is

$$dF_Q = (Q_z + Q_w)\, dx + i(Q_z - Q_w)\, dy, \qquad z = x + iy,$$

where both partial derivatives are evaluated at $(z, \bar{z})$. Hence

$$J_Q(z) = |Q_z(z, \bar{z})|^2 - |Q_w(z, \bar{z})|^2, \qquad \operatorname{rank} dF_Q(z) = 0 \iff Q_w = Q_z = 0. \tag{3.1}$$

We first isolate the local fact needed for the perturbation argument. It concerns real analytic maps, without any polynomial hypothesis.

Lemma 3.1. *Let $F : (\mathbb{R}^2, 0) \to (\mathbb{R}^2, 0)$ be real analytic, with an isolated zero at 0 and $\operatorname{rank} DF(0) = 1$. Let G be real analytic near 0, and suppose that $G(0) \notin \operatorname{im} DF(0)$. For every sufficiently small $t > 0$, the total number of nondegenerate zeros of $F + tG$ and $F - tG$ in a fixed neighborhood of 0 is at least two. This assertion remains valid when G varies in a sufficiently small finite-dimensional neighborhood.*

Proof. A real analytic change of source coordinates and an invertible real linear change of target coordinates put F in the form $F(x, y) = (x, g(x, y))$, with $Dg(0, 0) = 0$. Isolation gives

$$g(0, y) = cy^\ell + O(y^{\ell+1}), \qquad c \neq 0, \quad \ell \geq 2.$$

Write the components of the transformed G as G_1, G_2, and set $\beta = G_2(0, 0) \neq 0$. The first equation of $F \pm tG = 0$ has a unique solution $x = X_\pm(y, t) = O(t)$ near the origin. After substituting

$y = t^{1/\ell}u$ and dividing the second equation by t, the result converges in C^1 on bounded u-intervals to $cu^\ell \pm \beta$. For completeness, the mixed terms of g contain a factor x and have order at least two; after substitution they contribute $O(t^{1/\ell}) + O(t)$, as do their u-derivatives. This proves the claimed C^1 convergence.

If ℓ is odd, each limiting polynomial has one simple real zero. If ℓ is even, one of them has two simple real zeros. The implicit function theorem preserves those zeros and the nonvanishing Jacobian of the two-dimensional system. All estimates and simple-zero conditions are uniform for G sufficiently close to the given one. □

A constant perturbation need not preserve the number of isolated zeros. The following use of derivatives avoids that difficulty and also handles repeated factors. Coefficient neighborhoods refer to the usual finite-dimensional topology.

Lemma 3.2. *Let $Q \in \mathbb{C}[z, w]$ have finitely many zeros on $w = \bar{z}$. Every coefficient neighborhood of Q contains a polynomial R with*

$$\deg_w R \le \deg_w Q, \qquad \deg_z R \le \deg_z Q, \qquad N(R) \ge N(Q),$$

such that every zero of F_R is nondegenerate.

Proof. Let

$$S_Q = \{z : Q(z, \bar{z}) = Q_w(z, \bar{z}) = Q_z(z, \bar{z}) = 0\}, \qquad Q_\xi = Q + \xi_1 Q_w + \xi_2 Q_z.$$

By (3.1), S_Q is precisely the set of rank-zero zeros, and it is finite. At a zero of $Q_\xi(z, \bar{z})$ with $z \notin S_Q$, at least one of Q_w, Q_z is nonzero. Thus the differential with respect to $\xi \in \mathbb{C}^2$ is onto $\mathbb{C}$. The parametric transversality theorem [Hir76, Chapter 3, Section 2] shows that, for a residual set of parameters, all zeros outside S_Q are nondegenerate. We may impose this condition simultaneously on ξ and $-\xi$. An exhaustion by compact sets justifies the assertion on the whole plane. Every such polynomial has finitely many zeros: an infinite polynomial zero set contains an arc [SZ24, Theorem 2.7, p. 5], whereas an arc cannot consist of nondegenerate zeros outside a finite set.

We next choose parameters for which one sign does not decrease the number of zeros. At every rank-one zero of F_Q, the map $(a, b) \mapsto aQ_w + bQ_z$ is onto $\mathbb{C}$. We can therefore choose (a, b) so that the value of $aQ_w + bQ_z$ avoids the one-dimensional real image of dF_Q at every such zero. Only finitely many proper real linear subspaces are excluded. Lemma 3.1, applied in disjoint neighborhoods, shows that the two perturbations $Q \pm t(aQ_w + bQ_z)$ contribute at least two nondegenerate zeros in total near each rank-one zero. Rank-two zeros persist for both signs, and all points of S_Q remain zeros for both signs. The local conclusions hold in an open cone about the chosen direction. Consequently, an arbitrarily small parameter ξ can satisfy both these conclusions and the preceding transversality conditions. It then satisfies

$$N(Q_\xi) + N(Q_{-\xi}) \ge 2N(Q). \tag{3.2}$$

There is one more finite collection of restrictions on the direction. At $z_0 \in S_Q$, let H_k be the first nonzero homogeneous term of Q at the complex point $(z_0, \bar{z}_0)$; here $k \ge 2$. Choose (a, b) so that $a(H_k)_w + b(H_k)_z \not\equiv 0$ for every $z_0 \in S_Q$. Each excluded set is a proper complex linear subspace. For every nonzero sufficiently small parameter in a suitable cone,

$$\operatorname{ord}_{(z_0, \bar{z}_0)} Q_{\pm\xi} = k - 1.$$

Choose the sign in (3.2) with at least $N(Q)$ zeros and repeat the procedure.

After one step, every degenerate zero belongs to S_Q. At the points that remain rank zero the complex vanishing order drops strictly at every subsequent step. When the order reaches one, the point either is nondegenerate or is rank one; in the latter case the next step removes its degeneracy. Thus the procedure terminates in finitely many steps. Derivatives do not increase either degree. Choosing all successive perturbations sufficiently small keeps the final polynomial in the prescribed neighborhood of Q. □

We now pass from the affine plane to a compact surface. Set

$$X = \mathbb{CP}^1_z \times \mathbb{CP}^1_w, \qquad \tau(z, w) = (\bar{w}, \bar{z}), \qquad L = \operatorname{Fix} \tau.$$

Let $\alpha, \beta \in H^2(X; \mathbb{Z})$ be the positive generators pulled back from the first and second factors, respectively. Thus $\alpha^2 = \beta^2 = 0$ and $\int_X \alpha\beta = 1$. A polynomial of bidegree (a, b) defines a divisor with class $b\alpha + a\beta$.

The next elementary genericity statement records the hypotheses needed later; in particular, transversality at the fixed set is not left implicit.

Lemma 3.3. *Suppose $a, b \geq 1$. An arbitrarily small perturbation within the space of sections of $\mathcal{O}_X(b, a)$ can be chosen so that its zero curve C is smooth and connected, avoids (∞, ∞), and is transverse both to L and to τC.*

Proof. Here $\mathcal{O}_X(b, a)$ is the tensor product of the pullbacks of $\mathcal{O}_{\mathbb{CP}^1}(b)$ and $\mathcal{O}_{\mathbb{CP}^1}(a)$ from the two factors. The complete linear system separates first jets and separates any two distinct points. Its evaluation map is a submersion, so parametric transversality gives smoothness of C and transversality to L. On $X \setminus L$, the points x and τx are distinct, and the values of a section at those points can be prescribed independently. Apply the same theorem to the two-point evaluation map to obtain $C \pitchfork \tau C$ off L.

At a fixed point, failure of this last transversality would mean that the complex line $T_x C$ is invariant under the antilinear involution $d\tau_x$. An antilinear involution on a complex line has a nonzero fixed vector. Such a vector would belong to $T_x C \cap T_x L$, contradicting transversality to L. Hence transversality holds there as well. Avoiding (∞, ∞) excludes one coefficient hyperplane, namely the vanishing of the $w^a z^b$ coefficient.

Finally, a disconnected smooth divisor would have disjoint irreducible components. For component bidegrees (a_i, b_i) and (a_j, b_j), disjointness gives $a_i b_j + a_j b_i = 0$. All entries are nonnegative, so all components would have to belong to the same ruling, unless there is only one component. Since the total degrees are both positive, disconnection is impossible. □

The proof has a useful support-preserving form. Its importance is that an arbitrary generic perturbation in a larger coefficient space would not justify the sharper estimates for a smaller space.

Corollary 3.4. *Let $V \subset \mathbb{C}[z, w]$ be a finite-dimensional complex linear space stable under ∂_z and ∂_w. If $f \in V$ has finitely many zeros, then every neighborhood of f in V contains $\widetilde{f}$ with $N(\widetilde{f}) \geq N(f)$ and all zeros nondegenerate.*

Proof. Every perturbation in Lemma 3.2 is a complex linear combination of the current polynomial and its two derivatives, so it stays in V. All local and transversality arguments in that proof are

unchanged. Polynomial zero sets are semialgebraic, which supplies the same finiteness argument at intermediate steps. □

In spaces with a prescribed nonzero highest coefficient, the coefficient is unchanged by these derivative perturbations. A final small generic perturbation can therefore be made within the same degree class.

4 The linear bound from the genus of a surface

We prove the second term of (1.1). By Lemmas 3.2 and 3.3, it suffices to work with a smooth connected curve $C \subset X$ satisfying all the conclusions of Lemma 3.3, with bidegree (m, n). Set $N = \#(C \cap L)$. The point (∞, ∞) is absent, so this is the number of affine zeros under consideration.

Consider the smooth map

$$\Phi : X \longrightarrow \mathrm{Sym}^2(\mathbb{CP}^1) \simeq \mathbb{CP}^2, \qquad \Phi(z, w) = \{z, \bar{w}\}. \tag{4.1}$$

In homogeneous coordinates it is

$$\Phi([z_0 : z_1], [w_0 : w_1]) = [z_0\bar{w}_0 : z_1\bar{w}_0 + z_0\bar{w}_1 : z_1\bar{w}_1].$$

The fibers are precisely the τ-orbits. The restriction to C therefore identifies pairs of points, but its behavior at a fixed point requires a separate check.

Lemma 4.1. *The map $\Phi|_C$ is an immersion. Its only multiple points are transverse positive double points, and their number is*

$$D = \frac{m^2 + n^2 - N}{2}. \tag{4.2}$$

Proof. Off L, the map Φ is a local diffeomorphism. Near a point of L choose compatible affine coordinates and put

$$u = \frac{z + \bar{w}}{2}, \qquad v = \frac{z - \bar{w}}{2}.$$

A smooth change of target coordinates puts (4.1) in the form $(u, v) \mapsto (u, v^2)$. If J denotes the original complex structure on X, then $\ker d\Phi = J(TL)$ at L. Since TC is a complex line,

$$TC \cap TL = \{0\} \quad \Longrightarrow \quad TC \cap J(TL) = \{0\}.$$

Thus $\Phi|_C$ is immersive also at fixed points.

Two distinct points of C have the same image exactly when they are $x, \tau x$, with $x \in (C \cap \tau C) \setminus L$. Transversality of C and τC makes the image branches transverse. No fiber has more than two points. To check signs, identify the branch from τx with the branch of τC at x. The map τ reverses the complex orientation of this curve, whereas Φ reverses the ambient four-dimensional orientation off L. These two sign changes cancel, leaving the positive intersection sign of the complex curves C and τC.

Their intersection number is $m^2 + n^2$. Exactly N intersection points are fixed by τ, and the rest occur in pairs. This proves (4.2). □

We now resolve the double points provided by Lemma 4.1. Resolving a double point replaces two disks by an annulus. In local oriented coordinates the positive intersection is modeled on the

two coordinate planes in $\mathbb{C}^2$, and the replacement is the usual smoothing $xy = \delta$, rounded near the boundary. Thus the replacement is embedded and preserves the relative homology class. Performing it in disjoint balls produces a smooth closed connected oriented surface $\Sigma \subset \mathbb{CP}^2$.

By the adjunction formula [Har77, Chapter V, Proposition 1.5, p. 361],

$$g(C) = (m-1)(n-1).$$

Each annular replacement decreases the Euler characteristic by two. If $H \in H^2(\mathbb{CP}^2;\mathbb{Z})$ is the positive hyperplane class, restriction to the two factors gives $\Phi^* H = \alpha - \beta$. We have therefore proved the following two identities:

$$g(\Sigma) = (m-1)(n-1) + D, \qquad \langle H, [\Sigma] \rangle = \int_C (\alpha - \beta) = m - n. \tag{4.3}$$

The remaining input is the Thom genus theorem of Kronheimer and Mrowka [KM94, Theorem 1, author version, p. 1]: a smooth closed connected oriented surface embedded in $\mathbb{CP}^2$ and representing $d[\mathbb{CP}^1]$, $d > 0$, has genus at least $(d-1)(d-2)/2$. No holomorphic or symplectic hypothesis is required. Applying this theorem completes the proof of Theorem 1.2.

Proof. Proposition 2.2 gives the quadratic term. Reverse the orientation of the surface in (4.3); its degree is $d = n - m > 0$. The Thom theorem and (4.2) yield

$$(m-1)(n-1) + \frac{m^2 + n^2 - N}{2} \geq \frac{(n-m-1)(n-m-2)}{2}.$$

Rearranging gives $N \leq (4m+1)n - 5m$. The initial regularization and generic perturbation do not decrease the number of zeros, so this bound applies to the original P. □

The proper mapping degree gives a useful arithmetic restriction on extremal counts. It concerns the maximum, not every degenerate polynomial individually.

Corollary 4.2. *The maximum $M_{m,n}$ is attained by a polynomial with nonzero $w^m z^n$ coefficient and only nondegenerate zeros. Moreover,*

$$M_{m,n} \equiv n - m \pmod 2.$$

Proof. The class is nonempty, for instance by $P = w^m z^n + 1$, and its finite counts are bounded integers. A largest count therefore exists. Apply the two perturbation lemmas to a maximizer. Maximality forces the perturbed count to remain equal to $M_{m,n}$. If the highest coefficient is $c \neq 0$, then $F_P(z) = c\bar{z}^m z^n + O(|z|^{m+n-1})$ at infinity. Thus F_P is proper and has degree $n-m$. The signed regular-value count gives

$$N_+(P) - N_-(P) = n - m, \tag{4.4}$$

where N_+ and N_- count the zeros with positive and negative Jacobian. This is the ordinary degree formula [Hir76, Chapter 5, Section 1]. Combining (4.4) with $N(P) = N_+(P) + N_-(P)$ proves the parity assertion. □

5 The optimal asymptotic coefficient

The upper bound just proved has coefficient $4m+1$ in front of n. A family depending on wz supplies the matching coefficient. Its extremal count can be determined exactly, not only asymptotically.

We first record an elementary consequence of Descartes' rule of signs. For a sequence of nonzero real numbers, let V denote its number of sign changes. The argument is the coefficient-dominance principle used in the study of sign-independent real-rootedness; see [FNS17, Proposition 1 and its proof, author version, pp. 2, 9–10]. The form below permits gaps in the set of exponents, which will also be important for the harmonic construction.

Lemma 5.1. *Let $0 \le e_0 < \cdots < e_s$ be integers and let $f(r) = \sum_{i=0}^{s} a_i r^{e_i}$ with $a_i \in \mathbb{R}\setminus\{0\}$. Suppose there exist $0 < \rho_0 < \cdots < \rho_s$ such that*

$$|a_i|\rho_i^{e_i} > \sum_{j\neq i} |a_j|\rho_i^{e_j} \qquad (0 \le i \le s).$$

The numbers of positive and negative roots of f are, respectively,

$$V(a_0,\dots,a_s), \qquad V((-1)^{e_0}a_0,\dots,(-1)^{e_s}a_s).$$

Every nonzero real root of f is simple.

Proof. At ρ_i the sign of f is the sign of a_i. Every change of sign therefore supplies a different root in the corresponding open interval. Descartes' rule bounds the number of positive roots, counted with multiplicity, by exactly this number of sign changes. Thus all positive roots have been found and are simple. Apply the same argument to $f(-r)$ to treat negative roots. □

A single quantitative choice enforces all the hypotheses. If $0 < \mu \le |b_i| \le B$ and $0 < t < 1$, set

$$f_t(r) = \sum_{i=0}^{s} b_i t^{e_i^2} r^{e_i}, \qquad \rho_i = t^{-2e_i}.$$

At the ith radius one has

$$t^{e_i^2} f_t(\rho_i) = b_i + \sum_{j\neq i} b_j t^{(e_j-e_i)^2}. \tag{5.1}$$

The sum in (5.1) has absolute value at most sBt, so $sBt < \mu$ guarantees strict dominance. In particular, this applies to $b_i \in \{-1,1\}$ and $0 < t < 1/(s+1)$. We also use the immediate consequence of Descartes' rule that a polynomial with $s+1$ nonzero monomials has at most s positive roots, counted with multiplicity.

We next state the precise result for this family.

Proposition 5.2. *Let $k > 2m \ge 2$. Among the polynomials*

$$P(z,w) = B(wz) + z^k A(wz), \qquad \deg A = m, \quad \deg B \le m, \quad B(0) \neq 0,$$

with finitely many zeros on $w = \bar{z}$, the largest zero count is $(4m+1)k$. It is attained by an irreducible polynomial with only nondegenerate zeros.

Proof. Put $t = |z|^2$. No positive t can be a common zero of A and B, since that would give a circle of zeros. The possible squared radii are exactly the positive roots of

$$\Psi(t) = t^k A(t)\overline{A(\bar{t})} - B(t)\overline{B(\bar{t})}. \tag{5.2}$$

Every such radius gives k values of the argument of z. The two exponent ranges of (5.2) are $0, \dots, 2m$ and $k, \dots, k+2m$. They are disjoint, so Ψ has at most $4m+2$ nonzero monomials. The preceding sparse polynomial bound gives $N(P) \le (4m+1)k$.

Choose $0 < \delta < 1/(4m+4)$ and set

$$\begin{aligned} B(t) &= \sum_{j=0}^{m} (-1)^j \delta^{4j^2} t^j, \\ A(t) &= \sum_{j=0}^{m} (-1)^{m+1+j} \delta^{(k+2j)^2} t^j. \end{aligned} \tag{5.3}$$

Apply Lemma 5.1 to the ordered exponents $0, 2, \dots, 2m, k, k+2, \dots, k+2m$. The coefficients of $B(r^2) + r^k A(r^2)$ alternate in sign; it therefore has at least $2m+1$ distinct positive zeros. For $B(r^2) - r^k A(r^2)$, each of the two blocks has m changes of sign and there is no change between the blocks. It has at least $2m$ distinct positive zeros.

Applying the same lemma to each block separately shows that all m zeros of $B(r^2)$ with $r > 0$ lie between 1 and δ^{-4m}, whereas all m positive zeros of $A(r^2)$ lie between δ^{-2k} and $\delta^{-2(k+2m)}$. In each case these account for all roots of the degree-m polynomial in r^2 and are simple. Since $k > 2m$, the two intervals are disjoint. In particular, A and B are coprime and the two expressions with signs $+$ and $-$ have no common positive zero. Their product, up to sign, is $\Psi(r^2)$. The upper bound already proved forces exactly $4m+1$ simple positive roots of Ψ, and hence exactly $(4m+1)k$ zeros of $P(z, \bar{z})$.

At any such zero, $e^{ik\theta} = s \in \{-1, 1\}$. The radial derivative of the real part is $(B(r^2) + sr^k A(r^2))' \neq 0$, while the angular derivative of the imaginary part is $skr^k A(r^2) \neq 0$. The two other entries of this polar-coordinate Jacobian are zero. All zeros are nondegenerate.

Finally, choose any simple root b of B. Since $A(b) \neq 0$, $A(t)z^k + B(t)$ is Eisenstein at the prime $t - b$ in $\mathbb{C}[t][z]$. It is primitive because A, B are coprime, and is therefore irreducible in $\mathbb{C}[t, z]$. It remains irreducible after inverting z. The substitution $t = wz$ identifies this localization with $\mathbb{C}[w, z, z^{-1}]$. The polynomial in (5.3) satisfies $P(0, w) = 1$, so it has no factor z and is irreducible already in $\mathbb{C}[z, w]$. □

With $k = n - m$, Proposition 5.2 proves

$$(4m+1)(n-m) \le M_{m,n} \le (4m+1)n - 5m, \qquad n > 3m. \tag{5.4}$$

Dividing (5.4) by n proves the asymptotic assertion of Theorem 1.2. The difference between the two bounds is the constant $4m(m-1)$. For $m = 1$ it vanishes; together with Section 13, this recovers $M_{1,n} = 5n - 5$ for every $n \ge 2$.

6 Bounds that depend on the Newton support

Fix integers $1 \le m < r \le n$ and let $\mathcal{T}_{m,n,r}$ consist of polynomials with $[z^n]f \neq 0$ and support contained in

$$\{(i, j) \in \mathbb{Z}^2_{\ge 0} : j \le m,\ mi + rj \le mn\}. \tag{6.1}$$

Here $[z^n]f$ denotes the coefficient of $z^n w^0$. The convex hull of the allowed exponents is the trapezoid with vertices $(0,0),(n,0),(n-r,m),(0,m)$; for $r=n$ it is a triangle. The coefficient of w^m need not have the largest permitted degree. All estimates below include this possibility.

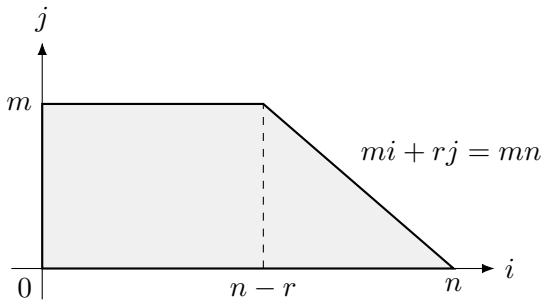


Figure 1: The prescribed support. The slope parameter r records how far the upper right corner is removed. The picture is schematic.

The upper bound records this sloping edge, rather than only the enclosing rectangle. Put $g=\gcd(m,r)$.

Theorem 6.1. *For $f\in\mathcal{T}_{m,n,r}$ the zero set of F is finite and*

$$N(F)\le \min\{n^2,\ (4m+1)n-m^2-4m-(m-1)r-g\}. \tag{6.2}$$

One endpoint of Figure 1 is particularly useful. Let $\mathcal{E}_{m,n}$ be the entire coefficient space

$$F(z)=cz^n+\sum_{\substack{i,j\ge 0,\ i+j\le n-1\\ j\le m}} c_{ij}z^i\bar z^j,\qquad c\neq 0. \tag{6.3}$$

The only restriction on its top homogeneous part is that it be holomorphic; the lower terms are neither real-valued nor harmonic in general. Denote its largest zero count by $E_{m,n}$. This is a finite integer by Theorem 6.1.

The asymptotic coefficient in this full space is sharp.

Theorem 6.2. *For $n>m\ge 1$,*

$$E_{m,n}\le \min\{n^2,\ (4m+1)n-2m(m+2)\}. \tag{6.4}$$

For $n>4m+1$,

$$E_{m,n}\ge (4m+1)n-4m(2m+1). \tag{6.5}$$

The lower bound is attained by a polynomial all of whose zeros are nondegenerate. For every fixed m,

$$\lim_{n\to\infty}\frac{E_{m,n}}{n}=4m+1. \tag{6.6}$$

Indeed, the lattice condition in (6.1) with $r=m+1$ is exactly (6.3). For $1\le j\le m$ it is equivalent to $i+j\le n-1$; for $j=0$ it is $i\le n$. This proves (6.4) from (6.2). The construction proving (6.5) is given in Section 6.5. The gap between these two linear expressions is $6m^2$. Thus (6.6) does not assert equality in (6.4) for each pair (m,n).

The other endpoint includes the total-degree class studied by Sète and Zur [SZ24, Section 5]. It is substantially smaller than (6.3).

Corollary 6.3. *If* $\deg p = n > m \geq 1$ *and* $\deg_{\mathrm{tot}} Q \leq m$, *then*

$$\#\{z : p(z) + Q(z, \bar{z}) = 0\} \leq \min\{n^2,\ (3m+2)n - m^2 - 4m - \gcd(m, n)\}. \tag{6.7}$$

Proof. Every mixed monomial in Q satisfies $mi + nj \leq n(i+j) \leq nm$. All analytic terms of p also satisfy this inequality. Apply Theorem 6.1 with $r = n$. □

In particular, the second estimate in (6.7) is not restricted to harmonic polynomials. This extension does not identify the maximum in the total-degree class with the harmonic maximum. Nor does the asymptotic result (6.6) apply to that smaller class: the allowed mixed terms in (6.3) can have degrees tending to n.

For comparison with prior work, the case of conjugate degree one is a rational harmonic equation. Its sharper degree-dependent bounds are due to Liesen and Zur [LZ18, Theorem 1.1, author version v2, p. 2]; sharpness with prescribed pole multiplicities follows from Lee and Makarov [LM14, Theorem C, p. 3]. We record this known endpoint in Section 6.6, rather than count it as a new result.

6.1 Genericity within the support restriction

The inequality $r > m$ makes cz^n the unique highest-total-degree term. Lemma 2.1, after scaling, gives finiteness and $N(F) \leq n^2$. The support in (6.1) is downward closed, so Corollary 3.4 applies without enlarging it. Thus we may preserve all zeros through regularization and then make a small generic perturbation in the same support.

A sufficiently small further perturbation preserves any finite set of nondegenerate zeros. We may consequently assume that the affine curve $C = \{f = 0\}$ is smooth and irreducible, that its finite intersections with τC are transverse, and that C is transverse to the fixed set of

$$\tau(z, w) = (\bar{w}, \bar{z}).$$

These are genuine generic conditions in the indicated sparse space. The monomials $1, z, w$ separate first-order jets at affine points and separate two different affine points. They give the required transversality both off and on the fixed set. Irreducibility is a nonempty algebraic open condition: $w^m + z^n + z$ belongs to the space and is irreducible by Eisenstein at z (the case $m = 1$ is immediate). We may also assume the leading coefficient in w has simple zeros, the next coefficient is nonzero there, and the polynomial on the sloping Newton edge has distinct nonzero roots. These are compatible open conditions. All subsequent arguments concern such a perturbation.

6.2 The normalization genus

Write

$$f(z, w) = \sum_{j=0}^{m} a_j(z) w^j, \qquad \deg a_j \leq \left\lfloor n - \frac{rj}{m} \right\rfloor.$$

Let $\widetilde{C}$ be the normalization of the closure of C in $X = \mathbb{CP}^1_z \times \mathbb{CP}^1_w$, and let γ be its genus. The projection $z : \widetilde{C} \to \mathbb{CP}^1$ has degree m.

The genus can be bounded directly from the discriminant; no nonsingularity of the curve at infinity is being assumed. Lemma 6.4 provides the precise estimate used in the surface argument.

Lemma 6.4. *For the generic curve just described,*

$$2\gamma \le (2m-2)n-(m-1)r-m-g+2. \tag{6.8}$$

Above (∞,∞) there are exactly g points of the normalization, with local parametrizations

$$u=t^{m/g}, \qquad v=c_\ell t^{r/g}(1+O(t)), \qquad c_\ell\neq 0, \quad u=1/z,\ v=1/w. \tag{6.9}$$

Proof. The discriminant $\operatorname{Disc}_w f$ is homogeneous of degree $2m-2$ in the coefficients a_j, and homogeneous of weight $m(m-1)$ when a_j has weight j. Therefore

$$\deg_z \operatorname{Disc}_w f \le (2m-2)n-r(m-1).$$

When $m=1$, the discriminant is 1 and this bound is zero. Over a finite value of z with $a_m\neq 0$, its order is the total ramification multiplicity, because the affine curve is smooth. At a zero of a_m, the branch at $w=\infty$ is unramified: in $v=1/w$ the equation is $a_m(z)+a_{m-1}(z)v+O(v^2)=0$, and both $a_m'(z)$ and $a_{m-1}(z)$ are nonzero there. The other branches are generically unramified there as well. Thus the finite ramification contribution is at most $(2m-2)n-r(m-1)$.

Near (∞,∞), multiply $f(1/u,1/v)$ by u^nv^m. Put $a=m/g$, $b=r/g$. Its terms of least (a,b)-weight form

$$c\prod_{\ell=1}^{g}(v^a-\lambda_\ell u^b), \qquad c\neq 0,$$

where the λ_ℓ are distinct and nonzero. All other terms have strictly larger weight. Substituting $u=t^a$, $v=t^bV$ and dividing by the least power of t gives a polynomial in V with m simple nonzero roots at $t=0$. The holomorphic implicit function theorem gives the local expansions. Grouping the roots by the action $t\mapsto e^{2\pi i/a}t$ yields exactly g branches and gives (6.9). There are no other points above $z=\infty$. Their total ramification is $g(a-1)=m-g$.

Riemann–Hurwitz now gives

$$2\gamma-2\le -2m+(2m-2)n-r(m-1)+(m-g),$$

which is (6.8). □

The right side of (6.8) is twice the number of interior lattice points in Figure 1. Only the displayed genus inequality is needed here.

6.3 The quotient surface and its end

The symmetric-product map

$$\Phi: X\longrightarrow \operatorname{Sym}^2(\mathbb{CP}^1)=\mathbb{CP}^2, \qquad \Phi(z,w)=\{z,\bar w\},$$

has fibers equal to the τ-orbits. In affine coordinates it is $[1:z+\bar w:z\bar w]$. Its restriction to the smooth affine curve is an immersion. To check this at a fixed point, use real coordinates $s=(z+\bar w)/2$, $t=(z-\bar w)/2$. The quotient has local form $(s,t)\mapsto(s,t^2)$ and kernel $J(T\operatorname{Fix}(\tau))$. A complex tangent line transverse to $T\operatorname{Fix}(\tau)$ is also transverse to its J-image. Away from the fixed set the quotient is a local diffeomorphism.

The ordinary projective closures of $f = 0$ and $f^* = 0$ have no common point at infinity and have degrees n, n. In generic position they therefore have n^2 distinct affine intersections. Of these, $N(F)$ are fixed and the rest occur in pairs. The affine image has

$$D_+ = \frac{n^2 - N(F)}{2} \tag{6.10}$$

ordinary double points. They are positive: the intersection of the two complex curves is positive, transport of the second sheet by the anti-holomorphic involution reverses its two-dimensional orientation, and Φ reverses the ambient four-dimensional orientation. The two changes cancel. There are no triple points, since fibers have size at most two.

If α, β are the positive cohomology generators of the two factors and H is the hyperplane class, then $\Phi^* H = \alpha - \beta$. The coordinate projections on $\widetilde{C}$ have degrees m, n. Thus the image homology class has degree $m - n$. Reversing the source orientation makes it positive, of degree $d = n - m$. This reversal does not change double-point signs.

The finite points of C on $w = \infty$ are smooth, unramified points with distinct images. Their partners under τ are not on C, since C has no point with $z = \infty, w \neq \infty$. They introduce no additional double points. It remains to treat (∞, ∞).

The required filling is elementary at the level of parametrized complex curves. We spell out why its double points have the correct sign.

Lemma 6.5. *The mirror torus link $T(a, -b)$, with $a, b > 0$, bounds $\gcd(a, b)$ immersed oriented disks in a four-ball, with only negative ordinary double points.*

Proof. The normalization of the complex curve $Y^a = X^b$ in a small ball is a union of $e = \gcd(a, b)$ disks, parametrized by $(X, Y) = (t^{a/e}, \zeta_\ell t^{b/e})$. Perturb these parametrizations by sufficiently small holomorphic polynomial maps. Generic such perturbations have nonzero derivative, only transverse double intersections, and no triple intersections. This follows from jet transversality: polynomial variations of sufficiently large degree interpolate the first jets at one point and the values at two or three distinct points. The respective bad conditions have the required real codimensions $4, 4, 8$. All transverse intersections are positive because the disks are complex curves. Near the boundary the unperturbed maps are embedded and transverse to the sphere; their perturbed inverse images of the ball remain disks and the boundary link remains isotopic to $T(a, b)$. Reflect one complex coordinate of the ambient ball. It reverses its orientation, turns every double point negative, and mirrors the boundary link. A collar realizing the boundary isotopy gives the claimed filling. □

We next identify the actual link, including a Newton edge with more than two monomials. Put $a = m/g$, $b = r/g$, so $b > a$. Near $\Phi(\infty, \infty)$ use coordinates

$$X = u + \bar{v}, \qquad Y = u\bar{v}.$$

By (6.9), $|v| = o(|u|)$ and $X = u(1 + o(1))$. For the branch associated with λ_ℓ the boundary on $|X| = \rho$ satisfies, after $Y = \rho^{1+b/a}\eta$,

$$\eta^a = \overline{\lambda_\ell} e^{-i(b-a)\theta} + o(1), \qquad X = \rho e^{i\theta}. \tag{6.11}$$

The convergence is in C^1 in the angular parametrizations. The roots of the limiting equations are distinct and nonzero, so the small error in (6.11) does not alter the braid type.

Move the g distinct nonzero constants $\overline{\lambda_\ell}$ through a path of distinct nonzero configurations to a scaled set of gth roots of unity. The displayed roots never collide, so this gives a link isotopy. The resulting union is

$$\eta^m = Ce^{-i(r-m)\theta}, \qquad C \neq 0,$$

which is $T(m, -(r-m))$. It has $\gcd(m, r-m) = g$ components, in agreement with the g normalization disks. A rounded bidisk in the (X, Y) chart is a four-ball whose boundary contains this braid; by taking ρ small, the curve meets its boundary only on this part. Formula (6.9) also shows that its inverse image consists of g disks. Their induced orientations agree with the complex-parameter orientations of the mirrored torus link. If the whole source is reversed, reverse the orientations of all filling disks as well.

Replace these disks by the fillings in Lemma 6.5, matching collars by the link isotopy. No source handle is introduced, all new double points are negative, and a replacement inside a ball does not change the global homology class. We obtain a closed connected immersed surface of genus γ, degree $d = n - m$, and exactly the positive double points in (6.10).

We use the immersed Thom inequality in the following precise form:

$$\gamma + D_+ \geq \frac{(d-1)(d-2)}{2}. \tag{6.12}$$

The version allowing only negative double points is stated in Orevkov [Ore24, Theorem 3.1, author version v2, p. 8], with attribution to Kronheimer–Mrowka [KM94]. Resolving each positive double point by an oriented annulus increases the genus by one, leaves the homology class unchanged, and reduces the general case to that version. Thus it is (6.12), not an inequality for a singular image, that applies.

Combining (6.8), (6.10), and (6.12) yields

$$N(F) \leq n^2 + (2m-2)n - (m-1)r - m - g + 2 - (n-m-1)(n-m-2).$$

The expression simplifies to the second bound in (6.2). Lemma 3.4 and the preservation of nondegenerate zeros under the subsequent small perturbation transfer it to every initial polynomial. This proves Theorem 6.1.

6.4 An exact count of zeros approaching infinity

Adding a higher analytic monomial changes both the leading homogeneous part and the degree at infinity. In the following situation every added zero can be counted, without an uncontrolled intermediate scale.

Lemma 6.6. *Let $k > m \geq 1$, $r > m$, and let $P \in \mathbb{C}[z, w]$ have degrees at most k, m in z, w, respectively, with $[z^k w^m]P = c \neq 0$. Suppose all zeros of $P(z, \bar{z})$ are nondegenerate. For all sufficiently small real $\varepsilon > 0$, the polynomial*

$$F_\varepsilon(z) = P(z, \bar{z}) + \varepsilon z^{k+r}$$

has exactly $N(P(z, \bar{z})) + r + m$ zeros, all nondegenerate. The $r + m$ new zeros approach infinity and have positive Jacobian.

Proof. The term $cz^k w^m$ is the unique term of total degree $k + m$. All old zeros are confined to a compact set, and each persists uniquely under the small perturbation. Set $t = \varepsilon^{1/(r-m)}$ and $z = \zeta/t$.

On compact ζ-sets,

$$t^{k+m}F_\varepsilon(\zeta/t) = \zeta^k(\zeta^r + c\bar{\zeta}^m) + O(t), \tag{6.13}$$

with C^1 convergence. The nonzero zeros of the limit satisfy

$$|\zeta|^{r-m} = |c|, \qquad (r+m)\arg\zeta = \arg(-c) \pmod{2\pi}.$$

There are exactly $r+m$. At such a zero the Jacobian of $\zeta^r + c\bar{\zeta}^m$ is

$$(r^2 - m^2)|\zeta|^{2r-2} > 0.$$

Multiplication by ζ^k preserves its sign. The implicit function theorem applied to (6.13) gives one simple positive zero near each.

For completeness, choose R so large that

$$|P(z,\bar{z}) - cz^k\bar{z}^m| \le \tfrac{1}{2}|c|\,|z|^{k+m} \quad (|z| \ge R).$$

Any zero of F_ε there must satisfy

$$\tfrac{1}{2}|c| \le \varepsilon|z|^{r-m} \le \tfrac{3}{2}|c|.$$

Its scaled coordinate $\zeta = tz$ therefore lies in a fixed compact annulus. Every limit point in that annulus is one of the above $r+m$ simple zeros; compactness and local uniqueness exclude further zeros there. On $|z| \le R$, compactness away from disjoint old-zero neighborhoods excludes all other zeros for small ε. This proves the exact count. □

The lemma applies to the full rectangle of exponents $i \le k, j \le m$. After adding z^{k+r}, this rectangle and the new exponent all belong to (6.1) with $n = k+r$, because $mi+rj \le mk+rm = mn$.

6.5 The optimal coefficient in the support-restricted space

The radial polynomial in Proposition 5.2, with analytic degree $k > 3m$, has $(4m+1)(k-m)$ nondegenerate zeros and a nonzero coefficient at $z^k w^m$. Its coefficients may be chosen real rational. We combine it with the preceding exact count at infinity.

Denote the largest zero count in $\mathcal{T}_{m,n,r}$ by $K_{m,n,r}$. The combination of the preceding lemmas proves a slightly more general asymptotic statement than Theorem 6.2.

Proposition 6.7. *If $n > r + 3m$ and $r > m$, then*

$$K_{m,n,r} \ge (4m+1)n - 4m(r+m). \tag{6.14}$$

For fixed m, r with $r > m$, $\lim_{n\to\infty} K_{m,n,r}/n = 4m+1$.

Proof. Put $k = n - r > 3m$. Lemma 5.2 gives $(4m+1)(k-m)$ zeros. Lemma 6.6 adds exactly $r+m$, giving

$$(4m+1)(n-r-m) + (r+m) = (4m+1)n - 4m(r+m).$$

The resulting polynomial lies in $\mathcal{T}_{m,n,r}$ by the observation at the end of Section 6.4. This proves (6.14); dividing it and (6.2) by n proves the limit. □

For $r = m+1$ this gives (6.5). Together with (6.4), it proves (6.6) and completes Theorem 6.2. All constructed zeros are nondegenerate, and the final ε can be chosen positive rational.

The distinction between the two support restrictions is substantive. For example, when $m \geq 2$ is fixed, Corollary 6.3 bounds the leading coefficient for $p + Q$ with $\deg_{\mathrm{tot}} Q \leq m$ by $3m + 2$, whereas Theorem 6.2 gives the strictly larger optimal coefficient $4m + 1$ in (6.3). Both classes have holomorphic highest homogeneous term. That condition alone is therefore insufficient to obtain the lower coefficient.

6.6 A known sharp endpoint and the remaining gap

There is a complete degree-dependent answer when $m = 1$. We include it to locate the new bounds relative to established sharp results.

Proposition 6.8. *For $2 \leq r \leq n$,*

$$K_{1,n,r} = 5n - 2r - 2. \tag{6.15}$$

In particular, $E_{1,n} = 5n - 6$ for every $n \geq 2$.

Proof. Write $f = p(z) - wq(z)$, with $\deg p = n$, $\deg q \leq n - r$. If $q \equiv 0$, the elementary analytic bound suffices. Otherwise, remove a greatest common divisor of degree s, leaving numerator and denominator degrees $a = n-s$ and $b \leq n-r-s$. The rational harmonic bound in [LZ18, Theorem 1.1, author version v2, p. 2] gives

$$N(f(z, \bar{z})) \leq s + 3a + 2b - 2 \leq 5n - 2r - 2.$$

Here $a - b \geq r \geq 2$; denominator zeros are not rational zeros because the reduced pair is coprime.

For equality, apply Lee–Makarov's Theorem C [LM14, p. 3] with rational degree n, one pole of order r at infinity, and $n - r$ simple finite poles. There are $n - r + 1$ distinct poles. Their theorem gives

$$\min\{3n + 2(n - r + 1) - 3, 5n - 5\} = 5n - 2r - 1$$

fixed points on the sphere. Infinity is one, leaving $5n - 2r - 2$ finite zeros. The numerator and denominator degrees are $n, n - r$. Regularity of an extremizer also follows from [LZ18, Theorem 3.5, author version v2, p. 7]. Taking $r = 2$ proves the last assertion. □

Proposition 6.8 is already implicit in those results and is not claimed here as new. It also shows that the upper bound in Theorem 6.1 is not always the finite-degree maximum: for $m = 1, r > 2$ it gives $5n - 6$, whereas (6.15) is smaller.

The finite-degree equalities proved later in Theorem 1.3 settle part of this range. The general expression in Theorem 6.1 is an upper bound, not a claim of equality for every triple (m, n, r).

7 A bound depending on the total-degree gap

Let $1 \leq m < n$, let $p \in \mathbb{C}[z]$ have degree n, and let $Q \in \mathbb{C}[z, w]$ have total degree at most m. We write

$$F(z) = p(z) + Q(z, \bar{z}), \qquad d = n - m, \qquad N(F) = \#\{z \in \mathbb{C} : F(z) = 0\}. \tag{7.1}$$

In this note the first variable of Q is the analytic variable. The restriction on Q is a *total-degree* restriction, not merely a bound on its degree in w. Thus the class in (7.1) contains every harmonic polynomial $p + \overline{q}$ with $\deg q \le m$, but it does not contain all polyanalytic polynomials of bidegree (m, n).

The classical harmonic estimate is n^2; see Wilmshurst [Wil98, pp. 2077–2079]. The Hermitian trace approach developed by Furchì gives $n^2 - 2$ when $m \le n - 2$, subject in his statement to a possible $n^2 - 1$ exception determined by two analytic coefficients [Fur25, Theorem 1.1, author version v2, p. 2]. The proof below removes that exception and permits mixed lower-degree terms. The estimate below takes account of the entire gap between the two degrees.

This is the main result of the present note.

Theorem 7.1. *Under the assumptions in (7.1), the zero set of F is finite and*

$$N(F) \le n^2 - 2\left\lfloor \frac{(n-m)^2}{4} \right\rfloor. \tag{7.2}$$

For $d = 2$, (7.2) is $n^2 - 2$. For $d = 3, 4, 5, 6$ the deficits from n^2 become $4, 8, 12, 18$, respectively. For large n with m fixed, this is still a quadratic estimate; it is not a replacement for the linear bound of Corollary 6.3. Rather, the two bounds are effective in different regimes. No claim of sharpness for every d is made.

The proof uses the standard Hermitian trace form, whose signature counts real solutions in the appropriate real structure. That correspondence is Furchì's Theorem 3.13 and Corollary 3.14 [Fur25, author version v2, pp. 9–10]; we include the required argument, including multiple complex roots. The additional ingredient is a totally isotropic subspace of dimension $\lfloor d^2/4 \rfloor$.

7.1 The quotient algebra and its Hermitian form

Divide F by the leading coefficient of p, so that p is monic. For a univariate polynomial a, put $a^{\#}(w) = \overline{a(\overline{w})}$. For $a \in \mathbb{C}[z, w]$, put

$$a^*(z, w) = \overline{a(\overline{w}, \overline{z})}.$$

The latter is a conjugate-linear involution. Define

$$f = p(z) + Q(z, w), \qquad f^* = p^{\#}(w) + Q^*(z, w), \qquad \mathcal{A} = \mathbb{C}[z, w]/(f, f^*). \tag{7.3}$$

With any graded monomial order, the leading monomials of f and f^* are z^n and w^n. Since these are relatively prime, the two polynomials form a Gröbner basis. Consequently

$$\{z^i w^j : 0 \le i, j < n\} \tag{7.4}$$

is a basis of $\mathcal{A}$ and $\dim_{\mathbb{C}} \mathcal{A} = n^2$. This proves, in particular, the finiteness assertion in Theorem 7.1. We identify a polynomial with its class when no ambiguity can arise.

For $a \in \mathcal{A}$, let M_a denote multiplication by a, and set

$$T(a) = \operatorname{Tr} M_a, \qquad \mathcal{H}(a, b) = T(ab^*). \tag{7.5}$$

The ideal in (7.3) is invariant under $*$, so these expressions are well defined. Moreover $T(a^*) = \overline{T(a)}$; hence $\mathcal{H}$ is Hermitian. The involution satisfies $\mathcal{H}(a^*, b^*) = \overline{\mathcal{H}(a, b)}$.

The signature is the positive index minus the negative index; the radical is the nullspace. The following form of the signature theorem retains all multiplicities throughout the algebraic argument.

Lemma 7.2. *The signature of $\mathcal{H}$ is $N(F)$. If R is the number of distinct common zeros of f, f^* and s is the number of two-element orbits of $(z,w) \mapsto (\bar{w}, \bar{z})$ on this set, then*

$$\operatorname{rank} \mathcal{H} = R, \qquad \nu_-(\mathcal{H}) = s, \qquad N(F) = R - 2s.$$

Here ν_- denotes the number of negative eigenvalues, counted with multiplicity.

Proof. Decompose the finite algebra into its local algebras. On a local algebra of length μ supported at x, multiplication by a has trace $\mu a(x)$: the difference from scalar multiplication is nilpotent. At a fixed point of the involution the resulting trace form has a single positive eigenvalue, with coefficient μ, and is zero on the maximal ideal. At a nonfixed pair x, x^*, the two local lengths agree; on their constant coordinates the matrix of the form is

$$\begin{pmatrix} 0 & \mu \\ \mu & 0 \end{pmatrix}.$$

It has one positive and one negative eigenvalue. The remaining local coordinates are in the radical. Different orbits are orthogonal. Adding these contributions proves the assertions and identifies the fixed points with the zeros of F. □

A single isotropic subspace can lose dimension in the radical. The involution provides precisely the control needed to handle this loss.

Lemma 7.3. *Let V be a finite-dimensional complex vector space with a Hermitian form H and a conjugate-linear involution J satisfying $H(Ju, Jv) = \overline{H(u,v)}$. If $U \subset V$ is totally isotropic and $U \cap JU = \{0\}$, then*

$$\operatorname{signature} H \le \dim V - 2 \dim U. \tag{7.6}$$

Proof. Put $r = \dim \operatorname{rad} H$ and $k = \dim(U \cap \operatorname{rad} H)$. The radical is J-invariant. The subspaces $U \cap \operatorname{rad} H$ and $J(U \cap \operatorname{rad} H)$ are disjoint, so $r \ge 2k$. In the nondegenerate quotient $V/\operatorname{rad} H$, the image of U is isotropic of dimension $\dim U - k$. Projection onto the negative eigenspace is injective on any isotropic subspace of a nondegenerate Hermitian space. Therefore $\nu_-(H) \ge \dim U - k$. It follows that

$$\operatorname{signature} H = \dim V - r - 2\nu_-(H) \le \dim V - 2\dim U - r + 2k \le \dim V - 2\dim U,$$

as required. □

We shall apply (7.6) to (7.5). No simplicity assumption on the roots of the complexified system will be needed.

7.2 An exact trace identity below degree $2d$

Write

$$Q(z,w) = \sum_{r+s \le m} c_{rs} z^r w^s.$$

For a univariate polynomial u, let

$$S(u) = \operatorname{Tr}_{\mathbb{C}[z]/(p)} M_u, \qquad D_r(u) = [z^{n-1}]\big(z^r u'(z) \bmod p(z)\big).$$

The remainder in this definition has degree less than n. Use $S^{\#}, D_r^{\#}$ for the corresponding functionals defined by $p^{\#}$ in the variable w. These are complex-linear functionals. In particular,

$$D_r(u) = 0 \quad \text{if } \deg u + r < n, \tag{7.7}$$

and the same assertion holds for $D_r^{\#}$. Also $S^{\#}(u^{\#}) = \overline{S(u)}$.

The next identity, rather than just a single vanishing trace, is the algebraic input to the proof.

Lemma 7.4. *If $u \in \mathbb{C}[z]$, $v \in \mathbb{C}[w]$, and $\deg u + \deg v < 2d$, then*

$$\begin{aligned} T(u(z)v(w)) = S(u)S^{\#}(v) \\ - \sum_{r+s\le m} c_{rs} D_r(u) S^{\#}(w^s v) \\ - \sum_{r+s\le m} \bar{c}_{rs} S(z^s u) D_r^{\#}(v). \end{aligned} \tag{7.8}$$

Proof. Introduce a real parameter t and replace Q, Q^* by tQ, tQ^* in (7.3); denote the multiplication trace by T_t. The standard basis (7.4) is unchanged for every t.

We first show that $T_t(z^a w^b)$ is affine in t when $a + b < 2d$. To compute a diagonal entry of the multiplication matrix, reduce $z^{a+i} w^{b+j}$ to standard monomials. A substitution using a lower term of p or $p^{\#}$ decreases total degree by at least one. A substitution using a term of tQ or tQ^* decreases total degree by at least $n - m = d$ and contributes one factor of t. A reduction path ending in $z^i w^j$ can decrease degree by only $a + b$. It therefore uses at most one term containing t. All reductions strictly decrease total degree, so they terminate, and each diagonal entry is affine in t. Linearity gives the same assertion for $T_t(uv)$.

Suppose first that p has distinct roots $\alpha_1, \dots, \alpha_n$. At $t = 0$, the roots of the complexified system are the simple grid points $(\alpha_a, \bar{\alpha}_b)$. Their derivatives satisfy

$$z'_{ab}(0) = -\frac{Q(\alpha_a, \bar{\alpha}_b)}{p'(\alpha_a)}, \qquad w'_{ab}(0) = -\frac{Q^*(\alpha_a, \bar{\alpha}_b)}{(p^{\#})'(\bar{\alpha}_b)}.$$

For small t, the trace is the sum of the values of uv at these n^2 simple roots. Differentiating this sum at zero gives (7.8), because Lagrange interpolation yields

$$D_r(u) = \sum_{a=1}^{n} \frac{\alpha_a^r u'(\alpha_a)}{p'(\alpha_a)}. \tag{7.9}$$

Indeed, the right side is the coefficient of z^{n-1} in the remainder of $z^r u'$ modulo p. The constant term in t is $S(u)S^{\#}(v)$, and the two sums in (7.8) are its first derivative. Affineness in t makes this first-order identity exact at $t = 1$.

Finally, all expressions in (7.8), with D_r defined by remainders, are polynomial functions of the coefficients of p, Q and their conjugates. The identity therefore extends from polynomials p with simple roots to all monic p. Formula (7.9) is used only in the proof on the dense simple-root locus. □

The threshold $2d$ is important. At total degree $2d$, two substitutions using the perturbation can contribute to a diagonal entry, and the trace need not be affine in Q. We do not use (7.8) at that threshold or beyond it.

7.3 A large isotropic subspace

For the moment assume that the leading moment matrices

$$\big(S(z^{a+b})\big)_{0\le a,b\le k}, \qquad 0 \le k < d, \tag{7.10}$$

are nonsingular. This is a nonempty Zariski-open condition on the coefficients of p. To see nonemptiness, take p with n distinct real roots: the matrices are positive definite for $k < n$, by evaluation of a nonzero real polynomial of degree at most k at those roots.

Apply Gram–Schmidt to $1, z, \dots, z^{d-1}$ for the complex bilinear pairing $(a, b) \mapsto S(ab)$. The result is a sequence of monic polynomials r_i of degree i, satisfying

$$S(r_i a) = 0 \quad (\deg a < i), \qquad 0 \le i < d. \tag{7.11}$$

This is bilinear orthogonality, not Hermitian orthogonality. Positivity is not assumed for the general complex polynomial p.

We can now construct the subspace needed in Lemma 7.3.

Proposition 7.5. *Under (7.10), the subspace*

$$U = \operatorname{span}_{\mathbb{C}}\{r_i(z)r_j^{\#}(w) : 0 \le j < i,\ i + j \le d - 1\} \subset \mathcal{A} \tag{7.12}$$

is totally isotropic for $\mathcal{H}$, satisfies $U \cap U^* = \{0\}$, and has complex dimension $\lfloor d^2/4 \rfloor$.

Proof. The products $r_i(z)r_j^{\#}(w)$ with $0 \le i, j < d$ are linearly independent in (7.4). The involution exchanges their two indices. The index sets defining U and U^* are disjoint, which proves $U \cap U^* = \{0\}$. Counting the pairs with $i > j$ and $i + j \le d - 1$ gives

$$\dim U = \begin{cases} t^2, & d = 2t, \\ t(t+1), & d = 2t+1, \end{cases} = \lfloor d^2/4 \rfloor. \tag{7.13}$$

Take indices $(i, j), (k, \ell)$ occurring in (7.12) and put

$$u = r_i r_\ell, \qquad v = r_j^{\#} r_k^{\#}.$$

Their total degree is at most $2d - 2$, so Lemma 7.4 applies. The constant term $S(r_i r_\ell)S^{\#}(r_j^{\#} r_k^{\#})$ is zero: by (7.11), both factors could be nonzero only if $i = \ell$ and $j = k$, contradicting $i > j$ and $k > \ell$.

Consider a term with coefficient c_{rs} in the first sum of (7.8). If $D_r(u) \neq 0$, then (7.7) implies $i + \ell + r \ge n$. Since $i + j \le n - m - 1$, it follows that

$$\ell - j \ge m + 1 - r, \qquad k - j \ge m + 2 - r > s.$$

Thus $j + s < k$, and (7.11) implies $S^{\#}(w^s r_j^{\#} r_k^{\#}) = 0$. If $D_r(u) = 0$, the same term is zero directly. Hence every term in the first sum vanishes.

For the second sum, a nonzero $D_r^{\#}(v)$ would imply $j + k + r \ge n$. Using $k + \ell \le n - m - 1$ and $i > j$, we obtain

$$j - \ell \ge m + 1 - r, \qquad i - \ell \ge m + 2 - r > s.$$

Now $\ell + s < i$, so $S(z^s r_i r_\ell) = 0$. Every term in the second sum also vanishes. By (7.5), the resulting identity $T(uv) = 0$ is precisely the required pairwise isotropy of U. □

Combining Lemmas 7.2 and 7.3 with Proposition 7.5, including (7.13), proves (7.2) under (7.10). Notice that the common zeros of f, f^* may have arbitrary multiplicities: the radical estimate in Lemma 7.3 has already accounted for them.

7.4 Removing the auxiliary moment condition

We finish the proof of Theorem 7.1. The coefficient space $p(z)+Q(z,w)$, $\deg p \le n$, $\deg_{\mathrm{tot}} Q \le m$, is stable under both derivatives. The highest analytic coefficient remains nonzero. Corollary 3.4 therefore gives an arbitrarily small perturbation in this space, without decreasing the zero count, whose zeros are all nondegenerate. A further small change of the coefficients of p preserves these zeros and satisfies the nonempty Zariski-open moment condition (7.10). The estimate proved using Proposition 7.5 applies to that last polynomial, and hence to the original one. This proves (7.2) with no assumption on multiplicities or moment matrices.

For a gap of two the bound is n^2-2, and it is attained by the real-valued construction in Theorem 10.1, with $m=n-2$. Consequently the total-degree statement from the earlier drafts is retained in its strongest form: for $\deg p = n \ge 3$ and $\deg_{\mathrm{tot}} Q \le n-2$, the maximum is exactly n^2-2. For larger gaps the formula (7.2) is an upper bound, not an asserted sharp formula. Combining it with (6.7) gives the two upper bounds in Theorem 1.4 for the harmonic subclass.

8 Harmonic polynomials with many zeros

We now turn to (8.1). Unlike the examples in Section 5, every polynomial constructed in this section is harmonic. The upper bound refines the Descartes argument of [Lun23, Section 4.1, author version, pp. 8–9]; the lower bound makes that argument exact on every line at once. The coefficients on different lines are not independent, and we keep track of their common angular dependence throughout.

The full harmonic maximum also equals the maximal valence in that class, since subtracting a constant preserves both degrees.

Lundberg's probabilistic construction gives $M^{\mathrm{harm}}_{m,n} \ge \lceil n\sqrt{m} \rceil$ by using the family

$$h(z) = \varepsilon z^n + q(z) + \overline{q(z)}, \qquad \deg q = m < n, \quad \varepsilon \in \mathbb{C}\setminus\{0\}. \tag{8.1}$$

His concluding question asks for a deterministic choice of q giving mn-order many zeros in this same family [Lun23, Section 4.2, author version, p. 9]. We determine its exact maximal zero count, denoted by $S_{m,n}$. Put $g=\gcd(m,n)$ and define

$$\Delta_{m,n} = \begin{cases} g, & \text{if } m/g \text{ and } n/g \text{ are both odd,} \\ 0, & \text{otherwise.} \end{cases} \tag{8.2}$$

The correction in (8.2) comes from the parity of the exponent gap and a count of equally spaced directions.

Theorem 8.1. *For every $1 \le m < n$,*

$$S_{m,n} = n(m+1) + \Delta_{m,n}. \tag{8.3}$$

The maximum is attained with $\varepsilon \in \mathbb{Q}_{>0}$ and $q \in \mathbb{Q}(i)[z]$, and all zeros of the extremizer are nondegenerate.

8.1 Real restrictions and the last exponent gap

For real polynomials the contribution of the final gap can be bounded more precisely than by two in all cases.

Lemma 8.2. *Let $N > M \geq 0$, let $a \in \mathbb{R} \setminus \{0\}$, and let $u \in \mathbb{R}[r]$ have degree at most M. Write $b = [r^M]u$ for the coefficient of r^M. The number of nonzero real roots of $ar^N + u(r)$, counted with multiplicity, is at most*

$$\begin{cases} M+1, & N-M \text{ odd}, \\ M + 2\mathbf{1}_{\{ab<0\}}, & N-M \text{ even}, \end{cases}$$

where the indicator $\mathbf{1}_{\{ab<0\}}$ equals one when $ab < 0$ and zero otherwise.

Proof. There is nothing to prove if $u = 0$. Otherwise, put $d = \deg u$ and list its nonzero coefficients in increasing exponent order. An odd gap between successive exponents contributes exactly one sign change in total to the coefficient lists of $u(r)$ and $u(-r)$. An even gap contributes zero or two sign changes, which is at most its size. The total contribution within u is therefore at most d.

Appending the degree-N coefficient contributes one when $N - d$ is odd. When $N - d$ is even, it contributes two precisely if the two leading coefficients have opposite signs, and otherwise zero. If $d = M$, this proves the assertion by Descartes' rule on the two half-lines. If $d < M$ and $N - M$ is odd, use $d+2 \leq M+1$. If $N-M$ is even, the case $d = M-1$ has odd last gap and gives $d+1 = M$; for $d \leq M-2$, use $d+2 \leq M$. These last two cases also explain why a vanishing coefficient b causes no exception. □

A rotation $z = e^{i\theta}\zeta$ with $\varepsilon e^{in\theta} > 0$ reduces (8.1) to $\varepsilon > 0$. The transformed polynomial $q(e^{i\theta}\zeta)$ still has degree m. After this normalization, the imaginary part of h vanishes exactly on the lines

$$\Lambda_j = e^{ij\pi/n}\mathbb{R}, \qquad 0 \leq j < n. \tag{8.4}$$

Write $q(z) = \sum_{k=0}^{m} a_k z^k$, with $a_m \neq 0$, and put $\theta_j = j\pi/n$. The restriction to Λ_j is the real polynomial

$$f_j(r) = \varepsilon(-1)^j r^n + 2\sum_{k=0}^{m} \operatorname{Re}(a_k e^{ik\theta_j}) r^k. \tag{8.5}$$

Its leading coefficient never vanishes. In particular, every member of (8.1) has a finite zero set, and the possible integer counts are bounded. Thus the maximum defining $S_{m,n}$ exists.

8.2 The exact upper bound

Assume first that $\operatorname{Re} a_0 \neq 0$. The origin is not a zero, so counts on the distinct lines (8.4) may be added. If $n - m$ is odd, Lemma 8.2 gives at most $m + 1$ roots per line. The resulting bound is $n(m + 1)$, and in this case $\Delta_{m,n} = 0$.

Suppose that $n - m = 2d$ is even, and put

$$G = \gcd(n, d), \qquad \ell = n/G, \qquad a_m = |a_m| e^{i\beta}.$$

The sign controlling the final gap in (8.5) is that of

$$(-1)^j \operatorname{Re}(a_m e^{imj\pi/n}) = |a_m| \cos\left(\beta - \frac{2\pi dj}{n}\right).$$

By Lemma 8.2,

$$N(h) \leq nm + 2B_\beta, \qquad B_\beta = \#\left\{0 \leq j < n : \cos\left(\beta - \frac{2\pi dj}{n}\right) < 0\right\}. \tag{8.6}$$

As j varies, the angles $2\pi dj/n$ traverse the vertices of a regular ℓ-gon, each with multiplicity G. An open semicircle contains at most $\ell/2$ such vertices when ℓ is even and at most $(\ell+1)/2$ when ℓ is odd. The first assertion follows by pairing antipodal vertices. For the second, $(\ell+3)/2$ consecutive vertices would span an angle $\pi+\pi/\ell$, which is longer than a semicircle. Therefore

$$2B_\beta \le \begin{cases} n, & \ell \text{ even}, \\ n+G, & \ell \text{ odd}. \end{cases} \tag{8.7}$$

Boundary vertices have zero cosine and are not counted, so no nonvanishing assumption on the intermediate coefficients is used.

We identify the parity alternative in (8.7) with (8.2). Write $m = gu$ and $n = gv$, where $g = \gcd(m,n)$ and $\gcd(u,v) = 1$. If u, v are both odd, then

$$G = \gcd\big(gv, g(v-u)/2\big) = g, \qquad \ell = v \text{ is odd}.$$

Here $\gcd(v,(v-u)/2) = 1$ because v is odd and coprime to u. Otherwise u, v have opposite parity. Since $n-m = g(v-u)$ is even, g is even, and

$$G = \frac{g}{2}\gcd(2v, v-u) = \frac{g}{2}, \qquad \ell = 2v \text{ is even}.$$

The last greatest common divisor is one because $v-u$ is odd and coprime to v. Equations (8.6) and (8.7) now give $N(h) \le n(m+1) + \Delta_{m,n}$.

It remains to treat $\operatorname{Re} a_0 = 0$. Each f_j in (8.5) is then divisible by r. Apply Lemma 8.2 to f_j/r, using $(N,M) = (n-1, m-1)$. The last-gap parity and the two relevant leading signs are unchanged, whereas each nonzero-root bound decreases by one. The common origin must be restored once, not n times. This actually gives the stronger estimate

$$N(h) \le n(m+1) + \Delta_{m,n} - n + 1.$$

The upper bound in Theorem 8.1 is therefore valid for all members of the family, including those with a zero at the origin, missing coefficients, or degenerate zeros.

8.3 Rational coefficients attaining the bound

The lower bound uses the same lines but forces every restriction to attain its Descartes bound. Fix $m < n$ and set

$$g = \gcd(m,n), \qquad \eta = \frac{1}{4n}, \qquad t = \frac{1}{16n(m+1)}, \qquad \gamma = 1 + i\eta,$$
$$\sigma = \begin{cases} -1, & \Delta_{m,n} = g \text{ and } n/g \equiv 1 \pmod 4, \\ 1, & \text{otherwise}. \end{cases} \tag{8.8}$$

Define

$$q(z) = \frac{\sigma\gamma}{2}\sum_{k=0}^{m} t^{k^2} z^k, \qquad p(z) = t^{n^2} z^n + q(z), \qquad h = p + \overline{q}. \tag{8.9}$$

All coefficients in (8.9) belong to $\mathbb{Q}(i)$, the two degrees are exactly m, n, and $h(0) = \sigma \neq 0$. The restriction of this h to Λ_j is

$$f_j(r) = (-1)^j t^{n^2} r^n + \sigma \sum_{k=0}^{m} \left(\cos\frac{kj\pi}{n} - \eta \sin\frac{kj\pi}{n}\right) t^{k^2} r^k. \tag{8.10}$$

The factors in parentheses are uniformly separated from zero:

$$\frac{1}{4n} \le \left|\cos\frac{kj\pi}{n} - \eta \sin\frac{kj\pi}{n}\right| < 2. \tag{8.11}$$

Indeed, a zero cosine gives absolute value exactly η. For a nonzero cosine, the angle's distance from a zero of cosine, modulo π, is at least $\pi/(2n)$. Hence its absolute value is at least $\sin(\pi/(2n)) \ge 1/n$. Subtracting a quantity of absolute value at most η gives at least $3/(4n)$. The upper bound in (8.11) follows from $1 + \eta < 2$.

The coefficient factor of r^n in (8.10) has absolute value one. Use (5.1) with the support $\{0, 1, \dots, m, n\}$, $\mu = 1/(4n)$, and $B = 2$. The sum of the other normalized monomials at every prescribed radius is bounded by

$$2(m+1)t = \frac{1}{8n} < \frac{1}{4n}.$$

Lemma 5.1 therefore applies simultaneously to every f_j. It gives exact positive and negative root counts, and all these roots are simple.

For each of the consecutive exponent pairs $(0,1), (1,2), \dots, (m-1, m)$, exactly one of the sign lists for $f_j(r)$ and $f_j(-r)$ changes sign. These pairs contribute exactly m roots on each line. If $n - m$ is odd, the final gap contributes one more, proving (8.3) in this case.

For $n - m = 2d$, retain the integers $G = \gcd(n, d)$ and $\ell = n/G$ from the upper-bound proof. Write $\gamma = |\gamma| e^{i\alpha}$, so $0 < \alpha = \arctan\eta < 1/(4n)$. The final gap contributes two roots on Λ_j precisely when

$$\sigma \cos\left(\alpha - \frac{2\pi dj}{n}\right) < 0, \tag{8.12}$$

and contributes none otherwise. When ℓ is even, antipodal pairing gives exactly $n/2$ negative signs. None is zero, by (8.11), so the final gaps contribute n roots in total.

When ℓ is odd, the preceding greatest-common-divisor calculation gives $G = g$ and $\ell = n/g$. At phase zero, a regular ℓ-gon has $(\ell-1)/2$ vertices in the open negative semicircle if $\ell \equiv 1 \pmod 4$, and $(\ell+1)/2$ if $\ell \equiv 3 \pmod 4$. This follows directly by counting integers strictly between $\ell/4$ and $3\ell/4$. Every vertex is at least $\pi/(2\ell)$ from the boundary of that semicircle. Since $\alpha < 1/(4n) < \pi/(2\ell)$, the phase shift does not change these counts. The sign σ in (8.8) selects $(\ell+1)/2$ negative cosines, each repeated g times. Thus (8.12) contributes $g(\ell+1) = n+g$ roots. Adding the nm roots supplied by the consecutive low exponents proves (8.3) in every case.

Finally, all zeros are nonzero and different lines meet only at the origin. At a zero $z = e^{i\theta_j} r_0$, choose the oriented local coordinates $z = e^{i\theta_j}(r + is)$. Along $s = 0$,

$$\partial_r \operatorname{Re} h = f_j'(r_0) \neq 0, \qquad \partial_r \operatorname{Im} h = 0, \qquad \partial_s \operatorname{Im} h = t^{n^2} n (-1)^j r_0^{n-1} \neq 0.$$

The real Jacobian is nonzero. This completes the proof of Theorem 8.1, including the exact count, rationality of the coefficients, and nondegeneracy of every zero.

8.4 Interpretation and an exact small example

The construction proves degree-uniform growth from below: $M_{m,n}^{\mathrm{harm}} \ge S_{m,n} \ge n(m+1)$. For fixed m, $S_{m,n}/n \to m+1$; for $m \to \infty$ the ratio $S_{m,n}/(mn)$ tends to one uniformly over $n > m$, because $0 \le \Delta_{m,n} \le m$. These statements answer Lundberg's deterministic question [Lun23, Section 4.2, author version, p. 9]. They do not identify the maximum for arbitrary analytic heads, nor the maximum over arbitrary target values of one fixed member of this restricted family. The

exact subclass count does not settle the best coefficient for fixed $m \geq 2$ in the unrestricted harmonic class.

The general construction uses very different coefficient scales. For one small pair, short rational coefficients suffice.

Example 8.3. The harmonic polynomial

$$h(z) = 10^{-8}z^6 - \operatorname{Re}\big((1+i)(z^2+10z+1)\big)$$

has exactly twenty nondegenerate zeros. Its analytic and conjugate degrees are 6 and 2, respectively.

Proof. Here $q(z) = -(1+i)(z^2+10z+1)/2$ and $p(z) = 10^{-8}z^6 + q(z)$. On Λ_j, with $\theta_j = j\pi/6$ and $0 \leq j \leq 5$, the equation is

$$f_j(r) = 10^{-8}(-1)^j r^6 - \big(\cos 2\theta_j - \sin 2\theta_j\big)r^2 - 10\big(\cos\theta_j - \sin\theta_j\big)r - 1 = 0.$$

The absolute values of the linear coefficients lie between $7/2$ and 14, and those of the quadratic coefficients between $7/20$ and $7/5$. These inequalities follow from $17/10 < \sqrt{3} < 9/5$. At the radii $1/100, 1, 50, 1000$, the constant, linear, quadratic, and degree-six monomials, respectively, dominate all the others. For the quadratic monomial the estimate is

$$\frac{7}{20}\,50^2 = 875 > 1 + 14\cdot 50 + 10^{-8}50^6 = \frac{3429}{4}.$$

For the other three monomials the same bounds give, respectively,

$$1 > \frac{14}{100} + \frac{7}{5\cdot 10^4} + 10^{-20}, \qquad \frac{7}{2} > 1 + \frac{7}{5} + 10^{-8},$$
$$10^{10} > 1 + 14\cdot 10^3 + \frac{7}{5}\cdot 10^6.$$

Lemma 5.1 makes the following counts exact; signs are listed in exponent order $0, 1, 2, 6$.

j	Coefficient signs	Positive roots	Negative roots	Total
0	$-\ -\ -\ +$	1	3	4
1	$-\ -\ +\ -$	2	2	4
2	$-\ +\ +\ +$	1	1	2
3	$-\ +\ +\ -$	2	2	4
4	$-\ +\ -\ +$	3	1	4
5	$-\ +\ -\ -$	2	0	2

The roots are simple, and $h(0) = -1$, so there is no overlap between lines. The tangential and normal derivative calculation above proves planar nondegeneracy. □

Example 8.3 attains $S_{2,6} = 20$; it is not asserted to maximize the unrestricted harmonic class. An independent exact Sturm check over $\mathbb{Q}(\sqrt{3})$ is included in the supplement as `verify_harmonic_20.py`. The analytic proof above does not depend on that computation.

9 A harmonic construction with a Chebyshev analytic part

The monomial-head family has now been counted exactly. Changing its analytic head yields a larger lower bound for the unrestricted harmonic class. Here T_j denotes the Chebyshev polynomial specified by $T_j(\cos\theta) = \cos(j\theta)$.

Theorem 9.1. *Fix $1 \le m < n$ and $a > 2$. For all sufficiently small $\varepsilon > 0$,*

$$h_\varepsilon(z) = \varepsilon T_n(z) - 2\,\mathrm{Im}\,T_m(z-a) \tag{9.1}$$

has exactly $(m+2)n - m - \gcd(m,n)$ distinct nondegenerate zeros.

The harmonic decomposition in this theorem is

$$p_\varepsilon(z) = \varepsilon T_n(z) + iT_m(z-a), \qquad q(z) = iT_m(z-a).$$

Thus both degrees are exact. Rational $a > 2$ and sufficiently small positive rational ε give coefficients in $\mathbb{Q}(i)$. In particular,

$$M_{m,n}^{\mathrm{harm}} \ge (m+2)n - m - \gcd(m,n). \tag{9.2}$$

For fixed m, the leading coefficient of this lower bound is $m+2$. It is a lower bound for the unrestricted harmonic class, not a general formula for its maximum.

We prove Theorem 9.1 by accounting for every zero on the nodal set of $\mathrm{Im}\,T_n$. The identity $T_n(\cos\zeta) = \cos(n\zeta)$ follows from the defining recurrence and holds for complex ζ.

9.1 The nonreal nodal curves

Every point of the upper half-plane has a unique representation

$$z = \cos(\theta - it) = \cos\theta\cosh t + i\sin\theta\sinh t, \qquad 0 < \theta < \pi, \quad t > 0. \tag{9.3}$$

For example, inverting $z = (\zeta + \zeta^{-1})/2$ selects the unique root outside the unit circle, whose imaginary part is positive. Its polar coordinates give (9.3). The coordinate map has nonzero derivative in this open half-strip. Since $\mathrm{Im}\,T_n(z) = \sin(n\theta)\sinh(nt)$, the upper-half-plane zero set consists of the disjoint regular curves

$$z_j(t) = \cos\theta_j\cosh t + i\sin\theta_j\sinh t, \qquad \theta_j = j\pi/n, \quad 1 \le j \le n-1. \tag{9.4}$$

The lower curves are their reflections; all remaining points of the nodal set lie on the real axis.

The curve of angle θ has real coordinate at height $y > 0$

$$X_\theta(y) = \cot\theta\sqrt{y^2 + \sin^2\theta}.$$

The upper nodal curves of $\mathrm{Im}\,T_m(z-a)$ have the same form translated by a, with angles $\phi_\ell = \ell\pi/m$, $1 \le \ell \le m-1$.

Lemma 9.2. *For $a > 2$, a curve of angle θ and a translated curve of angle ϕ have one transverse intersection if $\theta < \phi$, and none if $\theta \ge \phi$.*

Proof. An intersection is a solution of $X_\theta(y) - X_\phi(y) = a$. For each y, $X_\theta(y)$ is strictly decreasing in θ, so $\theta \geq \phi$ is impossible. If $\theta < \phi$, the difference at zero is $\cos\theta - \cos\phi \in (0, 2)$, and the difference tends to infinity. It is strictly increasing for $y > 0$, because

$$\frac{\partial}{\partial\theta}\frac{\cot\theta}{\sqrt{y^2+\sin^2\theta}} = -\frac{1}{\sin^2\theta\sqrt{y^2+\sin^2\theta}} - \frac{\cos^2\theta}{(y^2+\sin^2\theta)^{3/2}} < 0$$

and $X'_\theta(y) = y\cot\theta/\sqrt{y^2+\sin^2\theta}$. This proves existence, uniqueness, and transversality. □

Let k_j count the zeros of $\operatorname{Im} T_m(z_j(t) - a)$ for $t > 0$ on the curve (9.4). They are simple by Lemma 9.2, and

$$k_j = \#\{1 \leq \ell \leq m-1 : j/n < \ell/m\}, \qquad \sum_{j=1}^{n-1} k_j = \frac{(n-1)(m-1)-(g-1)}{2}, \quad g = \gcd(m,n). \tag{9.5}$$

For the sum, there are $g - 1$ equality pairs, and $(j, \ell) \mapsto (n - j, m - \ell)$ exchanges the two strict inequalities. No triple intersection is possible, since the curves within either upper family are disjoint.

9.2 Small levels at the two ends

Put $r(z) = T_m(z - a)$ and

$$A_j(t) = \frac{2\operatorname{Im} r(z_j(t))}{\cosh(nt)}, \qquad t > 0.$$

The zero equation for (9.1) on the upper curve is $A_j(t) = \varepsilon(-1)^j$, and on its lower reflection it is $A_j(t) = -\varepsilon(-1)^j$. Their combined count is the number of solutions of $A_j(t) = \pm\varepsilon$.

At zero,

$$A_j(t) = 2r'(\cos\theta_j)\sin\theta_j\, t + O(t^2),$$

with nonzero coefficient. The roots of T'_m lie in $(-1, 1)$ when $m \geq 2$, whereas $\cos\theta_j - a < -1$; for $m = 1$ the derivative is one. At infinity write $r = \sum_{k=0}^m b_k T_k$ with real coefficients. Then

$$\operatorname{Im} r(z_j(t)) = \sum_{k=1}^m b_k \sin(k\theta_j)\sinh(kt).$$

This is not identically zero, by its derivative at zero. Let κ_j be the largest index with nonzero coefficient in this sum. It satisfies $1 \leq \kappa_j \leq m < n$. For a nonzero real C_j,

$$\begin{aligned} A_j(t) &= C_j e^{-(n-\kappa_j)t}(1 + o(1)), \\ A'_j(t) &= -(n-\kappa_j)C_j e^{-(n-\kappa_j)t}(1 + o(1)). \end{aligned} \tag{9.6}$$

Thus A_j is eventually monotone and tends to zero. Keeping κ_j in this argument is important: it need not equal m when $\sin(m\theta_j) = 0$.

It follows that, for sufficiently small $\varepsilon > 0$, the two levels together have exactly $2k_j + 2$ simple solutions. Indeed, choose disjoint intervals around the k_j simple interior zeros in (9.5), a one-sided interval at zero, and a terminal half-line on which (9.6) is monotone. Each interior zero contributes one solution to each level. The initial and terminal intervals each contribute one solution to one of the two levels. The remaining set is compact and $|A_j|$ has a positive minimum there, so it contributes none. This argument proves both the exact count and the nonvanishing of A'_j at every solution.

9.3 Completion and nondegeneracy

On the real axis, (9.1) has exactly the n simple zeros of T_n. Their Jacobians are

$$|\varepsilon T_n'(x) + ir'(x)|^2 - |ir'(x)|^2 = \varepsilon^2 T_n'(x)^2 > 0.$$

At a nonreal zero, the derivative of the real part along its curve is a nonzero multiple of $A_j'(t)$. The imaginary part is zero along the curve and has transverse derivative $\varepsilon n(-1)^j \sinh(nt) \neq 0$. The coordinate map in (9.3) is regular, so the planar Jacobian is nonzero.

One common small positive threshold for ε works on the finitely many curves. Using (9.5), the exhaustive count is

$$\begin{aligned} N(h_\varepsilon) &= n + \sum_{j=1}^{n-1} (2k_j + 2) \\ &= n + (n-1)(m-1) - g + 1 + 2(n-1) \\ &= (m+2)n - m - g. \end{aligned}$$

This proves Theorem 9.1 and (9.2).

9.4 Comparison with the monomial-head construction

The preceding section gives

$$S_{m,n} = n(m+1) + \Delta_{m,n}, \qquad \Delta_{m,n} = \begin{cases} g, & m/g \text{ and } n/g \text{ both odd}, \\ 0, & \text{otherwise}, \end{cases} \qquad g = \gcd(m,n).$$

The count in (9.2) exceeds it by $n - m - g - \Delta_{m,n}$, which is nonnegative. If $\Delta = 0$, use $g \mid (n-m)$; if $\Delta = g$, then $(n-m)/g$ is a positive even integer. For fixed m, the difference is $n - O_m(1)$.

Wilmshurst's proposed estimate $3n - 2 + m(m-1)$ was disproved in general by Lee–Lerario–Lundberg [LLL15, author version v1, pp. 2–3]. The next consequence is a uniform family of counterexamples, not a claim to the first disproof of that conjecture.

Corollary 9.3. *If $2 \le m \le n-2$ and $(m,n) \neq (2,4)$, there is a harmonic polynomial of degrees (m,n) with more than $3n - 2 + m(m-1)$ distinct zeros.*

Proof. Subtract the proposed count from Theorem 9.1. The difference is $(m-1)(n-m-1)+1-g$. If $n - m \ge 3$, it is at least $(m-1)(n-m-2) > 0$, since $g \le m$. If $n - m = 2$, it is $m - \gcd(m,2)$, positive except when $m = 2$. □

Corollary 9.3 does not determine the corresponding harmonic maxima. For example, it gives $M_{2,5}^{\mathrm{harm}} \ge 17$ but not equality. The endpoint counts for $m = 1$ and $m = n-1$ in Theorem 9.1 recover already known sharp values. The rectangular equality range in Theorem 1.3 concerns a different coefficient space. In particular, its mixed-term constructions do not give harmonic extremizers; the alternatives $M_{2,7} \in \{49, 51, 53\}$ also remain unresolved.

The construction, Corollary 6.3, and Theorem 7.1 prove (1.3). The exact endpoints in Theorem 1.4 follow from [KS03, Gey08, Wil98]. For $m \ge 2$, the coefficient $3m + 2$ is at most $4m$; for $m = 1$, use the stronger classical $3n - 2$. This proves the uniform bound $mn \le M_{m,n}^{\mathrm{harm}} \le 4mn$ claimed in the introduction. No assertion that the Chebyshev lower bound is the exact maximum is needed.

10 Real-valued lower-degree perturbations

The zero-counting problem for $p(z) + \overline{q(z)}$ distinguishes the degrees of its analytic and anti-analytic parts. One particularly useful family is $\varepsilon z^n + q(z) + \overline{q(z)}$. In this family, the imaginary part vanishes on n lines, and one may count zeros by restricting the real part to those lines. Lundberg used this reduction to obtain the bound $n(m+2)$ and asked for deterministic constructions with order mn zeros; see [Lun23, Sections 4.1–4.2, author version, pp. 8–9].

Here we retain the monomial analytic term but allow the real-valued perturbation to be an arbitrary real polynomial, not necessarily a harmonic one. Throughout, $1 \le m < n$ and

$$F(z) = \varepsilon z^n + R(x,y), \qquad z = x + iy, \quad \varepsilon \in \mathbb{C} \setminus \{0\}, \quad R \in \mathbb{R}[x,y], \quad \deg R = m. \tag{10.1}$$

Write $N(F) = \#F^{-1}(0)$, counting distinct points. Let $A_{m,n}$ be the largest possible value of $N(F)$ in (10.1). A zero is *nondegenerate* if the real Jacobian of $F : \mathbb{R}^2 \to \mathbb{R}^2$ is nonzero there. Finiteness, including for singular zeros, will follow directly from the line restrictions.

The exact answer depends only on the parity of the degree gap.

Theorem 10.1. *For every $1 \le m < n$,*

$$A_{m,n} = n(m+1) + \begin{cases} 0, & n-m \text{ is odd}, \\ m, & n-m \text{ is even}. \end{cases} \tag{10.2}$$

The maximum is attained with $\varepsilon \in \mathbb{Q}_{>0}$, $R \in \mathbb{Q}[x,y]$, and all zeros nondegenerate.

There is a finer answer when the gap is even. Let H_m be the leading homogeneous part of R. Denote by $s(H_m)$ the number of distinct real projective zeros of H_m having odd multiplicity. Equivalently, this is the number of real linear factors occurring with odd exponent, with proportional factors identified. Factoring a real binary form shows that

$$0 \le s(H_m) \le m, \qquad s(H_m) \equiv m \pmod 2.$$

No condition is imposed on its other real or complex factors.

The next statement gives both the coefficient-dependent upper bound and its exact extremal meaning.

Theorem 10.2. *Suppose $n-m$ is even. Every polynomial in (10.1) satisfies*

$$N(F) \le n(m+1) + s(H_m). \tag{10.3}$$

For each integer s with $0 \le s \le m$ and $s \equiv m \pmod 2$, the maximum among such polynomials with $s(H_m) = s$ equals $n(m+1) + s$. It is attained with rational coefficients and only nondegenerate zeros.

Theorem 10.2 explains the extra term in (10.2): the leading real homogeneous form can change sign on at most m projective directions. In the harmonic subclass those directions are equally spaced, which imposes an additional arithmetic restriction. The exact difference from the harmonic subclass is recorded after the proof.

The two real-root counting ingredients have already been proved: Lemma 8.2 is the joint Descartes estimate, and Lemma 5.1 makes it exact under separated coefficients. We now use the fact that all leading coefficients of the line restrictions come from a single real homogeneous form.

10.1 The upper bound and the leading form

A rotation $z = e^{i\omega}\zeta$ with $\varepsilon e^{in\omega} = |\varepsilon|$ preserves the number of zeros, the degree of the real perturbation, and $s(H_m)$. We may therefore assume $\varepsilon > 0$. Put

$$\theta_j = \frac{j\pi}{n}, \qquad v_j = (\cos\theta_j, \sin\theta_j), \qquad L_j = e^{i\theta_j}\mathbb{R}, \qquad 0 \le j < n. \tag{10.4}$$

Every zero lies on one of these lines because $\Im F(z) = \varepsilon \Im z^n$. On L_j the equation is the real equation

$$f_j(r) = \varepsilon(-1)^j r^n + R(rv_j) = 0. \tag{10.5}$$

Each f_j has degree n and nonzero leading coefficient. In particular, there are finitely many zeros without any separate hypothesis.

Suppose first that $R(0,0) \neq 0$. The lines in (10.4) meet only at the origin, which is not a zero, so their root counts add. If $n - m$ is odd, Lemma 8.2 gives $N(F) \le n(m+1)$ immediately.

Now let $n - m$ be even. Define

$$B(H_m) = \#\{0 \le j < n : (-1)^j H_m(v_j) < 0\}.$$

Equations (10.5) and the estimate in Lemma 8.2 give

$$N(F) \le nm + 2B(H_m). \tag{10.6}$$

The missing ingredient is a constraint relating the leading coefficients on different lines.

Lemma 10.3. *If $n - m$ is even and H_m is a nonzero real homogeneous polynomial of degree m, then*

$$2B(H_m) \le n + s(H_m). \tag{10.7}$$

Proof. Regard $0, \dots, n-1$ as the vertices of a cycle, and mark the vertices counted by $B(H_m)$. Let E be the number of edges with both endpoints marked. At most $2(n - B(H_m))$ edges meet an unmarked vertex, so $E \ge 2B(H_m) - n$.

Each edge with two marked endpoints forces a sign change of $H_m(\cos\theta, \sin\theta)$ in the corresponding open interval of length π/n. For the last interval, use $H_m(v_n) = (-1)^m H_m(v_0)$ and $m \equiv n \pmod 2$; marking is then consistent across the endpoint of $[0, \pi]$. A sign change requires a real projective zero of odd multiplicity in that interval. Distinct edges give disjoint intervals, so $E \le s(H_m)$. This proves (10.7). Zeros at the sample directions do not create an exception: such vertices are unmarked, whereas an edge counted by E has nonzero values at both endpoints. □

Combining (10.6) and (10.7) proves (10.3) in Theorem 10.2, and hence the even case of Theorem 10.1, whenever the origin is not a zero.

If $R(0,0) = 0$, divide each restriction in (10.5) by r. The leading exponent becomes $n - 1$, the lower degree is at most $m - 1$, and the gap retains its parity. Count the nonzero roots with Lemma 8.2, then add the common origin only once. For odd gap the resulting bound is $nm + 1$. For even gap it is

$$n(m-1) + 2B(H_m) + 1 \le nm + s(H_m) + 1.$$

Both are strictly below the corresponding asserted maximum. This completes the upper-bound proofs for all coefficients and all multiplicities.

10.2 Simultaneous extremal constructions

We now attain the bounds. In the even case we prescribe an admissible integer s, with $0 \le s \le m$ and $s \equiv m \pmod 2$. Set

$$\eta = \frac{1}{4n}, \qquad L(x,y) = x + \eta y, \qquad \tau = \frac{1}{2(m+1)(8n)^m}.$$

For $0 \le k < m$, put $H_k = L^k$. If $n - m$ is odd, also put $H_m = L^m$. If $n - m$ is even, define

$$\begin{aligned} \phi_\nu &= \frac{(n - s + \nu + \frac{1}{2})\pi}{n}, \qquad 0 \le \nu < s, \\ H_m(x,y) &= -(x^2+y^2)^{(m-s)/2} \prod_{\nu=0}^{s-1} (\sin\phi_\nu\, x - \cos\phi_\nu\, y). \end{aligned} \tag{10.8}$$

An empty product equals one. This leading form has exactly s real projective zeros, all simple; the other displayed factor has no real projective zeros. Finally define

$$R_\tau(x,y) = \sum_{k=0}^{m} \tau^{k^2} H_k(x,y), \qquad F_\tau(z) = \tau^{n^2} z^n + R_\tau(\Re z, \Im z). \tag{10.9}$$

The perturbation has degree exactly m, and $F_\tau(0) = 1$.

10.2.1 One choice of scale works on every line

For the sample directions in (10.4),

$$\frac{1}{4n} \le |L(v_j)| < 2.$$

Indeed, if $\cos\theta_j = 0$, the left side is an equality. Otherwise $|\cos\theta_j| \ge \sin(\pi/(2n)) \ge 1/n$, so subtracting at most η still leaves at least $3/(4n)$. It follows that the values of $H_k(v_j)$ for $k < m$, and also for $k = m$ in the odd case, are nonzero and have absolute value between $(4n)^{-m}$ and 2^m.

For (10.8), every sine factor at v_j is nonzero and has absolute value at least $\sin(\pi/(2n)) \ge 1/n$. Since $|v_j| = 1$, it follows that $|H_m(v_j)| \ge n^{-s} \ge (4n)^{-m}$ and $|H_m(v_j)| \le 1$. Thus every line restriction of (10.9), on the support

$$0, 1, \dots, m, n,$$

has its unscaled coefficients bounded in absolute value between $\delta = (4n)^{-m}$ and $B_0 = 2^m$. The leading coefficient is $(-1)^j$. Our scale satisfies

$$(m+1)B_0\tau = \frac{\delta}{2} < \delta.$$

Lemma 5.1, with (5.1), therefore applies simultaneously to the restrictions on all n lines and to both halves of each line.

10.2.2 Exact count and nondegeneracy

The exponent pairs $(0,1), (1,2), \dots, (m-1,m)$ each contribute one to the sum of positive and negative sign variations, regardless of the coefficient signs. Their total contribution on each line is exactly m. If $n - m$ is odd, the last gap contributes one more. Thus (10.9) has exactly $n(m+1)$ zeros in the odd case.

In the even case, the last gap contributes two exactly when $(-1)^j H_m(v_j) < 0$. Let $d' = n - s$, which is even. In (10.8), $H_m(v_j)$ is negative for $0 \le j \le d'$. For $j > d'$, it changes sign at each step, because exactly one of the linear factors changes sign. Consequently the marked sample indices are the even indices in $0, \dots, d' - 1$, together with all indices in $d', \dots, n - 1$. The same description covers $s = 0$. Hence

$$B(H_m) = \frac{n - s}{2} + s = \frac{n + s}{2}.$$

Lemma 5.1 now gives exactly $nm + 2B(H_m) = n(m + 1) + s$ zeros. This proves attainment in Theorem 10.2. Taking $s = m$ proves the even case of (10.2).

All the real roots on each line are simple and nonzero. At such a root, the derivative of $\Re F_\tau$ in the direction of the line is nonzero. The derivative of $\Im F_\tau = \tau^{n^2} \Im z^n$ in that direction is zero, but its normal derivative is nonzero. These two facts imply a nonzero real Jacobian. This verifies nondegeneracy in the plane, rather than merely simplicity of a univariate restriction.

10.2.3 Rational coefficients

In the odd case the construction already has rational coefficients. In the even case the displayed coefficients of H_m are real algebraic numbers. Approximate the coefficients of each of its linear factors by rational numbers, keeping the positive quadratic factor unchanged. For sufficiently accurate approximations, the s real projective roots remain distinct, avoid the sample directions, and have the same signs at those directions. All strict dominance inequalities are also preserved. The count and nondegeneracy just proved therefore remain valid, and the leading form still satisfies $s(H_m) = s$. The resulting polynomial has coefficients in $\mathbb{Q}$. This proves all attainment assertions in Theorems 10.1 and 10.2.

10.3 Comparison with the restricted harmonic family

Equations (10.2) and (8.3) give

$$A_{m,n} - S_{m,n} = \begin{cases} 0, & n - m \text{ odd}, \\ m - \Delta_{m,n}, & n - m \text{ even}. \end{cases}$$

Thus arbitrary real-valued lower terms do not change the limiting coefficient $m + 1$ for fixed m, although they can change the bounded correction. This compares only the two monomial-head spaces, not the unrestricted harmonic and rectangular maxima.

10.4 A rational quartic with fourteen zeros

The first strict comparison above has a particularly simple witness. Set $t = 1/1000$ and consider

$$F(z) = t^{16} z^4 - t^4(2x^2 + 5xy + 2y^2) + t\left(x + \frac{y}{16}\right) + 1, \qquad z = x + iy. \tag{10.10}$$

All possible zeros lie on $y = 0$, $y = x$, $x = 0$, or $y = -x$. Use respectively the real parametrizations $r, (1 + i)r, ir, (-1 + i)r$. The restrictions are $at^{16}r^4 + bt^4r^2 + ctr + 1$, with the following exact coefficients:

Line	a	b	c	Number of real roots
$y = 0$	1	-2	1	4
$y = x$	-4	-9	$17/16$	2
$x = 0$	1	-2	$1/16$	4
$y = -x$	-4	1	$-15/16$	4

For the support $0, 1, 2, 4$, all unscaled coefficients have absolute value at least $1/16$ and at most 9. Since $3 \cdot 9t < 1/16$, Lemma 5.1 applies. The two Descartes counts give the last column, and all these roots are simple. The four lines meet only at the origin, where $F(0) = 1$, so there are exactly fourteen distinct zeros. Their real Jacobians are nonzero by the argument already given.

This example is not harmonic: its Laplacian is $-8t^4 \neq 0$. It proves $A_{2,4} = 14$, whereas Theorem 8.1 gives $S_{2,4} = 12$. It does not give an upper bound of twelve for an arbitrary harmonic polynomial with analytic degree four and anti-analytic degree two.

The supplied standard-library Python program independently verifies (10.10) by exact rational Sturm computations. It also checks square-freeness of every line restriction and all the strict dominance inequalities. Its finite combinatorial tests of (10.8) and (10.7) are diagnostic only; the proofs above cover all degrees.

11 Sparse extremizers and a published conjecture

The coefficient-count conjecture in [SZ24, Remark 4.4, p. 16] is false. We give the least-degree counterexample, an exact obstruction for binomials, and sharp zero counts in a full cyclic coefficient space. The size of the support is a different extremal question from the number of zeros at prescribed bidegree. Write $\nu(F)$ for the number of nonzero monomials of F.

11.1 Two monomials and the first possible counterexample

The needed lower bound for the coefficient count does not require a curve-theoretic argument.

Proposition 11.1. *Let $n \geq 2$. A degree-n polyanalytic polynomial with at most two nonzero monomials and finitely many zeros has at most $2n$ distinct zeros. This bound is attained.*

Proof. A single nonconstant monomial has only the origin as a zero. For two distinct monomials write

$$F(z) = \alpha z^{a_1} \bar{z}^{b_1} + \beta z^{a_2} \bar{z}^{b_2}, \qquad \alpha\beta \neq 0, \quad d_j = a_j + b_j, \quad k_j = a_j - b_j.$$

For $z = re^{i\theta} \neq 0$, the equation is

$$r^{d_1 - d_2} e^{i(k_1 - k_2)\theta} = -\beta/\alpha. \tag{11.1}$$

If $d_1 = d_2$, a nonzero solution generates a whole ray; hence finiteness excludes all nonzero solutions. If $d_1 \neq d_2$ and $k_1 = k_2$, a nonzero solution generates a whole circle. In the remaining case, (11.1) gives one positive radius and exactly $|k_1 - k_2|$ angles. Since d_1, d_2 are distinct integers with maximum n,

$$|k_1 - k_2| \leq d_1 + d_2 \leq 2n - 1.$$

Adding the origin when it is a zero proves the upper bound. The polynomial $z^n - \bar{z}^{n-1}$ has the origin and $2n - 1$ distinct unit roots as its zeros, and attains the bound. □

Let s_n be the least support size of a degree-n extremal polynomial. The exact low-degree values follow once Theorem 11.3 is proved.

Corollary 11.2. *One has $s_1 = 1$, $s_2 = 2$, $s_3 = 3$, and $s_4 = 3$.*

Proof. The examples z and $z^2 - \bar{z}$ give the first two values, and a single monomial cannot have four zeros. For degree three use

$$z^3 + z\bar{z} - \frac{1}{8}.$$

Its imaginary part vanishes on the three lines $e^{2\pi ij/3}\mathbb{R}$, $j = 0, 1, 2$. On each line the restriction is

$$r^3 + r^2 - \frac{1}{8} = \left(r + \frac{1}{2}\right)\left(r^2 + \frac{1}{2}r - \frac{1}{4}\right),$$

which has three different nonzero real roots. Thus there are nine distinct zeros. Proposition 11.1 excludes two monomials in degrees three and four; Theorem 11.3 supplies the quartic upper bound for s_4. □

In particular, the proposed inequality $s_n \geq n$ first fails at $n = 4$. This is a statement about the degree of the first counterexample, not a determination of s_n for all n.

Theorem 11.3. *The polynomial*

$$F(z) = z^4 + z\bar{z}^2 - \frac{1}{8}\bar{z} \tag{11.2}$$

has sixteen nondegenerate zeros. Among all degree-four polyanalytic polynomials with sixteen distinct zeros, the least number of nonzero coefficients is three.

There is also a uniform even-degree construction. In this statement the number of coefficients is an attained upper bound for the least support size; minimality is asserted only in the quartic case above.

Theorem 11.4. *For every even integer $n \geq 2$, there is a polynomial $P \in \mathbb{Q}[z, w]$ whose unique leading homogeneous term is a nonzero multiple of z^n, with $\deg_w P = n/2$, exactly $n/2 + 1$ nonzero coefficients, and n^2 nondegenerate zeros on $w = \bar{z}$.*

Section 11.1 gives the elementary obstruction for two monomials. Section 11.2 proves Theorem 11.3 and, in Proposition 11.5, classifies the entire complex one-parameter family containing it. Section 11.3 proves a sharp zero-count formula in a cyclically equivariant coefficient space and yields Theorem 11.4.

11.2 A trinomial with sixteen zeros

We first prove Theorem 11.3 directly. The parameter calculation afterward gives its full range of stability within the trinomial family.

Proof. More generally, for real t put

$$F_t(z) = z^4 + (|z|^2 - t)\bar{z}.$$

The origin is a zero, with $J_{F_t}(0) = -t^2$. For a nonzero zero write $z = re^{i\theta}$ and divide by $\bar{z}$. This gives

$$r^3 e^{5i\theta} + r^2 - t = 0. \tag{11.3}$$

Suppose $0 < t < 4/27$. The positive radii with $e^{5i\theta} = 1$ are the positive roots of $r^3 + r^2 - t$, which has exactly one such root. The radii with $e^{5i\theta} = -1$ are the positive roots of $r^2 - r^3 - t$. The latter function is negative at zero, has its unique positive maximum $4/27 - t$ at $2/3$, and tends to $-\infty$; hence it has exactly two positive roots. These three radii are different and simple, and each gives five angles in (11.3). Together with zero, there are exactly sixteen zeros.

For $t = 1/8$ they are

$$\{0\} \cup \left\{ \frac{\sqrt{5}-1}{4} e^{2\pi i j/5} : 0 \le j < 5 \right\} \cup \left\{ r e^{(2j+1)\pi i/5} : r \in \left\{ \frac{1}{2}, \frac{1+\sqrt{5}}{4} \right\}, \ 0 \le j < 5 \right\}. \quad (11.4)$$

At a nonzero zero put $\epsilon = e^{5i\theta} \in \{1, -1\}$. Direct differentiation and the relation $t = r^2 + \epsilon r^3$ give

$$J_{F_t}(z) = 5r^5(3r + 2\epsilon).$$

The positive-branch root has positive Jacobian. The two negative-branch radii lie strictly on opposite sides of $2/3$, so their Jacobians are nonzero as well. The origin is nondegenerate because $t \neq 0$. The sixteen points in (11.4) are therefore all nondegenerate. Formula (11.2) has three nonzero coefficients; Proposition 11.1 proves minimality. □

The same example admits an exact description for a complex parameter. For $\tau = a + ib$ define

$$F_\tau(z) = z^4 + z\bar{z}^2 - \tau\bar{z}, \qquad \mathcal{D}(a,b) = 4a^3 + (36a - 4)b^2 - 27(a^2 + b^2)^2. \quad (11.5)$$

The two exceptional nonzero parameters below correspond to a triple radial root.

Proposition 11.5. *For the family (11.5),*

$$N(F_\tau) = \begin{cases} 16, & \mathcal{D}(a,b) > 0, \\ 11, & \mathcal{D}(a,b) = 0, \quad \tau \notin \{0,\ 1/6 + i/(6\sqrt{3}),\ 1/6 - i/(6\sqrt{3})\}, \\ 6, & \text{otherwise.} \end{cases}$$

If $\mathcal{D}(a,b) \neq 0$, every zero is nondegenerate.

Proof. For $z = re^{i\theta} \neq 0$ the equation is $r^3 e^{5i\theta} = \tau - r^2$. With $s = r^2$, its positive radial solutions are precisely the positive roots of

$$\psi_\tau(s) = s^3 - s^2 + 2as - (a^2 + b^2). \quad (11.6)$$

Each positive root gives exactly five different angles, since $\tau - s \neq 0$ there. The discriminant of (11.6) is $\mathcal{D}(a,b)$.

If $\mathcal{D} > 0$, then $a > 0$: for $a \le 0$ all terms in the formula for $\mathcal{D}$ are nonpositive, with equality only at $a = b = 0$. The cubic has three different real roots, with sum 1, pairwise-product sum $2a > 0$, and product $a^2 + b^2 > 0$. They must all be positive. Indeed, if two were $-u, -v$ with $u, v > 0$, the third would be $1 + u + v$ and the pairwise-product sum would be $uv - (u + v)(1 + u + v) < 0$. Thus $N(F_\tau) = 1 + 5 \cdot 3$. If $\mathcal{D} < 0$, there is one real root. It is positive because $\psi_\tau(0) < 0$ and the leading coefficient is positive. This gives $1 + 5$ zeros.

If $\mathcal{D} = 0$ and $\tau \neq 0$, the same sign argument gives only positive real roots. There are two distinct ones unless the cubic is $(s - 1/3)^3$, which occurs exactly at the two listed nonzero exceptional parameters. These cases give eleven and six zeros, respectively. For $\tau = 0$, the sole positive root is $s = 1$, so again there are six zeros including zero.

At a nonzero zero a direct polar-coordinate computation gives

$$J_{F_\tau}(z) = 5s\psi_\tau'(s). \tag{11.7}$$

Thus every nonzero zero is nondegenerate when $\mathcal{D} \neq 0$. The origin has Jacobian $-|\tau|^2 \neq 0$ in these cases. This proves the last assertion using (11.7). □

11.3 A sharp count under cyclic symmetry

Fix $n \geq 2$, put $k = \lfloor n/2 \rfloor$, and let $\omega = e^{2\pi i/(n+1)}$. Consider all total-degree-n polynomials whose leading homogeneous part is cz^n, $c \neq 0$, and satisfy $F(\omega z) = \omega^{-1}F(z)$. Since a monomial $z^i\bar{z}^j$ transforms by ω^{i-j}, this full coefficient space is exactly

$$\mathcal{C}_n = \{F(z) = cz^n + \bar{z}R(|z|^2) : \ c \in \mathbb{C} \setminus \{0\}, \ R \in \mathbb{C}[s], \ \deg R \leq k-1\}.$$

The zero polynomial R is allowed. The parity of n changes the exact maximum in this space.

Theorem 11.6. *For every $n \geq 2$,*

$$\max_{F \in \mathcal{C}_n} N(F) = \begin{cases} n^2, & n \text{ even}, \\ n^2 - n - 1, & n \text{ odd}. \end{cases} \tag{11.8}$$

The maximum is attained with rational coefficients, exactly $k+1$ nonzero monomials, and all zeros nondegenerate.

We separate the amplitude count from the elementary construction that attains it. This also handles missing coefficients and multiple radial roots in the upper bound.

Lemma 11.7. *For $F = cz^n + \bar{z}R(|z|^2) \in \mathcal{C}_n$, let $Z_+(\Phi)$ be the number of distinct positive roots of*

$$\Phi(s) = |c|^2 s^{n-1} - R(s)\overline{R(\bar{s})} \in \mathbb{R}[s]. \tag{11.9}$$

Then $N(F) = 1 + (n+1)Z_+(\Phi)$. At a nonzero zero with $s = |z|^2$,

$$J_F(z) = (n+1)s\Phi'(s). \tag{11.10}$$

Proof. The origin is always a zero. Dividing the equation at a nonzero point $z = re^{i\theta}$ by $\bar{z}$ gives

$$cr^{n-1}e^{i(n+1)\theta} + R(r^2) = 0.$$

Its modulus equation is exactly (11.9). At a positive root, $R(s) \neq 0$, and the phase equation has exactly $n+1$ solutions. The polynomial Φ has degree $n-1$ and positive leading coefficient, since $2\deg R \leq 2k-2 < n-1$. Thus the zero count is finite.

For the Jacobian write $F = \bar{z}G$ off the origin, where $G = cr^{n-1}e^{i(n+1)\theta} + R(r^2)$. At a zero, multiplication by $\bar{z}$ contributes determinant s, while conversion from polar to Cartesian coordinates contributes $1/r$. Substituting $cr^{n-1}e^{i(n+1)\theta} = -R(s)$ into the two derivatives of G gives

$$J_F = (n+1)\big((n-1)|R(s)|^2 - 2s\Re(R'(s)\overline{R(s)})\big).$$

Using $|R(s)|^2 = |c|^2 s^{n-1}$ yields (11.10). □

For the lower bound we use a coefficient-dominance observation. It is just a quantitative way of making Descartes' rule of signs exact.

Lemma 11.8. *Let $0 \le e_0 < \cdots < e_q$ be integers and $\sigma_j \in \{-1, 1\}$. Let $t > 0$ satisfy $qt < 1$. Then*

$$p_t(r) = \sum_{j=0}^{q} \sigma_j t^{e_j^2} r^{e_j}$$

has exactly as many positive roots as there are sign changes in $(\sigma_0, \dots, \sigma_q)$. Every positive root is simple.

Proof. At $\rho_j = t^{-2e_j}$,

$$t^{e_j^2} p_t(\rho_j) = \sigma_j + \sum_{\ell \ne j} \sigma_\ell t^{(e_\ell - e_j)^2}.$$

The last sum has modulus at most $qt < 1$. Thus the values at the increasing radii ρ_j have the prescribed signs. Each sign change produces a different positive root. Descartes' rule, counting roots with multiplicity, gives the matching upper bound and forces all of these roots to be simple. The case $q = 0$ is immediate. □

We can now prove the sharp count and the even-degree sparsity claim.

Proof. If $R = 0$, then $N(F) = 1$. Otherwise let h be the order of vanishing of R at zero, so $0 \le h \le k - 1$. Factoring s^{2h} from Φ leaves a real polynomial with positive leading coefficient, negative constant term, and degree $D = n - 1 - 2h$. If n is even, D is odd, and there are at most $D \le n - 1$ different positive roots. If n is odd, D is even. There cannot be D different positive roots, since their product would make the constant term positive. Thus there are at most $D - 1 \le n - 2$. Lemma 11.7 proves the two upper bounds in (11.8), including all degeneracies and missing terms.

For attainment, choose the explicit rational parameter

$$t = \frac{1}{2(k+1)}, \qquad c = t^{(n-1)^2}, \qquad R_t(s) = \sum_{j=0}^{k-1} (-1)^j t^{4j^2} s^j,$$

and put

$$G_{n,t}(z) = t^{(n-1)^2} z^n + \bar{z} \sum_{j=0}^{k-1} (-1)^j t^{4j^2} |z|^{2j}. \tag{11.11}$$

The positive radial solutions are the positive roots of the two real polynomials

$$R_t(r^2) + c r^{n-1}, \qquad R_t(r^2) - c r^{n-1}.$$

Their common support is $0, 2, \dots, 2k - 2, n - 1$. In both coefficient lists the first k signs alternate, contributing $k - 1$ changes. Their last signs are opposite, so their last intervals contribute one change in total. Lemma 11.8, with $q = k$, shows that they have exactly k and $k - 1$ positive roots, in some order, all simple. They have no common positive root because their difference is $2cr^{n-1}$. Consequently, Φ has exactly $2k - 1$ positive simple roots. Lemma 11.7 now gives

$$N(G_{n,t}) = 1 + (n + 1)(2k - 1),$$

which is (11.8). Formula (11.10) proves nondegeneracy at every nonzero zero, and the origin has Jacobian -1. Finally, (11.11) has exactly $k + 1$ rational nonzero coefficients and conjugate degree k. □

For even n the construction (11.11) proves Theorem 11.4. It is useful to compare the parity classes:

n	2	4	6	8	10
$N(G_{n,t})$	4	16	36	64	100
$\nu(G_{n,t})$	2	3	4	5	6

The general zero bound shows that each listed even-degree count is extremal, not merely maximal among the displayed examples.

For completeness, the analogous odd-degree radial construction gives a uniform support estimate across both parities.

Corollary 11.9. *For every $n \geq 2$, there is a rational-coefficient polynomial of total degree n, with holomorphic leading term, $\lceil n/2 \rceil + 1$ nonzero coefficients, and n^2 nondegenerate zeros.*

Proof. For even n use Theorem 11.4. For $n = 2k+1$ choose a rational $0 < t < 1/(k+1)$ and set

$$H_{n,t}(z) = t^{n^2} z^n + \sum_{j=0}^{k} (-1)^j t^{4j^2} |z|^{2j}.$$

Its imaginary part confines the zeros to n lines. On one such line, with real coordinate r, its restriction is $\pm t^{n^2} r^n + \sum_{j=0}^{k} (-1)^j t^{4j^2} r^{2j}$. Applying Lemma 11.8 to its positive and negative half-lines gives $2k+1 = n$ simple nonzero real roots: the k even exponent gaps contribute $2k$ sign changes in total, and the final odd gap contributes one. The lines meet only at zero, which is not a root. Thus there are n^2 distinct zeros. The real part has nonzero tangential derivative and the imaginary part has nonzero normal derivative at every zero, proving nondegeneracy. There are $k+2$ nonzero coefficients. □

11.4 The earlier four-term quintic

For comparison with the first sparsity counterexample in the research record, we retain its explicit factorization. The even-degree construction above has superseded the earlier even-degree induction by saving one term. Set

$$F_5(z) = 99z^5 + 1031 z^2 \bar{z}^2 - 4132 z\bar{z} + 3200. \tag{11.12}$$

The pure-head intersection bound is Lemma 2.1; no new upper-bound argument is needed.

For clarity, the next proposition locates every zero, rather than merely counting them.

Proposition 11.10. *Let F_5 be given by (11.12), and set*

$$\rho_{\pm} = \frac{128 \pm 4\sqrt{34}}{99}.$$

Its zero set consists of the 25 distinct points

$$\left\{ \rho_{\pm} e^{2\pi i j/5} : 0 \leq j < 5 \right\} \cup \left\{ r e^{(2j+1)\pi i/5} : r \in \{1, 2, 10\},\ 0 \leq j < 5 \right\}. \tag{11.13}$$

Every zero is nondegenerate.

Proof. The real polynomial

$$\begin{aligned} f(s) &= 99s^5 + 1031s^4 - 4132s^2 + 3200 \\ &= (s+1)(s+2)(s+10)(99s^2 - 256s + 160) \end{aligned} \tag{11.14}$$

has five distinct nonzero real roots: $-1, -2, -10, \rho_-, \rho_+$. The quadratic factor has discriminant $64 \cdot 34 > 0$, and both its roots are positive.

Since $\operatorname{Im} F_5(z) = 99 \operatorname{Im} z^5$, every zero lies on one of the five lines $L_j = e^{ij\pi/5}\mathbb{R}$, $0 \le j < 5$. On L_j, with $z = se^{ij\pi/5}$ and $s \in \mathbb{R}$, the restriction of F_5 is $f(s)$ when j is even and $f(-s)$ when j is odd. Thus (11.14) supplies exactly five simple roots on every line. The lines meet only at 0, where $F_5(0) \neq 0$; hence there are exactly 25 zeros. Rewriting the signed real roots in polar coordinates gives (11.13).

At any of these zeros, the derivative of $\operatorname{Re} F_5$ along its line is nonzero. The derivative of $\operatorname{Im} F_5$ along that line is zero, whereas its derivative in the normal direction is nonzero, since $z \neq 0$. The two real gradients are therefore independent. □

The four monomials in (11.12) are distinct, and all four coefficients are nonzero. Proposition 11.10 consequently contradicts the coefficient-count assertion in [SZ24, Remark 4.4, p. 16] in degree five, with no additional hypothesis or limiting argument. The polynomial is not harmonic: its terms $z\bar{z}$ and $z^2\bar{z}^2$ are genuinely mixed.

The odd-degree radial family and the smaller even-degree support are already covered by Corollary 11.9. Thus no second odd-degree proof is needed here. For every $n \ge 2$,

$$s_n \le \lceil n/2 \rceil + 1.$$

Minimality in all higher degrees remains unproved; the exact low-degree values are those of Corollary 11.2.

12 A local four-zero splitting

We turn to the finite-degree equalities (1.2). The total-degree saturation lemma and the common-factor criterion proved earlier will exhaust each construction, rather than leaving an unproved global finiteness assumption. The following quadratic model is elementary.

Lemma 12.1. *If $\alpha, \beta \in \mathbb{C}$ and $|\alpha| < |\beta|$, then $u^2 + \alpha u - \beta\bar{u}$ has exactly four zeros, all nondegenerate.*

Proof. The origin has negative Jacobian $|\alpha|^2 - |\beta|^2$. A nonzero zero satisfies $|u + \alpha| = |\beta|$. If $|2u + \alpha| \le |\beta|$, the triangle inequality gives

$$|u + \alpha| \le \tfrac{1}{2}\big(|2u + \alpha| + |\alpha|\big) < |\beta|,$$

a contradiction. Every nonzero zero has positive Jacobian. The map is proper, all its zeros are isolated, and its degree is two because its highest homogeneous part is u^2. The signed zero count is two; hence there are three positive zeros and the one negative zero at the origin. □

A common perturbation can realize this model at every zero simultaneously.

Lemma 12.2. *Suppose $f = z^N +$ lower-total-degree terms has N^2 distinct zeros. Let $A, B \in \mathbb{C}[z, w]$ satisfy $\deg(Af - Bf^*) < 2N$ and*

$$|A(z, \bar{z})| < |B(z, \bar{z})| \quad \text{whenever } F_f(z) = 0.$$

For all sufficiently small positive ε, the polynomial

$$g_\varepsilon = f^2 + \varepsilon(Af - Bf^*) \tag{12.1}$$

has exactly $4N^2$ real-slice zeros, all nondegenerate and lying in arbitrarily small neighborhoods of the zeros of F_f.

Proof. At a zero z_0, Lemma 2.1 makes $\xi = F_f(z)$ a local real-analytic coordinate. On the real slice, $f^* = \overline{f}$. Substituting $\xi = \varepsilon u$ in (12.1) and dividing by ε^2 gives a C^1 limit on bounded u-sets equal to

$$u^2 + A(z_0, \bar{z}_0)u - B(z_0, \bar{z}_0)\bar{u}.$$

Lemma 12.1 and the implicit function theorem give four nondegenerate zeros near each of the N^2 seed zeros. The highest homogeneous part of g_ε is z^{2N}, so Lemma 2.1 excludes every additional zero. This also proves the assertion about neighborhoods, after taking ε sufficiently small. □

13 Explicit seeds and two successive squarings

Fix $a = 1 + 2i$, $b = 1 - 2i$, and $c = \sqrt{5}$. Define

$$\begin{aligned} &S_0 = 3, \qquad S_1 = z, \qquad S_2 = z^2 + 2bw, \\ &S_k = zS_{k-1} + bwS_{k-2} - ab^2S_{k-3} \quad (k \geq 3), \\ &H_k = S_k + (-a)^k. \end{aligned} \tag{13.1}$$

The S_k are the power sums of the roots of

$$\lambda^3 - z\lambda^2 - bw\lambda + ab^2 = 0. \tag{13.2}$$

We record both the roots and the support of these polynomials.

Proposition 13.1. *For every $k \geq 2$, the polynomial H_k has highest homogeneous part z^k, support contained in $i + 2j \leq k$, and exactly k^2 nonzero nondegenerate real-slice zeros. They are*

$$z = b(u^{-1} + v^{-1}) - auv, \qquad u^k = 1, \quad v^k = -1. \tag{13.3}$$

Its conjugate degree is $\lfloor k/2 \rfloor$. For $k = 2r + 1 \geq 3$,

$$[w^r]H_{2r+1} = (2r + 1)b^r z. \tag{13.4}$$

Proof. Assign weight one to z and weight two to w in (13.1). Induction gives the support condition and shows that the only term of total degree k is z^k. The highest-weight coefficients are $[w^r]S_{2r} = 2b^r$ and $[zw^r]S_{2r+1} = (2r + 1)b^r$. All weights occurring in S_k are congruent to k modulo three. Thus no constant multiple of w^r occurs in S_{2r+1}, which proves (13.4).

At a point (13.3), set $w = \bar{z} = a(u + v) - b/(uv)$. The three roots of (13.2) are

$$b/u, \qquad b/v, \qquad -auv.$$

Their kth power sum is $-(-a)^k$, so $H_k = 0$. To check distinctness, the numbers u, v are roots of $aT^3 - wT^2 - zT + b = 0$, whose third root is $-b/(auv)$. The kth power of the third root is neither 1 nor -1: otherwise b/a would be a root of unity. But $b/a = (-3-4i)/5$ and its sum with its inverse is $-6/5$, not a rational algebraic integer. Thus u and v can be uniquely recovered, proving that all k^2 points are distinct. Lemma 2.1 gives completeness and nondegeneracy. If their sum z were zero, the three equal-modulus roots of (13.2) would form an equilateral triple. Then v/u would be a cube root of unity, contrary to $(v/u)^k = -1$. Hence no zero is the origin. □

The following strict inequality provides the auxiliary polynomial for squaring an odd-degree seed. No optimality of its constant is needed.

Lemma 13.2. *If $|\eta_1| = |\eta_2| = |\eta_3| = 1$, $\eta_1\eta_2\eta_3 = 1$, and $s = \eta_1 + \eta_2 + \eta_3 \neq 0$, then, for every $k \geq 3$,*

$$\left|\eta_1^k + \eta_2^k + \eta_3^k - 3\mathbf{1}_{3|k}\right| < (2k-1)|s|. \tag{13.5}$$

Proof. For $k = 3$, Newton's identities give $\sum \eta_j^3 - 3 = s(s^2 - 3\bar{s})$, while

$$s^2 - 3\bar{s} = \tfrac{1}{2}\sum_{j<\ell}(\eta_j - \eta_\ell)^2, \qquad \sum_{j<\ell}|\eta_j - \eta_\ell|^2 = 9 - |s|^2.$$

The desired constant is therefore bounded by $9/2 < 5$.

Put $R = |s| < 1$ temporarily. The cubic $T^3 - sT^2 + \bar{s}T - 1$ has no repeated root in this disk. A repeated nonunit root would require its reciprocal conjugate with the same multiplicity, hence degree at least four. A repeated unit root would give $s = 2\eta + \eta^{-2}$ and $|s| \geq 1$. Starting at $s = 0$ and continuing along the segment to s, the roots remain on the unit circle: the reciprocal conjugate of a continued simple root is a root with the same initial value, hence agrees with it. With ω_j the three cube roots of unity, we may consequently write

$$\eta_j = \omega_j e^{i\delta_j}, \qquad |\delta_j| \leq \tfrac{2}{3}\arcsin R < \tfrac{\pi}{3}, \qquad \sum \delta_j = 0.$$

The bound follows from $|\eta_j^3 - 1| \leq 2R$; the three possible arcs are disjoint and continuation assigns one root to each arc.

If $h = \max|\delta_j|$, then

$$\sum|\delta_j| = 2h, \quad \tfrac{3}{2}h^2 \leq \sum\delta_j^2 \leq 2h^2, \quad \left|\sum\omega_j\delta_j\right|^2 = \tfrac{3}{2}\sum\delta_j^2.$$

Taylor's estimate $|e^{ix} - 1 - ix| \leq x^2/2$ gives $R \geq 3h/2 - h^2$. Since $\sum\omega_j^k = 3\mathbf{1}_{3|k}$, it follows that

$$\left|\sum\eta_j^k - 3\mathbf{1}_{3|k}\right| \leq 2kh \leq \frac{2k}{3/2 - h}R. \tag{13.6}$$

For $k \geq 4$ not divisible by three, the trivial bound is three. It is enough to consider $R \leq 3/(2k-1) \leq 3/7$. The inequality $\sin x \geq x - x^3/6$ at $x = 9/20$ implies $\arcsin(3/7) < 9/20$, so $h < 3/10$. Expression (13.6) is strictly below $(5k/3)R < (2k-1)R$. For $3 \mid k$, $k \geq 6$, the trivial bound is six. On $R \leq 6/(2k-1) \leq 6/11$, the same sine estimate at $x = 3/5$ gives $h < 2/5$. Now (13.6) is strictly below $(20k/11)R \leq (2k-1)R$. Together with the already treated case $k = 3$, these estimates prove (13.5). □

To apply this estimate at a seed zero, choose $\rho^3 = -ab^2$. The three roots of (13.2), divided by ρ, have modulus one, product one, and nonzero sum z/ρ. At the origin their unnormalized kth power sum is $3\rho^k \mathbf{1}_{3|k}$. Thus Lemma 13.2 gives

$$|S_k(z,\bar{z}) - S_k(0,0)| < (2k-1)c^{k-1}|z| \quad \text{at every zero of } H_N, \quad k \geq 3. \tag{13.7}$$

The first square covers every even conjugate degree on the new boundary. A second square supplies another complete residue class.

Proposition 13.3. *For every even $m \geq 2$, and for every $m \geq 5$ with $m \equiv 1 \pmod 4$, there is a polynomial in $\mathcal{E}_{m,2m+2}$ with $(2m+2)^2$ nondegenerate zeros and coefficients in $\mathbb{Q}(i)$.*

Proof. Let $L = 2r+1$, $r \geq 1$, and set $H = H_L$,

$$B(z) = Lb^r z, \qquad A(z,w) = \begin{cases} w^2, & r = 1, \\ S^*_{r+1}(z,w) - S^*_{r+1}(0,0), & r \geq 2. \end{cases}$$

The polynomial A is monic of degree $r+1$ in w, with total degree $r+1$. At every zero of H, inequality $|A| < |B|$ follows from (13.7) if $r \geq 2$. If $r = 1$, the three equal-modulus roots of (13.2) are not all equal, so $0 < |z| < 3c$ and $|w|^2 < 3c|z| = |B(z)|$.

By (13.4), the coefficients of w^{2r+1} in AH and BH^* are both B. Hence $D = AH - BH^*$ has conjugate degree at most $2r$ and total degree at most $3r+2 < 2L$. Lemma 12.2 shows that

$$f_\delta = H^2 + \delta D$$

has $(2L)^2$ nondegenerate zeros for every sufficiently small positive δ. Its leading w^{2r} coefficient tends to $B^2 \neq 0$, so its conjugate degree is exactly $2r$. This proves the assertion for even m.

For the second square, put $B_\delta = [w^{L-1}]f_\delta$ and $U = A^2$. The polynomial U is monic in w of degree $L+1$ and has total degree $L+1 < 2L$. All zeros of f_δ tend to zeros of H, and $B_\delta \to B^2$ coefficientwise. The strict inequality $|A| < |B|$ on the finite seed set consequently gives

$$|U| < |B_\delta| \quad \text{at every zero of } f_\delta$$

when δ is sufficiently small. In

$$f_\delta^2 + \eta(U f_\delta - B_\delta f_\delta^*)$$

the terms of w-degree $2L$ cancel. All remaining terms have conjugate degree at most $2L-1$, and the correction has total degree below $4L$. For fixed sufficiently small δ, Lemma 12.2 gives $(4L)^2$ nondegenerate zeros for all sufficiently small positive η. This is the pair $m = 2L-1 = 4r+1$, $n = 4L = 2m+2$. All parameters may be chosen rational. If the conjugate degree in this second construction is smaller than m, adding an arbitrarily small nonzero rational multiple of w^m makes it exact and preserves all $(2m+2)^2$ zeros; Lemma 2.1 again excludes additional zeros. □

14 A squaring criterion that persists under iteration

The remaining odd conjugate degrees require more than a second isolated application of Lemma 12.2. We now keep track of the next coefficient after the leading cancellation. This coefficient allows an auxiliary inequality to pass to every subsequent square.

Suppose $N = 2m + 2$ and

$$f = z^N + \text{terms of total degree below } N, \qquad \deg_w f = m.$$

Let A have total degree below N and be monic in w of degree $N - m = m + 2$. Define

$$\begin{aligned} B &= [w^m]f, \qquad C = [w^{m-1}]f, \qquad D = [w^{N-m-1}]A, \\ e &= [w^{N-1}]f^*, \qquad d = C + B(D - e). \end{aligned} \tag{14.1}$$

The quantities B, C, D, d are polynomials in z. The coefficient e is a constant because the only term of total degree N in f^* is w^N. The role of d is the exact identity

$$[w^N](Af - Bf^*) = 0, \qquad [w^{N-1}](Af - Bf^*) = d. \tag{14.2}$$

The following proposition includes the condition needed for a second application, not just the first one.

Proposition 14.1. *Suppose f has N^2 nondegenerate zeros. Let $S = w + s(z)$, with $\deg S < N$, and suppose that at every zero of F_f,*

$$|A| < |B|, \qquad |S|(|A| + |B|) < |d|. \tag{14.3}$$

For all sufficiently small positive ε, put

$$g = f^2 + \varepsilon(Af - Bf^*), \qquad A_g = Sf^*.$$

Then g has highest homogeneous part z^{2N}, conjugate degree exactly $N-1$, and $(2N)^2$ nondegenerate zeros. With B_g, C_g, D_g, e_g, d_g defined from g, A_g as in (14.1), every zero of F_g satisfies

$$|A_g| < |B_g|, \qquad |w|(|A_g| + |B_g|) < |d_g|. \tag{14.4}$$

Proof. Since $2m = N - 2$, identities (14.2) give

$$B_g = [w^{N-1}]g = \varepsilon d, \qquad C_g = [w^{N-2}]g = B^2 + \varepsilon[w^{N-2}](Af - Bf^*). \tag{14.5}$$

The second inequality of (14.3) forces d to be nonzero at every seed zero, hence not identically zero. Thus g has conjugate degree $N - 1$. Both correction terms have total degree below $2N$, so Lemma 12.2 gives exactly $(2N)^2$ nondegenerate zeros. All of them lie in arbitrarily small fixed neighborhoods of the seed zeros.

The polynomial $A_g = Sf^*$ is monic in w of degree $N+1$ and has total degree below $2N$. These are precisely the required degree conditions for the new pair $(m', N') = (N - 1, 2N)$. At a zero of g, put $x = F_f(z)$. The equation $x^2 + \varepsilon(Ax - B\bar{x}) = 0$ implies

$$|x| \le \varepsilon(|A| + |B|),$$

also when $x = 0$. On a sufficiently small neighborhood of the seed set, the strict second inequality of (14.3) still holds. All zeros of g lie there for small ε. Consequently,

$$|A_g| = |S||x| \le \varepsilon|S|(|A| + |B|) < \varepsilon|d| = |B_g|.$$

This proves the first inequality in (14.4).

For the second one, the next coefficient of A_g is $D_g = e + s(z)$, whereas $e_g = [w^{2N-1}]g^*$ stays bounded as $\varepsilon \to 0$. Equations (14.5) therefore yield the coefficientwise limit

$$d_g = C_g + B_g(D_g - e_g) = B^2 + O(\varepsilon). \tag{14.6}$$

Choose small compact neighborhoods of the finite seed set on which $|B|$ is bounded below by a positive constant. On these neighborhoods all coefficient polynomials occurring above are uniformly bounded. At zeros of g, the bound on x gives $|A_g| + |B_g| = O(\varepsilon)$, while (14.6) bounds $|d_g|$ below by a positive constant. The coordinate $w = \bar{z}$ is bounded there. Taking ε smaller proves the second inequality of (14.4). □

Thus after one successful step the next choice of S is simply w. The construction does not require a parameter that works for infinitely many degrees at once. For each finite number of steps, the parameters are chosen successively, each sufficiently small after the preceding ones have been fixed. If the initial data are rational, all these parameters may be chosen positive and rational.

We next give the finite starting data for the residue class not covered by Proposition 13.3. The degree-eight polynomial f_0 is specified exactly in Table 5. It was used as an isolated extremizer in the preceding degree-$(3, 8)$ construction. The additional polynomials A_0, S_0 in Tables 6 and 7 are what allow that isolated example to propagate.

All three tables, together with the rational boxes in the supplementary files, establish the following finite assertion.

Proposition 14.2. *The polynomial f_0 has highest homogeneous part z^8, conjugate degree three, and 64 nondegenerate zeros. The polynomial A_0 has total degree five and is monic of degree five in w; $S_0 = w + s_0(z)$ with $\deg s_0 = 3$. With*

$$B_0 = [w^3]f_0, \quad C_0 = [w^2]f_0, \quad D_0 = [w^4]A_0, \quad e_0 = [w^7]f_0^*, \quad d_0 = C_0 + B_0(D_0 - e_0),$$

every zero of F_{f_0} satisfies

$$|A_0| < \frac{3}{50}|B_0|, \qquad |S_0|(|A_0| + |B_0|) < \frac{7}{10}|d_0|. \tag{14.7}$$

Proof. The degree assertions are read from the coefficient tables. There are 64 disjoint rational boxes of radius 10^{-30} in each real coordinate. The program `verify_seed.py` verifies in each box a contraction and inclusion bound strictly below 10^{-12}, using only rational arithmetic. It proves the existence of a nondegenerate zero in each box. Lemma 2.1 shows that these are all the zeros.

The program `verify_auxiliary.py` proves the two inequalities in (14.7) on the whole of every box, not just at its center. It also checks that B_0 and d_0 do not vanish in any box. Appendix B gives the rational bounds and every comparison used by the two programs, as well as the exact input tables. The supplied output records that all comparisons pass. □

Because the margins in (14.7) are strictly smaller than one, they imply (14.3) with room to spare. The finite certificate can now be used in every degree generated by the iteration, without any further numerical root finding.

Corollary 14.3. *For every integer $j \geq 0$, there is a polynomial with highest homogeneous part $z^{2^{j+3}}$, exact conjugate degree $2^{j+2} - 1$, and 2^{2j+6} nondegenerate zeros. Its coefficients may be taken in $\mathbb{Q}(i)$.*

Proof. Start with f_0, A_0, S_0 from Proposition 14.2. Proposition 14.1 changes (m, N) to $(N-1, 2N)$ and supplies the next auxiliary polynomial. At all later steps take $S = w$. The initial values are $(m, N) = (3, 8)$, and induction gives the asserted degrees. The number of zeros is the square of the total degree at every step. Rationality follows from successive rational choices of the small positive parameters. □

15 Odd-degree composition and the complete boundary

Repeated squaring changes $m+1$ by a factor of two. The next construction changes it by any odd factor, while preserving the relation $N = 2m+2$. It is essential here to use the support of the outer polynomial, rather than only its total degree.

Proposition 15.1. *Suppose $f = z^N +$ lower-total-degree terms has conjugate degree exactly m, where $N > 2m$, and has N^2 nondegenerate zeros. For every odd integer $q = 2s+1 \geq 3$, there is a polynomial with highest homogeneous part z^{qN}, conjugate degree exactly $m + sN$, and $(qN)^2$ nondegenerate zeros. Rational coefficients may be retained when $f \in \mathbb{Q}(i)[z, w]$.*

Proof. Use the seed H_q from Proposition 13.1 and put

$$F_t(z, w) = t^q H_q\big(f(z, w)/t, f^*(z, w)/t\big), \qquad t > 0. \tag{15.1}$$

This is a polynomial: for each monomial $U^i V^j$ of H_q, the factor t^{q-i-j} has a nonnegative exponent. The unique term of total degree qN is z^{qN}, since the unique term of total degree q in H_q is U^q.

The support condition $i + 2j \leq q$ bounds the conjugate degree of a composed term by

$$mi + Nj \leq mq + (N - 2m)j \leq m + sN.$$

Since $N > 2m$, equality can occur only at $(i, j) = (1, s)$. Its coefficient in H_q is $qb^s \neq 0$ by (13.4). The corresponding highest w-coefficient in (15.1) is a nonzero scalar times $[w^m]f$. Thus the conjugate degree is exactly $m + sN$; no other term can cancel it.

On the real slice, (15.1) vanishes precisely when $F_f(z)/t$ is one of the q^2 nondegenerate zeros of $H_q(U, \bar{U})$. Near each zero of F_f, the map F_f is a real-analytic local diffeomorphism. For small t, the q^2 distinct target values therefore have q^2 distinct preimages in that neighborhood. Each preimage is nondegenerate, by the chain rule. This gives q^2N^2 zeros, and Lemma 2.1 proves completeness. A sufficiently small positive rational t preserves rational coefficients. □

We can now remove every parity exception on the boundary.

Proposition 15.2. *For every integer $m \geq 2$, there is a polynomial in $\mathcal{E}_{m,2m+2}$ with conjugate degree exactly m and $(2m+2)^2$ nondegenerate zeros. All coefficients may be taken in $\mathbb{Q}(i)$.*

Proof. Even m and $m \equiv 1 \pmod 4$, $m \geq 5$, are covered by Proposition 13.3. If $m \equiv 3 \pmod 4$, write uniquely

$$m + 1 = 2^{j+2}q, \qquad j \geq 0, \quad q \geq 1 \text{ odd}.$$

Corollary 14.3 supplies $m_0 = 2^{j+2} - 1$ and $N_0 = 2^{j+3} = 2(m_0 + 1)$. For $q = 1$ it already gives the desired pair. For $q = 2s+1 \geq 3$, Proposition 15.1 gives

$$m_0 + sN_0 = q(m_0 + 1) - 1 = m, \qquad qN_0 = 2m + 2.$$

The three cases exhaust all integers $m \geq 2$. □

The congruence decomposition is only an organization of the construction. The conclusion of Proposition 15.2 is a single extremal statement for all $m \geq 2$. In particular, the construction now covers $(m, n) = (5, 12), (7, 16), (9, 20), (11, 24), (15, 32)$, without separate searches for polynomials in those degree pairs.

16 From total degree to rectangular bidegree

The preceding sections construct the n^2 bounded zeros. A local perturbation after inversion supplies the additional m^2 zeros required by the rectangular intersection number.

Proposition 16.1. *Suppose $n > m \geq 2$ and there is a polynomial in $\mathcal{E}_{m,n}$ with n^2 nondegenerate zeros. Then there is a polynomial of exact bidegree (m, n) with n^2+m^2 nondegenerate zeros. Rational coefficients may be retained.*

Proof. Normalize the highest homogeneous part of the given polynomial f to z^n. If necessary, first add a sufficiently small nonzero rational multiple of w^m to make its conjugate degree exactly m. This retains all n^2 zeros, and Lemma 2.1 shows that no other zero occurs. Write

$$H_m^*(u, v) = v^m + Q_m(u, v), \qquad Q_m(u, v) = \sum_{i+j \leq m-1} q_{ij} u^i v^j.$$

Proposition 13.1 gives m^2 nonzero nondegenerate zeros of this real-slice polynomial; its constant term q_{00} is nonzero. For $t > 0$ put

$$g_t(z, w) = f(z, w) + \sum_{i+j \leq m-1} q_{ij} t^{m-i-j} z^{n-i} w^{m-j}. \tag{16.1}$$

All exponents are nonnegative. The term $q_{00} t^m z^n w^m$ is nonzero, so the bidegree is exactly (m, n). As $t \to 0$, the coefficients tend to those of f, and its n^2 nondegenerate zeros persist.

Set $G_t(u, v) = u^n v^m g_t(1/u, 1/v)$. Every nonleading monomial of f has total degree at most $n - 1$, so its inverted contribution has total degree at least $m + 1$. Consequently,

$$t^{-m} G_t(t\zeta, t\eta) \longrightarrow \eta^m + Q_m(\zeta, \eta) = H_m^*(\zeta, \eta)$$

in C^1 on compact real-slice sets. Each of the m^2 nonzero nondegenerate zeros of this limit gives a simple zero of G_t under the scaling. Inverting back yields m^2 zeros of g_t escaping to infinity, distinct from the n^2 bounded zeros.

These $n^2 + m^2$ nondegenerate zeros exceed $(n - 1)^2 + (m - 1)^2$. Lemma 2.3 forces g_t, g_t^* to be coprime and bounds their entire real-slice zero set by $n^2 + m^2$. The count is therefore exact, and all zeros are nondegenerate. Formula (16.1) permits a sufficiently small rational t. □

The two maxima in the main theorem can now be identified together.

Proof. If $m < n \leq 2m + 1$, then $\lfloor n/2 \rfloor \leq m$. Proposition 13.1 puts H_n in $\mathcal{E}_{m,n}$ with n^2 nondegenerate zeros. A sufficiently small rational w^m perturbation makes the conjugate degree exact when necessary. If $n = 2m + 2$, use Proposition 15.2. In either case, Lemma 2.1 gives $E_{m,n} = n^2$. Proposition 16.1 then gives $M_{m,n} \geq n^2 + m^2$, and Lemma 2.3 supplies the reverse inequality. □

The next degree pair beyond the range for $m = 2$ is $(2, 7)$. No step above constructs a 53-zero polynomial in that pair, nor does it rule one out. The proof should therefore be used to enlarge the established exact range, not to assert a sharp formula in all bidegrees. Likewise, the constructed polynomials contain mixed monomials; they do not supply matching extremizers for the harmonic problem.

17 A cubic construction on the next diagonal

A cubic local model gives a new infinite family, beyond the range (1.2). For this section the maximum in the pure analytic-head space is again denoted by $E_{m,n}$.

Theorem 17.1. *For every integer $r \geq 1$,*

$$E_{3r,6r+3} = (6r+3)^2, \qquad M_{3r,6r+3} = (6r+3)^2 + 9r^2. \tag{17.1}$$

The extrema have exact conjugate degree $3r$, rational real coefficients, and only nondegenerate zeros.

The proof of the pure-head equality is given next. The rectangular equality then follows from Proposition 16.1; its auxiliary seed can be chosen over $\mathbb{Q}(i)$. To retain real coefficients, one may instead construct the auxiliary seed inductively from $z - 1$ using a small real term sz^j added to the conjugate of the preceding seed. The nonzero model $\zeta^j + \bar{\zeta}^{j-1} = 0$ adds $2j-1$ positive simple zeros at infinity, precisely as in [SZ24, proof of Theorem 4.1, pp. 12–14]. The old constant is nonzero, so zero is not introduced. This gives j^2 nonzero simple zeros at each stage and justifies the real-coefficient assertion.

17.1 A uniform nine-zero cubic

Set $\kappa = 9/100$. The size of the perturbation in the next lemma is independent of its argument.

Lemma 17.2. *For every $e \in \mathbb{C}$ with $|e| < \kappa$, the polynomial*

$$H_e(v) = v^3 + |v|^2 + e\bar{v} - \kappa$$

has exactly nine zeros, all nondegenerate.

Proof. We first exclude singular zeros. Such a zero has $v \neq 0$. Put $v = \rho e^{i\theta}$ and $x = \rho^2$. The zero equation yields

$$e = \frac{\kappa - v^3 - |v|^2}{\bar{v}}.$$

Substituting this into $J = |3v^2 + \bar{v}|^2 - |v + e|^2$ shows that $J = 0$ implies

$$\cos(3\theta) = \frac{\kappa^2 - x^2 - 8x^3}{2x^{3/2}(3x + \kappa)}. \tag{17.2}$$

The requirement $|\cos(3\theta)| \leq 1$ is equivalent to

$$(4x^3 - x^2 - 2\kappa x - \kappa^2)(16x^3 - x^2 + 2\kappa x - \kappa^2) \leq 0. \tag{17.3}$$

For $x \geq 191/500$, both factors are positive. The second is positive since $16x - 1 > 0$ and $2x - \kappa > 0$. The first is strictly increasing there: its derivative is $12x^2 - 2x - 2\kappa > 0$, and its value at $191/500$ is $5871/31250000$. Thus (17.3) forces $x < 191/500$.

A second substitution of (17.2), this time in $|e|^2$, gives the exact identity

$$|e|^2 = \kappa^2 + \frac{(x-\kappa)^2(2-\kappa-5x)}{\kappa+3x}. \tag{17.4}$$

Since $(2-\kappa)/5 = 191/500$, the last term is nonnegative. Equation (17.4) contradicts $|e| < \kappa$.

At $e = 0$, every zero lies on one of the three lines $e^{2\pi ij/3}\mathbb{R}$. On each line the equation is $s^3 + s^2 - \kappa = 0$. This real cubic has three distinct nonzero real roots because its critical values at $-2/3$ and 0 are $4/27 - \kappa > 0$ and $-\kappa < 0$. Thus H_0 has nine zeros. Along the segment from 0 to e all zeros are nondegenerate, as just proved. They remain in a fixed compact disk because the cubic term dominates the lower terms uniformly on that segment. The implicit function theorem, compactness, and Lemma 2.1 show that their number remains nine. □

The radius in Lemma 17.2 cannot be enlarged. Indeed, take $|v|^2 = \kappa$ and $\cos(3\arg v) = -\sqrt{\kappa}$, and set $e = -v^3/\bar{v}$. Then $|e| = \kappa$, $H_e(v) = 0$, and $J_{H_e}(v) = 0$. Thus any larger open parameter disk contains a polynomial with a singular zero.

The form needed in the construction is obtained by a translation and a complex dilation. We state the coefficients explicitly.

Corollary 17.3. *Let $a, b, \delta \in \mathbb{C}$, where $b \neq 0$ and $|\delta| < \kappa|b|^2$. Put*

$$d = \frac{a^2 + b\bar{a}}{3}, \qquad e = \frac{ba}{3} + \delta, \qquad g = \frac{a^3}{27} + \frac{b|a|^2}{9} + \frac{\delta\bar{a}}{3} - \kappa|b|^2 b. \tag{17.5}$$

Then $u^3 + au^2 + bu\bar{u} + du + e\bar{u} + g$ has exactly nine nondegenerate zeros.

Proof. Set $v = u + a/3$. The polynomial becomes $v^3 + b|v|^2 + \delta\bar{v} - \kappa|b|^2 b$. Choose λ with $\lambda^3 = |b|^2 b$, so $|\lambda| = |b|$, and put $v = \lambda y$. After division by $|b|^2 b$, the resulting polynomial is

$$y^3 + |y|^2 + \frac{\delta\bar{\lambda}}{|b|^2 b}\bar{y} - \kappa.$$

The conjugate-linear coefficient has modulus $|\delta|/|b|^2$. Lemma 17.2 applies. Both changes of variables and the nonzero target multiplication preserve nondegeneracy. □

17.2 The general construction

Fix $r \geq 2$, and write

$$N = 2r + 1, \qquad \epsilon = \frac{1}{100(r+1)^N}, \qquad Q(z, w) = \prod_{j=1}^{r}(zw - \epsilon^2 j^2), \qquad f = z^N + Q. \tag{17.6}$$

All coefficients are rational and real, and $Q^* = Q$. Let

$$a = [(zw)^{r-1}]Q = -\epsilon^2 \sum_{j=1}^{r} j^2. \tag{17.7}$$

We first record precise information about the seed.

Lemma 17.4. *The polynomial f in (17.6) has exactly N^2 nondegenerate zeros. Every zero satisfies $|z| \geq 3\epsilon/4$.*

Proof. Consider

$$\phi(s) = s^N + \prod_{j=1}^{r}(s^2 - \epsilon^2 j^2).$$

Under $s = \epsilon x$, its rescaled form is $\epsilon x^N + \prod_{j=1}^{r}(x^2 - j^2)$. At the endpoints of each of the $2r$ disjoint intervals centered at $\pm j$ with radius $1/4$, the product has magnitude at least $7/16$: one factor has magnitude at least $(1/4)(7/4)$, and every other factor has magnitude at least $(3/4)(7/4) > 1$. Meanwhile $|\epsilon x^N| \le 1/100$. The endpoint signs are therefore unchanged, and there is a sign-changing root in each interval.

These $2r$ roots are simple. A sign-changing root of multiplicity at least three would already consume at least $2r + 2$ degrees. The last root of the real polynomial of degree $2r + 1$ is real and distinct from them. More explicitly, the coefficient of s^{N-1} is one, so the sum of all roots is -1. The sum of the absolute values of the $2r$ small roots is at most

$$2r\epsilon(r + 1) \le \frac{1}{50}.$$

The remaining root belongs to $[-51/50, -49/50]$ and is simple. In particular all roots have magnitude at least $3\epsilon/4$.

Let $\omega = e^{2\pi i/N}$. Because N is odd, the N lines $\omega^j\mathbb{R}$ are distinct. On each of them $F_f(s\omega^j) = \phi(s)$. Moreover, $\operatorname{Im} F_f(z) = \operatorname{Im} z^N$, so every zero lies on one of these lines. Their only common point is zero, which is not a zero of f. Thus there are exactly N^2 zeros. At each one, the derivative of the real part along its line is nonzero, whereas the imaginary part has zero tangential derivative and nonzero normal derivative. This proves nondegeneracy. □

The next coefficients are chosen to remove, rather than merely bound, the forbidden conjugate powers. Define

$$\begin{aligned} A &= w^{r+1}, \qquad B = -z^r, \qquad \Delta = -\frac{a}{3} z^{r-1} w^r, \\ D &= \frac{w^{2r+2} - z^N}{3}, \qquad E = -\frac{z^r w^{r+1} + a z^{r-1} w^r}{3}, \\ L &= \frac{z^r w + a z^{r-1}}{27} - \frac{z^r}{9}. \end{aligned} \tag{17.8}$$

Put

$$\begin{aligned} T_1 &= f(Af + Bf^*), \\ T_2 &= Df + Ef^*, \\ T_3 &= Q\left(-\frac{w^{r+2}}{27} + \frac{w^{r+1}}{9}\right) + \left(\kappa - \frac{a}{9}\right) z^{2r} w^r + Lf^*. \end{aligned} \tag{17.9}$$

Finally, for $t > 0$, define

$$P_t = f^3 + tT_1 + t^2 T_2 + t^3 T_3. \tag{17.10}$$

These are explicit polynomial expressions; no coefficient interpolation is required.

We verify the degree conditions before counting zeros.

Lemma 17.5. *For $r \ge 2$, every T_j in (17.9) has total degree less than $3N$ and conjugate degree at most $3r$. In addition, $\deg_w T_3 \le 2r$.*

Proof. The two top w-coefficients of Q are z^r and az^{r-1}. In $Af + Bf^*$ the coefficient of w^{2r+1} cancels, so its w-degree is at most $2r$. Multiplication by f gives $\deg_w T_1 \le 3r$. The total degree of T_1 is at most $5r + 3 < 3N$.

For T_2, the coefficients of w^{3r+2} cancel between Df and Ef^*. The coefficients of w^{3r+1} are $az^{r-1}/3$ and $-az^{r-1}/3$, and also cancel. The term $z^N w^{2r+2}/3$ has conjugate degree at most $3r$ precisely because $r \geq 2$. All other terms have the same bound. The total degree is at most $4r+3 < 3N$.

In T_3, the coefficient of w^{2r+2} cancels between $-Qw^{r+2}/27$ and $z^r w f^*/27$. The coefficient of w^{2r+1} also cancels, using the remaining two terms of L. Thus $\deg_w T_3 \leq 2r$, and its total degree is at most $3r+2 < 3N$. □

To identify the local limit, set

$$G = \frac{A^3}{27} + \frac{BAA^*}{9} + \frac{\Delta A^*}{3} - \kappa B^2 B^* = \frac{w^{3r+3}}{27} - \frac{z^N w^{r+1}}{9} + \left(\kappa - \frac{a}{9}\right) z^{2r} w^r. \tag{17.11}$$

The following exact identity will also explain why the reduction used for the constant term has the required effect:

$$T_3 - G = \frac{w^{r+1}}{9} f + \left(L - \frac{w^{r+2}}{27}\right) f^*. \tag{17.12}$$

It follows by substituting $f = z^N + Q$ and $f^* = w^N + Q$.

We can now complete the local count without requiring an exact unperturbed nine-zero model at every seed zero.

Proposition 17.6. *For every fixed $r \geq 2$, the polynomial P_t belongs to $\mathcal{E}_{3r,3N}$, has exact conjugate degree $3r$, and has exactly $9N^2$ nondegenerate zeros for all sufficiently small $t > 0$.*

Proof. The highest homogeneous part of f^3 is z^{3N}. By Lemma 17.5, this is also the highest homogeneous part of P_t. The coefficient of $z^{3r}w^{3r}$ is one at $t = 0$ and hence remains nonzero for sufficiently small t.

Fix a seed zero z_0. By Lemma 17.4, $\xi = F_f(z)$ is a local real-analytic coordinate; write its inverse as $z = \psi(\xi)$. Set $\xi = tu$ and divide (17.10) by t^3. Using (17.8), (17.11), and (17.12), the resulting maps converge in C^1 on compact u-sets to

$$u^3 + a_0 u^2 + b_0 u\bar{u} + d_0 u + e_0 \bar{u} + g_0,$$

where $a_0 = A(z_0, \bar{z}_0)$, $b_0 = B(z_0, \bar{z}_0)$ and the last three coefficients are given by (17.5), with $\delta_0 = \Delta(z_0, \bar{z}_0)$. In particular $b_0 \neq 0$.

Let $S_r = \sum_{j=1}^{r} j^2$. Equations (17.7) and (17.8), together with the modulus bound in Lemma 17.4, give

$$\frac{|\delta_0|}{|b_0|^2} = \frac{\epsilon^2 S_r}{3|z_0|} \leq \frac{4\epsilon S_r}{9} \leq \frac{1}{225} < \kappa. \tag{17.13}$$

Here $S_r \leq (r+1)^3$ and $\epsilon S_r \leq (r+1)^{2-2r}/100 \leq 1/100$. The uniform estimate (17.13) permits the use of Corollary 17.3 at every seed zero.

Each limit has nine nondegenerate zeros, and the real implicit function theorem gives nine nearby zeros of P_t. The seed neighborhoods can be chosen disjoint, and there are finitely many of them, so one common positive threshold for t works for all N^2 seed zeros. We have constructed $9N^2 = (3N)^2$ distinct nondegenerate zeros. Lemma 2.1 excludes all further zeros and proves the count. □

The construction has real rational coefficients whenever t is rational. No explicit numerical lower bound for the allowed smallness threshold is needed to obtain rational extremizers.

17.3 The first parameter

The preceding proof uses $2r+2 \le 3r$ and cannot simply be specialized to $r=1$. The following separate choice handles that case while using the same uniform cubic lemma.

Set

$$f = z^3 + zw - \frac{1}{8}, \qquad A = w^2, \qquad B = -z, \qquad \Delta = \frac{w - z^2}{12} - \frac{zw^2}{3},$$

and put

$$D = \frac{w^4 - z^3}{3}, \qquad E = \frac{w - z^2}{12} - \frac{2zw^2}{3}.$$

Define $T_1 = f(Af + Bf^*)$ and

$$\begin{aligned} T_2 = & -\frac{2}{3}z^5w^3 - z^6w - \frac{1}{3}z^4w + \frac{1}{24}z^3w \\ & - \frac{1}{3}z^8 - \frac{1}{3}z^6 + \frac{1}{6}z^5 + \frac{1}{24}z^3 - \frac{1}{192}z^2. \end{aligned} \tag{17.14}$$

$$T_3 = \frac{z^2w^2}{27} - \frac{w^2}{36} - \frac{23z^2w}{300} - \frac{zw}{108} + \frac{7z}{288} + \frac{1}{1728}. \tag{17.15}$$

Let $P_t = f^3 + tT_1 + t^2T_2 + t^3T_3$.

The identities required for the local limit can be checked without solving for any roots. Equation (17.14) satisfies

$$T_2 - Df - Ef^* = \frac{w - z^2}{3}f^2 + \frac{w - 2z^2}{3}ff^*. \tag{17.16}$$

Also, the difference between (17.15) and

$$G = \frac{w^6}{27} - \frac{2z^3w^2}{9} - \frac{z^4}{36} + \frac{53z^2w}{450}$$

belongs to (f, f^*). Indeed, use $z^3 = w^3 = 1/8 - zw$ to reduce G; its remainder is exactly T_3. This G is $A^3/27 + BAA^*/9 + \Delta A^*/3 - \kappa B^2B^*$.

Here the seed zeros are $s\omega^j$, where $\omega = e^{2\pi i/3}$ and

$$s \in \left\{ -\frac{1}{2}, \frac{-1+\sqrt{5}}{4}, \frac{-1-\sqrt{5}}{4} \right\}.$$

They are nondegenerate by the same line-restriction argument as in Lemma 17.4. At such a zero,

$$\frac{|\Delta|}{|B|^2} = \left| \frac{2}{3}s^2 + \frac{1}{3}s - \frac{1}{12} \right| = \frac{1}{12} < \kappa.$$

For $s = -1/2$ the expression inside the absolute value is $-1/12$; for the other two values it is $1/12$.

In the coordinate $\xi = F_f(z) = tu$, identity (17.16) contributes only an $O(t)$ term after the rescaling used in Proposition 17.6. The same is true for the difference $T_3 - G$. The limit is therefore again the cubic in Corollary 17.3, with nine nondegenerate zeros. Each of the nine seed zeros supplies nine zeros of P_t.

Direct inspection of T_1, (17.14), and (17.15) shows conjugate degree at most three and total degree at most eight. Thus $P_t \in \mathcal{E}_{3,9}$ and Lemma 2.1 bounds its number of zeros by 81. The constructed 81 zeros are all nondegenerate, and the conjugate degree is exactly three for sufficiently small t, since $[z^3w^3]f^3 = 1$. This proves the first assertion of (17.1) also for $r = 1$.

Applying Proposition 16.1 with $m = 3r$ and $n = 6r + 3$ completes the proof of (17.1), and hence the second family in Theorem 1.3. All choices of sufficiently small parameters are made in a finite sequence; density of the rationals permits each chosen positive parameter to be rational. No uniform numerical value for all r is claimed.

18 Two perturbations that increase the bidegree

The preceding constructions prescribe many zeros at once. There is also value in modifying an existing example while retaining its zeros. We give two such modifications. The first increases both degrees and preserves the deficit from the quadratic bound. The second increases only the analytic degree, under a local contact condition.

18.1 Adding a circle

The basic observation is that a real circle becomes a rational curve in the complexified surface. Restrictions to this curve allow enough freedom to prescribe the tangential equation of a perturbation.

Proposition 18.1. *Suppose a polynomial of bidegree (m,n), with $n > m \geq 1$, has at least N nondegenerate zeros. There is a polynomial of bidegree $(m+1,n+1)$ with finitely many zeros, all nondegenerate, and with at least $N + 2(m+n+1)$ zeros.*

Proof. An arbitrarily small generic perturbation preserves the chosen N zeros and makes P and

$$P^*(z,w) = \overline{P(\bar{w},\bar{z})}$$

coprime. Choose a circle with complexified equation

$$L_0(z,w) = (z-a)(w-\bar{a}) - R^2 = 0, \qquad R > 0,$$

whose complexification avoids all intersections of $P=0$ and $P^*=0$. This is possible: each of their finitely many intersection points excludes a proper real algebraic subset of the parameters (a,R). Require also that the circle avoid the selected zeros. With

$$z = a + Rt, \qquad w = \bar{a} + R/t,$$

put $d = m+n$. Avoiding the zeros of the highest w-coefficient at $z=a$ and of the highest z-coefficient at $w=\bar{a}$ gives

$$P|_{L_0} = t^{-m}A(t), \qquad \deg A = d, \qquad A(0) \neq 0.$$

For a polynomial V of degree at most e define its reversal, with respect to the specified degree e, by $V^{\#_e}(t) = t^e\overline{V(1/\bar{t})}$. The avoidance condition gives $\gcd(A, A^{\#_d}) = 1$. In particular, P is nonzero on the real circle $|t|=1$.

The restrictions of polynomials Q of bidegree at most $(m+1,n+1)$ are precisely

$$Q|_{L_0} = t^{-(m+1)}B(t), \qquad \deg B \leq d+2.$$

Indeed, the monomials in $z-a$ and $w-\bar{a}$ span every Laurent power from $-(m+1)$ to $n+1$. Consider the real linear map

$$B \longmapsto \operatorname{Im}(Q\overline{P})\big|_{|t|=1}.$$

Its target is the real vector space of trigonometric polynomials of degree at most $d+1$, which has dimension $2d+3$. Its kernel is specified by

$$BA^{\#_d} = B^{\#_{d+2}}A.$$

Coprimality implies $B = AS$ with $\deg S \le 2$, and the remaining condition is $S = S^{\#_2}$. Thus

$$S(t) = \gamma + \lambda t + \bar{\gamma} t^2, \qquad \gamma \in \mathbb{C}, \quad \lambda \in \mathbb{R}.$$

The kernel has real dimension three, while the domain has dimension $2d + 6$. The map is therefore onto.

Choose Q so that the target function is $\sin((d+1)\theta)$, where $t = e^{i\theta}$. Then $\operatorname{Im}(Q/P)$ has exactly $2(d+1)$ simple zeros on the circle. For real $\varepsilon \neq 0$ put

$$P_\varepsilon = L_0 P + \varepsilon Q.$$

Every selected old zero persists. Near the circle, division by P reduces the equation to

$$u + \varepsilon \operatorname{Re}(Q/P) = 0, \qquad \operatorname{Im}(Q/P) = 0, \qquad u = L_0.$$

At $\varepsilon = 0$ the Jacobian in normal and tangential coordinates is nonzero at each of the $2(d+1)$ simple circle zeros. They therefore give that many new nondegenerate zeros for sufficiently small nonzero ε. Their neighborhoods can be chosen disjoint from those of the old zeros. A final generic perturbation gives exact bidegree $(m+1, n+1)$ and a finite, entirely nondegenerate zero set, without losing any of these zeros. □

The increment in Proposition 18.1 agrees exactly with the increment of the bidegree intersection number. This gives a useful monotonicity statement for the unknown exact maxima.

Corollary 18.2. *For $n > m \ge 1$ and every integer $r \ge 0$,*

$$M_{m+r,n+r} \ge M_{m,n} + 2r(m + n + r).$$

In particular, if $M_{m,n} = m^2 + n^2$, then $M_{m+r,n+r} = (m+r)^2 + (n+r)^2$ for every $r \ge 0$.

Proof. Lemma 3.2 gives a nondegenerate extremal polynomial. Iterate Proposition 18.1; the sum of its r increments is $2r(m + n + r)$. This is also the difference between the two quadratic bounds. Proposition 2.2 then proves the equality assertion. □

18.2 Increasing only the analytic degree

Perturbations by a pole can create several nearby zeros of a rational harmonic function. A detailed analysis of that phenomenon is given by Sète, Luce, and Liesen [SLL15, Theorem 3.1, author version, p. 7]. The following polynomial modification has the same local origin, but keeps the degree in w fixed and gives an exact global count.

Proposition 18.3. *Let $n > m \ge 1$, $q \ge 2$, and*

$$P(z, w) = wA(z, w) - z^q B(z), \qquad A(0,0)B(0) \neq 0.$$

Suppose that P has bidegree (m, n), its $w^m z^n$ coefficient is nonzero, and all zeros of F_P are nondegenerate. For every sufficiently small real $\varepsilon > 0$, the polynomial

$$P_\varepsilon(z, w) = (wz - \varepsilon)A(z, w) - z^{q+1} B(z) = zP(z, w) - \varepsilon A(z, w) \tag{18.1}$$

has bidegree $(m, n+1)$, all its zeros are nondegenerate, and

$$N(P_\varepsilon) = N(P) + 2q + 1.$$

Proof. The nonzero highest corner coefficient makes F_P proper, so its isolated zeros form a finite set. The origin is a negative nondegenerate zero of F_P, since its linear part is $A(0,0)\bar{z}$. Every nonzero old zero remains nondegenerate for $zF_P(z)$ and therefore persists under (18.1).

In a fixed small disk about zero, divide by $A(z,\bar{z})$ and set $b(z) = B(z)/A(z,\bar{z})$. The local equation becomes

$$|z|^2 - \varepsilon - z^{q+1}b(z) = 0, \qquad b(0) \neq 0. \tag{18.2}$$

Write $t = \sqrt{\varepsilon}$ and $z = tre^{i\theta}$. For $t, r > 0$, this is equivalent to

$$\begin{aligned} r^2 - 1 - t^{q-1}r^{q+1}\,\mathrm{Re}\big(e^{i(q+1)\theta}b(tre^{i\theta})\big) &= 0, \\ \mathrm{Im}\big(e^{i(q+1)\theta}b(tre^{i\theta})\big) &= 0. \end{aligned}$$

At $t = 0$ there are exactly $2q+2$ solutions: $r = 1$ and $(q+1)\theta + \arg b(0) \in \pi\mathbb{Z}$. The Jacobian in (r,θ) at each is

$$2(q+1)\,\mathrm{Re}\big(e^{i(q+1)\theta}b(0)\big) \neq 0.$$

The implicit function theorem gives $2q+2$ distinct nondegenerate zeros for small positive t.

To exclude other local zeros, shrink the disk so that $|b(z)| \leq C$ and $C|z|^{q-1} \leq 1/2$. Equation (18.2) implies

$$\big||z|^2 - \varepsilon\big| \leq \tfrac{1}{2}|z|^2, \qquad \sqrt{2/3} \leq |z|/\sqrt{\varepsilon} \leq \sqrt{2}.$$

Every possible scaled limit lies in this compact annulus and solves the limiting system. Local uniqueness at its $2q+2$ solutions therefore accounts for all zeros in the disk.

Finally, if $c \neq 0$ is the $w^m z^n$ coefficient of P, then

$$P_\varepsilon(z,\bar{z}) = c\bar{z}^m z^{n+1} + O(|z|^{m+n}) \quad (|z| \to \infty),$$

uniformly for ε in a bounded interval. Thus zeros cannot escape to, or enter from, infinity. On a fixed compact set outside disjoint neighborhoods of the old zeros, zF_P is bounded away from zero. There are no other new zeros. Replacing the old origin by $2q+2$ zeros proves the exact increment $2q+1$. The highest corner coefficient also proves the asserted bidegree. □

The contact condition in Proposition 18.3 is restrictive; it cannot be imposed on an arbitrary extremal polynomial by a change of coordinates. The linear bound measures that restriction.

Corollary 18.4. *Under the hypotheses of Proposition 18.3,*

$$N(P) \leq \min\{m^2 + n^2,\ (4m+1)n - m - 2q\}.$$

Proof. Apply Theorem 1.2 to P_ε, and subtract $2q+1$ from its linear bound. Apply Proposition 2.2 to P for the other term. □

For $q > 2m$, Corollary 18.4 improves the general linear bound by $2(q-2m)$. The local Jacobian signs in the proof of Proposition 18.3 alternate around the small circle. Of the $2q+2$ new local zeros, $q+1$ are positive and $q+1$ are negative. Since the old origin was negative, the total increments in the positive and negative counts are $q+1$ and q, respectively. Their difference agrees with the increase of the mapping degree by one.

19 Earlier near-extremal constructions and their limits

The following radial and angular construction was an intermediate step. Its count is correct, but for $m \geq 2$ it is not the maximum: the exact range proved above supplies two additional zeros. We retain the proof because its three distinct scales make the limitation of that ansatz transparent.

Proposition 19.1. *For every $m \geq 1$ there is a polynomial of bidegree $(m, 2m+2)$ with exactly $5m^2 + 8m + 2$ zeros, all nondegenerate.*

Proof. Set $k = m + 2$, and define

$$\theta_j = \frac{j\pi}{k} \quad (0 \leq j < k), \qquad \phi_\nu = \frac{(\nu + 1/2)\pi}{k} \quad (1 \leq \nu \leq m), \qquad \ell_\nu(x, y) = \sin\phi_\nu\, x - \cos\phi_\nu\, y.$$

For later use put

$$a_{\nu j} = \sin\frac{(\nu + 1/2 - j)\pi}{k}, \qquad L_j = \prod_{\nu=1}^{m} a_{\nu j}. \tag{19.1}$$

None is zero, $L_0 > 0$, and $\operatorname{sgn} L_j = (-1)^{j-1}$ for $1 \leq j < k$. Indeed, in the latter product exactly $j - 1$ factors are negative. Let

$$\sigma = \sin\frac{\pi}{2k}, \qquad R_\nu = \frac{\nu\sigma}{4m}, \qquad c_\nu = (k+1)^\nu, \qquad A(\rho) = \prod_{\nu=1}^{m}(R_\nu^2 - \rho).$$

The real plane polynomial

$$F_t(z) = z^k A(|z|^2) + t^2 \prod_{\nu=1}^{m}\bigl(|z|^2 + \ell_\nu(\operatorname{Re} z, \operatorname{Im} z) - c_\nu t^2\bigr) \tag{19.2}$$

is the restriction of a polynomial of bidegree $(m, m+k)$. Its imaginary part is $A(|z|^2)\operatorname{Im} z^k$. Thus zeros lie on the m circles $|z| = R_\mu$ or the k lines $z = e^{i\theta_j}s$, $s \in \mathbb{R}$.

On $|z| = R_\mu$, each factor of the product in (19.2) gives the line $\ell_\nu = c_\nu t^2 - R_\mu^2$, which cuts the circle transversely in two points for small t. Different factors give different points. To see this at $t = 0$, the nonzero common intersection of $|z|^2 + \ell_\nu = 0$ and $|z|^2 + \ell_\eta = 0$ has modulus $|\cos((\phi_\nu - \phi_\eta)/2)| \geq \sin(3\pi/(2k)) > \sigma$, whereas $R_\mu \leq \sigma/4$. The other common intersection is 0. There are no overlaps when $m = 1$ either. These separations persist for small t. The circle zeros also avoid the k lines because $|a_{\nu j}| \geq \sigma > R_\mu$. The circles contribute exactly $2m^2$ distinct zeros.

On the jth line the real equation is

$$f_{j,t}(s) = (-1)^j s^k A(s^2) + t^2 \prod_{\nu=1}^{m}(s^2 + a_{\nu j}s - c_\nu t^2) = 0. \tag{19.3}$$

Its degree is $3m + 2$. There are $2m$ simple real roots near $\pm R_\nu$. At scale $s = t^2 u$,

$$t^{-(2m+2)} f_{j,t}(t^2 u) \longrightarrow \prod_{\nu=1}^{m}(a_{\nu j}u - c_\nu).$$

This gives m distinct nonzero real roots. Distinctness follows from $\sigma \leq |a_{\nu j}| \leq 1$, $\sigma \geq 1/k$, and $c_{\nu+1}/c_\nu = k + 1$. At scale $s = tu$,

$$t^{-k} f_{j,t}(tu) \longrightarrow u^m\bigl((-1)^j A(0)u^2 + L_j\bigr).$$

Since $A(0) > 0$, (19.1) gives two nonzero real roots for $j \geq 1$, and two nonreal roots for $j = 0$. All are simple. The complex implicit function theorem preserves the last two roots, and real coefficients preserve reality of the real simple roots. The three scales are disjoint and account for all $3m + 2$ roots of (19.3). Its number of real roots is therefore $3m$ for $j = 0$ and $3m + 2$ for every other j.

The lines meet only at 0, where $F_t(0) \neq 0$, and their zeros are disjoint from the circle zeros. Their total contribution is $k(3m+2)-2$. Thus (19.2) has exactly $2m^2+(m+2)(3m+2)-2 = 5m^2+8m+2$ zeros. On the lines, simplicity of the restricted real equation and the nonzero normal derivative of the imaginary equation give a nonzero Jacobian. On the circles, the same conclusion follows from their transverse intersections with the product factors and from $\operatorname{Im} z^k \neq 0$. These checks also show directly that all zeros are isolated and nondegenerate. □

For $m = 2$, Proposition 19.1 gives 38 zeros. Section 20 will exhibit a different polynomial with 40. The loss of two in this construction is therefore not a universal obstruction for the bidegree problem. It is, however, an exact obstruction in the total-degree class considered next.

For $m \geq 2$, Theorem 1.3 supersedes this lower bound by $M_{m,2m+2} = m^2 + (2m + 2)^2$. In particular the two missing points in this specific construction were not a universal obstruction. The first total-degree-gap bound n^2-2 and its trace proof likewise remain valid, but Theorem 7.1 retains more information when the gap is larger.

20 Exact certificates and the remaining extremal problem

Constructions with many zeros may involve coefficients for which floating-point root finding is unreliable. A list of approximations alone does not prove either that the roots exist or that the list is complete. We use rational coefficients and exact contraction estimates. This serves the same purpose as the certified counts in [HLLM15, Section 1.3, author version, p. 3], although the verification below uses only rational arithmetic and the contraction mapping theorem.

20.1 Two explicitly certified polynomials

The coefficient arrays are printed in Appendix A. The supplementary files give the rational centers and independently executable verification programs.

Theorem 20.1. *The polynomial P_{40} in Appendix A has bidegree $(2, 6)$ and exactly 40 zeros on $w = \bar{z}$, all nondegenerate, with 22 positive and 18 negative zeros. The polynomial P_{49} has bidegree $(2, 7)$ and exactly 49 such zeros, all nondegenerate, with 27 positive and 22 negative zeros. Both polynomials are irreducible over $\mathbb{C}$.*

We give the verification argument in detail before drawing extremal conclusions. Every inequality in this argument is checked using exact fractions by the accompanying programs; numerical approximations are not used in the verification.

For either polynomial put $P^*(z, w) = \overline{P(\bar{w}, \bar{z})}$. The programs first prove coprimality over $\mathbb{C}$. Multiply the coefficients by their common denominator and reduce modulo 101, sending i to 10, whose square is -1 in $\mathbb{F}_{101}$. Writing $P = \sum_j p_j(z)w^j$, the tests are

$$\gcd(p_0, p_1) = 1 \quad \text{in } \mathbb{F}_{101}[z], \qquad \gcd(P(0, w), P^*(0, w)) = 1 \quad \text{in } \mathbb{F}_{101}[w].$$

The relevant leading coefficients are checked to remain nonzero. The first nonzero resultant excludes a common factor depending only on z. The second shows that the resultant in w of P, P^* is not identically zero, by specialization at $z = 0$; it excludes any common factor of positive w-degree. Nonvanishing modulo 101 implies nonvanishing of the Gaussian integer resultant and hence of the characteristic-zero resultant. The two curves are therefore coprime and have at most $m^2 + n^2$ complex intersections, counted with multiplicity.

For the real zeros, define

$$F(x, y) = \big(\operatorname{Re} P(x + iy, x - iy),\ \operatorname{Im} P(x + iy, x - iy)\big).$$

A certificate box has rational center c and is $B_c = c + [-r, r]^2$, where $r = 10^{-20}$. At its center the verifier computes, exactly,

$$A = DF(c)^{-1}, \qquad T(v) = v - AF(v), \qquad \eta = \|AF(c)\|_\infty.$$

Here vector norms are maximum norms and matrix norms are the induced maximum-row-sum norms. Write $P(z, w) = \sum_{i,j} a_{ij} z^i w^j$, put $R = |c_1| + |c_2| + 2r$, and define

$$H = \sum_{i+j\geq 2} (i + j)(i + j - 1)(|\operatorname{Re} a_{ij}| + |\operatorname{Im} a_{ij}|)R^{i+j-2}.$$

Every second partial derivative of either real component of F is bounded in absolute value by H on B_c. Consequently

$$\|DF(v) - DF(c)\|_\infty \leq 4rH, \qquad \|DT(v)\|_\infty \leq q := 4r\|A\|_\infty H.$$

For every box the exact inequalities verified are

$$q + \eta/r < 10^{-7}. \tag{20.1}$$

In particular, T is a contraction and $\|T(v) - c\|_\infty \leq \eta + qr < r$ for $v \in B_c$. It maps the closed box into its interior, so that box contains a unique zero. The estimate $\|I - A\,DF(v)\|_\infty < 1$ also shows that its Jacobian is invertible. All boxes are checked to be pairwise disjoint.

For P_{40} there are 40 such boxes. They exhaust the complex intersection bound $2^2 + 6^2 = 40$, and hence certify the exact count. For P_{49} there are 49 real boxes and four additional boxes for the complex system $P = P^* = 0$. In the latter boxes we regard (z, w) as four real coordinates and apply the same contraction argument to $(\operatorname{Re} P, \operatorname{Im} P, \operatorname{Re} P^*, \operatorname{Im} P^*)$. The Hessian bound uses $R = \max\{|\operatorname{Re} z_c| + |\operatorname{Im} z_c|, |\operatorname{Re} w_c| + |\operatorname{Im} w_c|\} + 2r$ and the contraction constant $q = 16r\|A\|_\infty H$. These boxes also satisfy (20.1). The four boxes are checked to avoid $w = \bar{z}$, and all 53 boxes are disjoint after the real boxes are embedded in four coordinates. They exhaust $2^2 + 7^2 = 53$ complex intersections. Exactly 49 intersections are fixed, proving the asserted real count.

The signs of the real Jacobians at the rational centers are computed exactly. They cannot change within a box because the Jacobian is invertible there, and yield the stated orientation counts. Irreducibility of P_{40} follows from Corollary 2.4. For P_{49} it follows from the next proposition, whose proof does not assume that irreducibility. This completes the count and nondegeneracy part of Theorem 20.1.

20.2 The reducible case and exact consequences

Let $M_{2,7}^{\mathrm{red}}$ denote the maximum finite zero count among reducible polynomials of bidegree $(2, 7)$.

Proposition 20.2. *One has* $M_{2,7}^{\mathrm{red}} = 41$.

Proof. If there is an irreducible factor of w-degree two, it occurs once and all other factors depend only on z. Thus $P = h(z)R(z,w)$ with $s = \deg h \geq 1$, $b = \deg_z R$, and $s + b = 7$. Proposition 2.2 gives

$$N(P) \leq s + 4 + b^2 = b^2 - b + 11 \leq 41 \qquad (0 \leq b \leq 6).$$

In every other case, including repeated factors, write $P = h(z)R_1(z,w)R_2(z,w)$ with both R_j of w-degree one. Put $d_j = \deg_z R_j$ and $s = \deg h$, so $s + d_1 + d_2 = 7$. Theorem 1.2 for $(1, d_j)$ when $d_j \geq 2$, and Proposition 2.2 for $d_j = 0, 1$, give the uniform estimate $N(R_j) \leq 5d_j + 1$. Therefore

$$N(P) \leq s + 5d_1 + 1 + 5d_2 + 1 = 37 - 4s \leq 37.$$

These cases prove the upper bound. To attain it, choose a outside the zero set of P_{40} and use $(z - a)P_{40}(z,w)$. □

Since $49 > 41$, Proposition 20.2 supplies the remaining irreducibility assertion in Theorem 20.1. It also shows why products of lower-degree examples cannot reach the best known count at $(2,7)$.

The exact equality cases from these certificates have been subsumed by Theorem 1.3; the 49-zero example remains a lower bound, not a proof that $M_{2,7} = 49$.

21 A fixed rational fourteen-term extremizer of bidegree $(3,9)$

The family in Section 17 already proves $M_{3,9} = 90$ and also proves the stronger pure-head assertion $E_{3,9} = 81$. The following separate, fourteen-term family is retained because it has a certified fixed rational parameter and an independently checkable local-to-global proof. Its total degree is twelve, so it is not an element of $\mathcal{E}_{3,9}$. Section 2 recalls why the same bound holds whenever the real zero set is finite, without a coprimality assumption.

Here the intersection bound is attained in bidegree $(3,9)$.

Theorem 21.1. *The exact maximum is*

$$M_{3,9} = 90.$$

It is attained by a polynomial with rational real coefficients and ninety nondegenerate zeros.

The construction is explicit. Put $c = 1/8$ and define

$$f = z^3 + zw - c, \qquad f^* = w^3 + zw - c, \qquad A = z^2w^2, \qquad B = -z^3. \tag{21.1}$$

Set

$$T_1 = f(Af + Bf^*), \tag{21.2}$$

and

$$\begin{aligned} T_2 = {} & -\frac{w^3}{1536} - \frac{7}{24}z^8w^2 + \frac{103}{192}z^7w + \frac{11}{384}z^4w - \frac{11}{12288}zw \\ & + \frac{13}{48}z^9 - \frac{83}{1536}z^6 - \frac{23}{6144}z^3 + \frac{13}{98304}. \end{aligned} \tag{21.3}$$

The final correction is

$$T_3 = -\frac{1}{8}z^9w^3 + \frac{169}{6912}z^2w^2 - \frac{29}{3072}zw + \frac{67}{98304}. \tag{21.4}$$

For a positive real parameter t, let

$$P_t = f^3 + tT_1 + t^2T_2 + t^3T_3. \tag{21.5}$$

The exact cancellations in (21.2) are important: although its unexpanded expression appears to allow w^4, its conjugate degree is only three. Table 2 records the full coefficient array of (21.5) and provides another definition of the family.

The following stronger statement gives both an analytic construction and a fixed rational example.

Theorem 21.2. *There exists $t_0 > 0$ such that, for every $0 < t < t_0$, P_t has exact bidegree $(3, 9)$ and exactly ninety nondegenerate zeros. The same conclusion holds for $t = 10^{-8}$. In both statements the numbers of positive and negative zeros are 48 and 42, respectively.*

The coefficient identities in Section 21.2 lead to the analytic small-parameter proof in Section 21.3. The particular value 10^{-8} is verified independently in Section 21.4; the analytic proof does not assume that an unspecified smallness threshold exceeds this number. Theorem 21.1 follows from Theorem 21.2 and Proposition 2.2. This settles a single bidegree beyond the range $n \le 2m+2$; it is not a formula for arbitrary $M_{m,n}$.

Table 2: The nonzero coefficients $[z^i w^j]P_t$. At $t = 10^{-8}$ all fourteen displayed coefficients are nonzero.

i	j	$[z^i w^j]P_t$
9	3	$-t^3/8$
9	0	$(13t^2 + 48)/48$
8	2	$-t(7t - 24)/24$
7	1	$(103t^2 - 192t + 576)/192$
6	3	t
6	0	$-(83t^2 - 192t + 576)/1536$
5	2	$-(5t - 12)/4$
4	1	$(11t^2 + 96t - 288)/384$
3	3	$-(t - 8)/8$
3	0	$-(23t^2 + 96t - 288)/6144$
2	2	$(169t^3 + 108t - 2592)/6912$
1	1	$-(116t^3 + 11t^2 - 576)/12288$
0	3	$-t^2/1536$
0	0	$(67t^3 + 13t^2 - 192)/98304$

21.1 A cubic with nine zeros

The elementary model used at both scales is the seed in (21.1).

Lemma 21.3. *The function $h(u) = u^3 + |u|^2 - 1/8$ has exactly nine nonzero, nondegenerate zeros. Six have positive Jacobian and three have negative Jacobian. If*

$$\omega = e^{2\pi i/3}, \qquad r_1 = -\frac{1}{2}, \qquad r_2 = \frac{-1+\sqrt{5}}{4}, \qquad r_3 = \frac{-1-\sqrt{5}}{4},$$

its zero set is $\{r_a \omega^j : 1 \le a \le 3,\ 0 \le j < 3\}$.

Proof. The imaginary part is $\operatorname{Im} u^3$, so every zero lies on $\bigcup_{j=0}^{2} \omega^j \mathbb{R}$. On each of these three lines the equation is $r^3 + r^2 - 1/8 = 0$, whose factorization is

$$r^3 + r^2 - \frac{1}{8} = \left(r + \frac{1}{2}\right)\left(r^2 + \frac{1}{2}r - \frac{1}{4}\right).$$

The three roots displayed in the statement are distinct and nonzero. The three lines meet only at zero, which is not a root. At a real root r, the Jacobian is $3r^3(3r+2)$; it is negative for $r = -1/2$ and positive at the other two roots. Rotating the source by ω preserves the function and the Jacobian sign. □

We shall also need translated and rescaled copies of this model. No inequality between the quadratic coefficients is required.

Lemma 21.4. *Let $a, b \in \mathbb{C}$ with $b \neq 0$. The real-analytic polynomial*

$$\mathcal{M}_{a,b}(u) = u^3 + au^2 + bu\bar{u} + \frac{a^2 + b\bar{a}}{3}u + \frac{ba}{3}\bar{u} + \frac{a^3}{27} + \frac{b|a|^2}{9} - \frac{1}{8}|b|^2 b \tag{21.6}$$

has exactly nine nondegenerate zeros, six positive and three negative.

Proof. Choose $\lambda \neq 0$ such that $|\lambda| = |b|$ and $\lambda^3 = |b|^2 b$. With $v = (u + a/3)/\lambda$, direct expansion gives

$$\mathcal{M}_{a,b}(u) = |b|^2 b \left(v^3 + |v|^2 - \frac{1}{8}\right).$$

Both the affine change of source and multiplication of the target by a nonzero complex number preserve nondegeneracy and Jacobian signs. Apply Lemma 21.3. □

21.2 Two polynomial identities

The corrections (21.3)–(21.4) were chosen to prescribe the linear and constant terms of (21.6) at every seed zero without increasing the bidegree. Lemma 21.5 states all the algebra needed for the construction.

Let $I = (f, f^*) \subset \mathbb{Q}[z, w]$. Define

$$D = \frac{A^2 + BA}{3}, \qquad E = \frac{BA}{3}, \qquad G = \frac{A^3}{27} + \frac{BA^2}{9} - \frac{1}{8}B^2 B^*. \tag{21.7}$$

Here $A^* = A$. At a real slice point this is precisely the occurrence of $\bar{a}$ and $|b|^2 b$ in (21.6).

The first identity is a statement about first derivatives, not just values.

Lemma 21.5. *The polynomials above satisfy*

$$T_2 - Df - Ef^* \in I^2, \tag{21.8}$$

and

$$T_3 - G \in I. \tag{21.9}$$

Moreover, with $\widehat{P}_t(u,v) = u^9 v^3 P_t(1/u, 1/v)$,

$$t^{-3}\widehat{P}_t(tu, tv) \longrightarrow v^3 + uv - \frac{1}{8} \tag{21.10}$$

in C^1 *on every compact subset of* $\mathbb{C}^2$ *as* $t \to 0^+$.

Proof. For (21.8), an explicit identity is

$$T_2 - Df - Ef^* = U(f^*)^2 + Vff^* + Wf^2, \tag{21.11}$$

where

$$\begin{aligned} U &= -\frac{z^3}{3}, \\ V &= \frac{z^2w^2}{3} - \frac{z^4w}{3} + \frac{zw}{24} - \frac{z^3}{12} + \frac{1}{192}, \\ W &= \frac{3}{8}z^2w^2 + \frac{7}{192}zw + \frac{13}{48}z^3 + \frac{5}{1536}. \end{aligned}$$

Expanding the right side of (21.11), with (21.1) and (21.7), gives exactly (21.3).

For (21.9), reduce by the two relations $z^3 = 1/8 - zw$ and $w^3 = 1/8 - zw$. Their leading monomials are coprime, so these are a Gröbner basis in a graded order. The remainder of $T_3 - G$ is zero. This assertion may also be checked as a one-variable calculation: at $f = f^* = 0$, put $s = zw$. Then

$$z^3 = w^3 = \frac{1}{8} - s, \qquad s^3 = \left(\frac{1}{8} - s\right)^2,$$

and reduce $G + z^9w^3/8$ to degree at most two in s. The remainder is $169s^2/6912 - 29s/3072 + 67/98304$. The two-variable remainder calculation proves the ideal membership, and in particular its required value consequence.

For (21.10), consider a term $t^k z^i w^j$ of (21.5). After inversion, rescaling, and division by t^3, it becomes $t^{k+9-i-j}u^{9-i}v^{3-j}$. All these powers of t are nonnegative. The terms with exponent zero are z^9 in f^3, tz^8w^2 in tT_1, and $-t^3z^9w^3/8$ in t^3T_3. They give $v^3 + uv - 1/8$. All remaining terms have a positive integral power of t, which proves convergence together with all derivatives on compact sets. □

The accompanying verifier checks (21.11) by polynomial arithmetic over $\mathbb{Q}$, and checks (21.9) by the explicit degree-reducing substitutions in the proof. The construction below uses these identities, not a numerical approximation to them.

21.3 Eighty-one finite zeros and nine zeros at infinity

We prove the small-parameter part of Theorem 21.2. By Table 2, every term of P_t has $i \le 9$ and $j \le 3$, and its corner coefficient is $-t^3/8$. Thus its bidegree is exactly $(3, 9)$ for every $t > 0$.

21.3.1 The finite clusters

Let z_0 be any of the nine zeros of F_f. Lemma 21.3 and the real inverse function theorem give a real-analytic local inverse $z = \psi(\xi)$ of $\xi = f(z, \bar{z})$, with $\psi(0) = z_0$. On the real slice, $f^*(z, \bar{z}) = \bar{\xi}$. Set

$$a = A(z_0, \bar{z}_0), \qquad b = B(z_0, \bar{z}_0).$$

The seed zeros are nonzero, so $b = -z_0^3 \neq 0$.

Substitute $z = \psi(tu)$ and divide $P_t(z, \bar{z})$ by t^3. The first two terms of (21.5) become

$$u^3 + A(\psi(tu), \overline{\psi(tu)})u^2 + B(\psi(tu), \overline{\psi(tu)})u\bar{u}.$$

By (21.8), the third term converges to $D(z_0, \bar{z}_0)u + E(z_0, \bar{z}_0)\bar{u}$: its error is $O(t)$ in C^1 on bounded sets. Indeed, the remainder belongs to $(\xi, \bar{\xi})^2$. By (21.9), the last term converges to $G(z_0, \bar{z}_0)$. Hence

$$t^{-3}P_t\big(\psi(tu), \overline{\psi(tu)}\big) \longrightarrow \mathcal{M}_{a,b}(u) \tag{21.12}$$

in C^1 on compact subsets of the local u-plane.

By Lemma 21.4, the right side of (21.12) has nine simple zeros, each of which persists for small positive t. Choose their neighborhoods disjoint, and choose the nine seed neighborhoods disjoint as well. There are finitely many of these choices, so a single positive threshold for t works for all of them. We obtain $9 \cdot 9 = 81$ distinct nondegenerate zeros, all in a fixed bounded set. At this point only a lower count is needed; no assertion about other scales is being made.

21.3.2 The unbounded cluster

Restrict (21.10) to $v = \bar{u}$. Its limit is

$$\bar{u}^3 + |u|^2 - \frac{1}{8} = \overline{h(u)}.$$

It has the same nine nonzero simple zeros as h. Apply the real implicit function theorem in disjoint neighborhoods of these nine points. For small positive t this gives nine simple zeros of $\widehat{P}_t(tu, t\bar{u})$ with u bounded away from zero. They correspond to zeros of P_t under

$$z = \frac{1}{tu}.$$

The corresponding $|z|$ tend to infinity. They are thus distinct from all eighty-one bounded zeros. Inversion is holomorphic off zero; multiplication by a nonvanishing complex factor at a zero also preserves the sign and nonvanishing of its real Jacobian. All nine new zeros are nondegenerate.

21.3.3 Completeness and signs

We have constructed ninety distinct nondegenerate zeros. By Lemma 2.3, P_t, P_t^* are coprime, and their bidegree intersection number is $9^2 + 3^2 = 90$. This proves that the zero set is finite and contains no additional zeros.

For the signs, the seed has six positive and three negative zeros. At a positive seed zero, (21.12) supplies six positive and three negative zeros; at a negative seed zero, the signs are reversed by the local coordinate f. The bounded clusters therefore contain $6{\cdot}6{+}3{\cdot}3 = 45$ positive and $6{\cdot}3{+}3{\cdot}6 = 36$ negative zeros. The model at infinity is the conjugate of h, so it contributes three positive and six negative zeros. The total is 48 positive and 42 negative zeros, as stated in Theorem 21.2.

21.4 An exact certificate for a fixed rational parameter

This section proves the assertion in Theorem 21.2 for $t = 10^{-8}$ independently of any unspecified smallness threshold. The certificate contains ninety centers in $10^{-100}\mathbb{Z}^2$ and uses the common box radius $r = 10^{-40}$. The polynomial has the fourteen rational coefficients in Table 2, evaluated at this t. The file `bidegree_3_9_certificate.json` records these data exactly.

Here is the inclusion argument checked for every box. Let

$$V(x,y) = \big(\operatorname{Re} P_t(x+iy, x-iy), \operatorname{Im} P_t(x+iy, x-iy)\big)$$

and let c_0 be the rational center of a box $\mathcal{B} = c_0 + [-r,r]^2$. All entries of $V(c_0)$ and $DV(c_0)$ are rational. Put

$$L = DV(c_0)^{-1}, \qquad \eta = \|LV(c_0)\|_\infty.$$

Writing $P_t(z,w) = \sum a_{ij} z^i w^j$, define the rational number

$$H = \sum_{i+j\ge 2} |a_{ij}|(i+j)(i+j-1)\big(|(c_0)_x| + |(c_0)_y| + 2r\big)^{i+j-2}. \tag{21.13}$$

Every second real derivative of either component of V is bounded by (21.13) on $\mathcal{B}$. Thus

$$\|DV(v) - DV(c_0)\|_\infty \le 4rH \qquad (v \in \mathcal{B}).$$

For $T(v) = v - LV(v)$, let $q = 4r\|L\|_\infty H$. The exact verifier checks

$$q + \frac{\eta}{r} < 10^{-12} \tag{21.14}$$

for each of the ninety boxes. It follows that T maps each box strictly into itself and is a contraction there. Its unique fixed point is a zero of V. Moreover, $\|I - LDV(v)\|_\infty < 1$, so DV is invertible throughout the box.

The verifier also checks that the boxes are pairwise disjoint, that the center Jacobian signs are 48 positive and 42 negative, and that the coefficient data agree exactly with (21.5). All comparisons in (21.14) use integer fractions. The resulting ninety simple zeros force coprimality by Lemma 2.3, and therefore exhaust the zero set.

The certificate is independently executable with the Python standard library:

```
python verify_bidegree_3_9.py
```

No numerical root finder is called by this program. The role of approximation was only to locate the rational boxes; existence, nondegeneracy, and completeness follow from exact inequalities and the intersection argument.

22 Audit of the successive claims and the remaining questions

The research record contains a sequence of improving bounds, several changes of coefficient space, and both analytic and computer-assisted constructions. The following ledger states what is retained. A claim superseded by a stronger theorem is not thereby false; conversely, sharpness inside a restricted family is not evidence of sharpness in a larger one.

22.1 The original problem and assertions that must not be conflated

Problem 1.1 asks only for a coefficient-uniform bound linear in n for fixed m. Theorem 1.2 proves that statement. Its asymptotic coefficient is sharp by Proposition 5.2. Determining $M_{m,n}$ for every pair is a strictly stronger problem. The polynomial part of Eremenko's note is addressed; the elliptic systems on its later pages are not solved by any argument in this note.

The finite equality range in Theorem 1.3 and the cubic family in (17.1) do not cover $(m,n) = (2,7)$. The certified example of Theorem 20.1, the intersection bound, and the parity result in Section 4 imply precisely

$$49 \le M_{2,7} \le 53, \qquad M_{2,7} \in \{49, 51, 53\}.$$

None of the analytic constructions or exact certificates supplied here chooses among these three possibilities. In particular, the candidate formula $M_{m,n} = \min\{m^2+n^2, (4m+1)n - 5m\}$ remains a question outside the proved equality range, not a theorem inferred from small examples.

The maximum in the harmonic class is a different question again. Its best proved general bounds here are (1.3). For each fixed $m \ge 2$ they leave a gap between the constructed coefficient $m+2$ and the upper coefficient $3m+2$. Neither existence of a limit of $M^{\mathrm{harm}}_{m,n}/n$ nor its value for arbitrary $m \ge 2$ is proved. The suggested lower bound $2m(n-1)$ cannot hold throughout all $m < n$: for $n = m+1$ and $m \ge 3$, it exceeds the classical upper bound n^2. For fixed m and sufficiently large n, that observation supplies no obstruction, so it does not settle the corresponding asymptotic question.

22.2 A claim-by-claim ledger

Assertion in the successive drafts	Audited disposition and proof in this note
The initial quadratic estimate m^2+n^2	Correct for finite zero sets, including common conjugate factors. Proposition 2.2 gives a direct factorwise proof independent of regularization.
The intermediate bounds $6mn-5m$ and $5m(n-1)$	Valid but superseded by (1.1); they are not advertised as the strongest result. A constant perturbation alone is not sufficient for the reduction.
The rectangular coefficient $4m+1$	Sharp asymptotically. The radial family is generally not harmonic; its mixed terms are essential to that particular construction.
The first equality band $m < n \le 2m+1$, and individual 40-, 64-, and 73-zero examples	Correct; subsumed by the larger band $m \ge 2$, $m < n \le 2m+2$. The general proof of the extra boundary uses the certified degree-eight input in Appendix B.
The earlier $(m, 2m+2)$ construction with two fewer zeros	Its exact count is correct, as proved in Section 19, but the deficit is not universal. It is suboptimal for every $m \ge 2$.
The explicit 90-zero family at $(3,9)$	Correct, including the fixed parameter 10^{-8}; Section 21. It has total degree twelve and does not itself prove $E_{3,9} = 81$.
The later cubic construction	Gives the stronger pure-head equality and an infinite rectangular family: (17.1). The exceptional first parameter $r = 1$ is treated separately rather than substituted into an invalid degree inequality.

Assertion in the successive drafts	Audited disposition and proof in this note
The 49-zero example at $(2, 7)$	Exactly 49, not merely at least 49, by certification of all 53 complex intersections. It does not prove extremality in the full rectangle. The reducible maximum is separately 41.
The monomial-head harmonic formula	Exact for $S_{m,n}$ only. The zero count is not invariant under arbitrary imaginary translations of the target within that restricted family.
The Chebyshev construction	Correct exact count for the constructed family; supplies a lower bound, not a general harmonic maximum. The resonant directions at infinity are included in its proof.
The real-valued perturbation formula	Exact for $A_{m,n}$, including its leading-form refinement. Allowing an independent imaginary lower part or an arbitrary analytic head changes the problem.
The harmonic linear improvement and the Newton-support extension	The former is a special case of (6.7). The proof requires the negative-double-point version of the genus inequality and a separate check of the link at infinity; the embedded statement alone is not enough.
The $n^2 - 2$ total-degree statement	Correct and sharp for a gap of two. Theorem 7.1 strengthens its upper bound for larger gaps; general equality for those gaps is not claimed.
The quintic sparsity counterexample	Correct, but the trinomial quartic gives an earlier and stronger counterexample. Degree four and three terms are both optimal for the first failure.
The even- and odd-degree sparse families	Attain the quadratic zero bound. Their number of terms is an upper bound for s_n, not a proved minimum in every degree.
The cyclic symmetry formula	Exact for the entire cyclic coefficient space, including complex radial coefficients. It is not a formula for all polyanalytic polynomials with the same degrees.
Circle addition, contact splitting, and zeros at infinity	Correct deformation tools, with the degree conditions explicitly stated. They do not guarantee the existence of an extremal starting polynomial with an arbitrarily prescribed contact order.
The rational degree-one endpoints	Already covered by [KN06, LSL15, LZ18, LM14]; they are included for completeness, not claimed as new.
The use of Descartes signs and Hermitian traces	Standard antecedents are cited. The simultaneous extremal counts, the particular low-degree trace identity, and the isotropic-space estimate are proved rather than attributed to the general framework alone.

22.3 The exact certificate inputs

The logical role of computation is limited and explicit. The symbolic proofs of the rectangular upper bound, its asymptotic coefficient, the support and gap bounds, the harmonic and real-valued counts, and the cubic family do not depend on floating-point searches. The completion of the entire boundary $n = 2m + 2$ uses one finite rational input: the 64-zero polynomial and two auxiliary inequalities in Appendix B. All subsequent degrees in that argument follow by analytic iteration and odd-degree composition.

Other exact certificates provide explicit finite-parameter examples, not new logical assumptions for those analytic theorems. The reproduction archive contains the original rational inputs, independently executable programs, and the logs of the reruns performed during this consolidation.

Its machine-readable manifest records the SHA-256 digest of each file. Hashes identify the actual input; they do not replace a proof of the verification algorithm, which is given in Section 20 and the appendices.

The following actual checks were completed. Their finite parameter ranges are stated to avoid treating tests as universal proofs.

Check	Content and scope
Rectangular 40 and 49	All real boxes; the four additional nonfixed complex boxes for 49; modular coprimality tests; orientation signs.
Pure-head 36; pure-head 64 and rectangular 73	Exact contraction inequalities on every rational box.
Full-band auxiliary input	All 64 seed boxes and both auxiliary inequalities throughout each box, not just at its center.
Fourteen-term 90	Both polynomial-ideal identities and all 90 boxes at $t = 10^{-8}$.
Harmonic sign counts	2016 degree pairs and 11466 sparse coefficient-sign patterns; diagnostic checks in addition to the all-degree proof.
Harmonic 20 and Chebyshev 17	Exact Sturm calculations over the indicated real quadratic fields.
Degree-gap trace calculation	20 coefficient sets, 470 trace identities, and 149 isotropic matrix entries, using exact arithmetic.
Newton formulas	7006 integer triples, checking lattice, genus, and constant expansions; not a substitute for the surface proof.
Real-valued construction	Exact quartic check; the rerun used the reduced exhaustive ranges of 1225 degree/form triples and 8184 binary cycles. The default larger exhaustive run was not completed and is not recorded as passed.
Sparse constructions	Exact quintic factorization; odd-degree Sturm tests; recurrence identities and cyclotomic quotient checks; the quartic and cyclic checks from the final sparse supplement.
Cubic family	The cubic discriminant identities, 1860 rational endpoint comparisons, cancellation checks for $2 \le r \le 8$, and every special $r = 1$ correction.

22.4 What the source audit does and does not establish

A source locator is meaningful only together with a version. For example, page 16 of the published Sète–Zur article contains Remark 4.4; the corresponding arXiv version has different pagination. Likewise, Lazebnik–Lundberg is cited as its 2026 published article, whereas the Lee–Makarov pole-prescription theorem is cited as the specified 2014 preprint. Lu–Wang is a 2025 publication despite its 2026 arXiv posting. No DOI was identified in the checked sources for the Peretz–Schmid conference article or the Passare–Rojas–Shapiro journal article; no DOI has been manufactured from a journal's usual URL pattern.

The most consequential source distinctions are these: the polynomial conjecture is on the first page of Eremenko's note, not only in a MathOverflow discussion; the stronger orientation-reversing conjecture in [LLL15] remains separate; and average Gaussian zero counts do not settle maximal

valence. The current public status notice in [Ere26b] records a solution of the qualitative estimate, but does not by itself establish attribution or validate every sharper formula in this note. This review found no cited theorem that makes all of the new constructions and bounds redundant. That statement is not an exhaustive priority search through unpublished work.

22.5 Conclusion

The distinction between the resolved growth problem and the unresolved finite-degree maximum should remain visible in any later manuscript. The exact results for smaller coefficient spaces are substantial in their own right; they need not be overstated as answers to larger questions.

Conclusion. Fixed conjugate degree gives a sharp asymptotically linear bound for the full rectangular class. Several finite-degree ranges and natural subfamilies have exact extrema, with complete analytic or explicit computer-assisted proofs. The full bidegree formula and the optimal fixed-degree harmonic coefficient remain undetermined here.

A Rational coefficients and verification files

The following arrays completely specify the two polynomials in Theorem 20.1. With row index ℓ and column index j, write

$$P(z,w) = D^{-1}\sum_{\ell,j}(u_{\ell j} + iv_{\ell j})z^{\ell}w^{j}.$$

All entries are integers, not rounded decimal approximations.

The polynomial P_{40}

Here $D = 10^{10}$ and $0 \le \ell \le 6$, $0 \le j \le 2$.

Real numerators

ℓ	$u_{\ell 0}$	$u_{\ell 1}$	$u_{\ell 2}$
0	9989148	−1437879431	42110613567
1	−1488721221	85880077969	44531370007
2	41495821383	181048684005	−8945073549
3	127109446221	24463634439	−54584415742
4	70927784116	−90270783729	−13319637790
5	−30642879880	−58978230217	12977971008
6	171904696	8861758778	10000000000

Imaginary numerators

ℓ	$v_{\ell 0}$	$v_{\ell 1}$	$v_{\ell 2}$
0	2386743	−536302879	19849756729
1	−170102520	20587381168	26848909014
2	1758707049	27892403146	9999884021
3	−18670288098	19443937659	−6680374702
4	−10494108981	26168926864	−13516927630
5	7007756920	1936949230	−10640650833
6	4236054207	1869520737	0

The polynomial P_{49}

Here $D = 10^3$ and $0 \le \ell \le 7$, $0 \le j \le 2$.

Real numerators

ℓ	$u_{\ell 0}$	$u_{\ell 1}$	$u_{\ell 2}$
0	−42	−176	−139
1	−227	933	−432
2	−179	−1457	737
3	794	3712	353
4	−2958	3048	−1571
5	7455	1348	−1396
6	−448	−5235	−166
7	−68	158	1000

Imaginary numerators

ℓ	$v_{\ell 0}$	$v_{\ell 1}$	$v_{\ell 2}$
0	−33	−479	−2287
1	−262	−1987	653
2	−1501	−8252	965
3	−4456	4656	3427
4	−7424	1665	−1101
5	1057	8157	−195
6	−1669	−264	−2045
7	363	697	0

Reproducibility

The two data files

```
sharp_2_6_40_certificate.json
sharp_2_7_49_certificate.json
```

contain these same integer coefficients, all rational box centers, their common denominator 10^{40}, and the radius 10^{-20}. The second file also contains the four nonreal centers. Place each data file beside its corresponding verifier and run

```
python verify_sharp_2_6_40.py
python verify_sharp_2_7_49.py
```

Only the Python standard library is used. The arithmetic in the verification is performed with exact fractions and finite fields. The scripts reject execution with Python's optimization option, since disabling assertions would invalidate the verification. They check all hypotheses used in Section 20, including bidegree, coprimality, invertibility of the center Jacobians, box separation, contraction, and orientation. The second script additionally checks separation from the fixed plane and exhausts all 53 complex intersections. The supplement includes the verification logs and SHA-256 hashes of the four certificate files.

The harmonic example of Section 8 has a separate verifier, `verify_harmonic_20.py`, which requires SymPy and uses exact arithmetic over $\mathbb{Q}(\sqrt{3})$. The supplementary program `check_`

`harmonic_signs.py` checks the finite sign counts for all $1 \le m < n \le 64$ using integer and rational arithmetic. This last check is a regression test, not a substitute for the proof for arbitrary degrees. Neither harmonic construction depends on numerical root finding.

B The finite rational certificate

This appendix makes the computer-assisted input in Proposition 14.2 reproducible. The only starting polynomial requiring certification is

$$f_0(z,w) = z^8 + 10^{-18} \sum_{j=0}^{3} \sum_{i=0}^{7-j} (a_{ij} + ib_{ij}) z^i w^j, \tag{B.1}$$

with integers from Table 5. The two new auxiliary polynomials are specified by

$$A_0(z,w) = 10^{-8} \sum (\alpha_{ij} + i\beta_{ij}) z^i w^j, \qquad S_0(z,w) = 10^{-8} \sum (\sigma_{ij} + i\tau_{ij}) z^i w^j, \tag{B.2}$$

using Tables 6 and 7. Every omitted coefficient is zero. In particular, the coefficient of w^5 in A_0 and the coefficient of w in S_0 are exactly one, not approximations.

B.1 Existence of the 64 seed zeros

The file `E3_8_64_certificate.json` contains the integer coefficients in (B.1) and 64 centers $c_\nu = (X_\nu/10^{60}, Y_\nu/10^{60})$. Put

$$B_\nu = c_\nu + [-\rho, \rho]^2, \qquad \rho = 10^{-30}.$$

The verifier checks that the boxes are pairwise disjoint and avoid the origin. For

$$V(x,y) = (\operatorname{Re} f_0(x+iy, x-iy), \operatorname{Im} f_0(x+iy, x-iy)),$$

write $J_\nu = DV(c_\nu)$, $K_\nu = J_\nu^{-1}$, and $\eta_\nu = \|K_\nu V(c_\nu)\|_\infty$. All these quantities are rational. If $f_0 = \sum a_{ij} z^i w^j$, set

$$R_\nu = |c_{\nu,x}| + |c_{\nu,y}| + 2\rho, \quad H_\nu = \sum_{i+j \ge 2} (i+j)(i+j-1)(|\operatorname{Re} a_{ij}| + |\operatorname{Im} a_{ij}|) R_\nu^{i+j-2}.$$

Every second real partial derivative of either component of V is bounded by H_ν on the box. Thus

$$\|DV(v) - DV(c_\nu)\|_\infty \le 4\rho H_\nu, \qquad q_\nu = 4\rho \|K_\nu\|_\infty H_\nu.$$

For every box, the exact program checks

$$q_\nu + \eta_\nu/\rho < 10^{-12}. \tag{B.3}$$

It follows from (B.3) that $T_\nu(v) = v - K_\nu V(v)$ maps B_ν strictly into itself and is a contraction. Its unique fixed point is a zero of V. Moreover, $\|I - K_\nu DV(v)\|_\infty < 1$ on the whole box, so the Jacobian is invertible there. This proves nondegeneracy as well as existence. The orientation counts are 36 positive and 28 negative, consistent with degree eight. They are not used as a substitute for the contraction certificate.

B.2 The auxiliary inequalities on entire boxes

For a polynomial $P = \sum p_{ij} z^i w^j$, a box center $c = (x, y)$, and $R = |x| + |y| + 2\rho$, define the rational number

$$L_P = \sum_{i+j\geq 1} (i+j)(|\operatorname{Re} p_{ij}| + |\operatorname{Im} p_{ij}|)R^{i+j-1}, \qquad \delta_P = 2\rho L_P.$$

Each real component of $P(z,\bar{z}) - P(c,\bar{c})$ has absolute value at most δ_P in the box. If $P(c,\bar{c}) = u + iv$, the rational numbers

$$\begin{aligned} \ell_P &= \max(0, |u| - \delta_P)^2 + \max(0, |v| - \delta_P)^2, \\ u_P &= (|u| + \delta_P)^2 + (|v| + \delta_P)^2 \end{aligned}$$

therefore satisfy $\ell_P \leq |P(z,\bar{z})|^2 \leq u_P$ throughout the box. The symbol u_P here denotes a bound, not a polynomial variable.

Using the exact polynomials B_0, C_0, D_0, e_0, d_0 in Proposition 14.2, the verifier checks, in every box,

$$\ell_{B_0} > 0, \quad \ell_{d_0} > 0, \quad u_{A_0} < \left(\frac{3}{50}\right)^2 \ell_{B_0}, \quad \left(\frac{53}{50}\right)^2 u_{S_0} u_{B_0} < \left(\frac{7}{10}\right)^2 \ell_{d_0}. \tag{B.4}$$

The first modulus inequality in (14.7) follows immediately. It implies $|A_0| + |B_0| < (53/50)|B_0|$, so the last comparison in (B.4) gives the second modulus inequality. This proves the exact finite hypotheses needed for the iteration, including nonvanishing of B_0 and d_0.

B.3 Verification files and scope

The programs `verify_seed.py` and `verify_auxiliary.py` verify the seed boxes and the auxiliary inequalities, respectively. Both JSON input files are supplied. Running the wrapper

```
python verify_all.py
```

checks all degree conditions, the 64 root boxes, and every comparison in (B.3) and (B.4). It uses Python's standard-library `Fraction` type for every mathematical comparison. The auxiliary verifier refuses to run with Python's optimization flags that would disable its assertions. Informational decimal ratios printed at the end do not enter the proof. The complete exact run has been performed and its output is included.

The verifier supplies only the finite assertion in Proposition 14.2. Proposition 14.1, Corollary 14.3, and Proposition 15.1 are the proofs in all higher degrees. No extrapolation from finite numerical experiments is involved.

Table 5: Exact integers in (B.1). The coefficient of z^8 is one; every coefficient not specified here or by that term is zero.

i	j	a_{ij}	b_{ij}
0	0	325069001102644	67963801878226
1	0	450588204151335	-808429770939430
2	0	14923899097622392	-5215338168864671
3	0	-4193995275619317	-5607909027127370
4	0	-2285437933666029	1137114477688244
5	0	-256070329168631650	47437728771068840
6	0	39523562708151494	64053725075318900
7	0	-5060183014488084	-65745045532666410
0	1	4177129780392874	-1023578682503584
1	1	-3393291315201153	1031467504051507
2	1	1430170989562624	19247381823938400
3	1	-344982929090280050	106363039160048020
4	1	34205364104215395	44197369842432850
5	1	-33034647699973135	-122080649425548670
6	1	2542780791248466000	-411360496357260800
0	2	1072493974482530	2260721271437080
1	2	-87741687798084990	29388032582878587
2	2	-8388736302391480	-16179666096049084
3	2	-44302401572865020	-131522667442191320
4	2	1973476089687808700	-614554296250387700
5	2	88015603513326930	68500967995269020
0	3	-1048236220001323	-1254186823454144
1	3	-12625398733567206	-19057910533194857
2	3	449481778934072500	-191881909553323800
3	3	84177992300826730	54870976261917030
4	3	7730029550774110	45273336743991960

Table 6: Exact integers defining A_0 in (B.2); the denominator is 10^8.

i	j	α_{ij}	β_{ij}
0	0	-365212	-192995
1	0	-3524147	-1005031
2	0	-1938007	-2579631
3	0	-1503160	4757283
4	0	-1735632	-1234227
5	0	959672	20316700
0	1	-2286231	-3022935
1	1	3964573	2244562
2	1	38217049	14420232
3	1	17236977	41263280
4	1	1994439	-18949910
0	2	-11343430	-4404664
1	2	21665754	32721321
2	2	-12502311	3057533
3	2	-11767481	12691914
0	3	5185651	-3728884
1	3	133662149	42271688
2	3	8234474	7491773
0	4	18517574	20072885
1	4	-12040293	25284423
0	5	100000000	0

Table 7: Exact integers defining S_0 in (B.2); the denominator is 10^8.

i	j	σ_{ij}	τ_{ij}
0	0	-10030551	-894598
1	0	6150734	-2984218
2	0	-134397669	-948638
3	0	15813796	87467028
0	1	100000000	0

C Two additional fixed-parameter certificates

The following examples were used at intermediate stages of the research. Their extremal values are subsumed by Theorem 1.3, but the shorter fixed-parameter representatives remain useful for independent reproduction. These are polyanalytic, not harmonic, examples.

First put

$$\alpha = 3 - 6i, \qquad \beta = -4 + 32i, \qquad H = z^3 + \alpha z w + \beta,$$

and define

$$D = z^3 w^2 + \beta w^2 - 45 z^2 w + (204 + 72i) z, \qquad F_{36} = H^2 + \frac{1}{1000} D.$$

The explicit identity $D = w^2 H - \alpha z H^*$ cancels the forbidden highest conjugate term. The resulting polynomial has total degree six, unique highest homogeneous term z^6, and conjugate degree two.

The file `sharp_even_diagonal_36_certificate.json` and its verifier certify 36 disjoint simple real-slice zeros by the contraction argument of Section 20. There are 21 positive and 15 negative zeros. The pure-head bound proves that these exhaust the zero set, giving a fixed rational representative of $E_{2,6} = 36$.

Second, use precisely the polynomial f_0 of (B.1), and put

$$G_{73} = f_0 + (3 + 6i)10^{-20} z^7 w^2 - (4 + 32i)10^{-60} z^8 w^3.$$

The companion certificate `M3_8_73_certificate.json` gives 73 disjoint simple real-slice zeros, 39 positive and 34 negative. Since $73 > (8-1)^2 + (3-1)^2 = 53$, the transverse-zero criterion in Proposition 2.3 forces G_{73} and G_{73}^* to be coprime. Their bidegree intersection number is 73, so the certified zeros are the complete zero set. Thus this is a fixed rational representative of $M_{3,8} = 73$.

The programs for these two assertions are `verify_sharp_even_diagonal_36.py` and `verify_extremal_3_8.py`. Each is supplied with its original JSON input in the reproduction archive. The first also checks the coefficient formula above; the second verifies both the 64- and the 73-zero inputs. No approximation is used in the contraction comparisons.

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